



THE UNIVERSITY OF BRITISH COLUMBIA

Mass Appraisal Report: Residential Properties, Marilu City

BUSI 444 Case Study Examination

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May 11, 2025

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LETTER OF TRANSMITTAL

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November 30, 2017

The Assessor

City of Marilu

Marilu City

Re: Mass Appraisal Report for Residential Properties, Marilu City – 2018 Assessment Roll

Dear Assessor

In accordance with terms of our engagement, I am pleased to transmit the accompanying mass appraisal report. This report details the development, calibration, and testing of valuation models for selected residential properties within Marilu City. The primary objective of this assignment was to provide your office with a statistically sound and defensible model to estimate the full market value of the fee simple interest in these properties as of the valuation date, July 1, 2017, for use in establishing the 2018 Assessment Roll. The sales data analyzed primarily spanned the period from March 2017 to October 2017.

The scope of work undertaken involved the development of two distinct mass appraisal models:

1. A market-adjusted Cost Model.
2. A Sales Comparison Model utilizing Multiple Regression Analysis (MRA).

Both models were developed using the provided database of property sales and characteristics for specified neighbourhoods within Marilu City. The process included comprehensive data screening, variable transformation, model specification, rigorous calibration, and thorough statistical testing against recognized mass appraisal standards, including those promoted by the International Association of Assessing Officers (IAAO).

A comparative summary of the final performance of both models on the full dataset is as follows:

- **Cost Model:** Achieved an overall median Assessment Sales Ratio (ASR) of 1.00 and a Coefficient of Dispersion (COD) of 6.89%. After final adjustments for specific neighbourhood ASRs (Neighbourhoods 731 and 734), the model demonstrated excellent equity across all tested stratifications.
- **Sales Comparison (MRA) Model:** Achieved an overall median ASR of 1.01 and a COD of 6.89%. The MRA model showed strong predictive power ($R\text{-squared} \approx 0.86$) on the full set and excellent generalization. Final equity adjustments for Neighbourhoods 713 and 731 were applied based on ASR analysis on the full dataset.

Both models performed exceptionally well, meeting IAAO standards for level of assessment and uniformity. The CODs of 6.89% for both models indicate a very high degree of uniformity in the valuations produced.

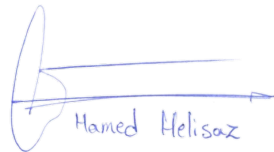
Based on a comprehensive review of the development process, statistical performance, and theoretical underpinnings, the **Sales Comparison MRA model is recommended** for use in establishing the 2018 Assessment Roll for the subject properties. The MRA model's direct reliance on market sales data, its robust statistical properties, its demonstrated predictive accuracy on unseen data, and its excellent final equity across various property characteristics make it the preferred choice. While the Cost Model also performed to a high standard, the MRA approach is generally favored in mass appraisal when sufficient sales data is available, as it most directly reflects market behaviour.

The accompanying report provides detailed documentation of all methodologies, analyses, statistical results, and the final specifications for both models. It also includes recommendations for ongoing model maintenance and monitoring.

Thank you for the opportunity to undertake this assignment. I trust this report will meet the needs of your office in preparing the 2018 Assessment Roll.

Sincerely,

Hamed Helisaz

A handwritten signature in blue ink, consisting of a stylized 'H' followed by the name 'Hamed Helisaz'.

**BUSI 444 – Advanced Computer Assisted Mass Appraisal
Final Examination Case Study**

DECLARATION

**THIS DECLARATION MUST BE SIGNED AND INCLUDED WITH YOUR EXAMINATION
SUBMISSION**

Name: Hamed Helisaz

Student Number: 4305475

I warrant that I am the sole author of this Case Study Examination; I have written all narrative sections and created all aspects of the model on my own. I have not knowingly plagiarized or carried out any other form of academic misconduct and have submitted this examination in good faith.



(Signature)

May 11th, 2025

1 Introduction

1.1 Definition of the Appraisal Problem

This report details the development, calibration, and testing of mass appraisal models designed to estimate the market value of residential properties within specified neighbourhoods of Marilu City. The primary intended user of this comprehensive analysis is the Assessor of Marilu City. The fundamental purpose of this undertaking is to equip the Assessor's office with reliable, statistically sound, and defensible valuation models suitable for establishing the 2018 Assessment Roll, as mandated by provincial guidelines requiring assessments based on full market value. The findings, detailed methodologies, statistical results, and specific recommendations contained herein are intended to provide a clear pathway for the Assessor. This includes not only selecting and applying the most appropriate model developed but also undertaking any further actions deemed necessary to refine the resulting assessments, thereby ensuring equity, uniformity, and accuracy across the appraised property base consistent with best practices and the standards promoted by the International Association of Assessing Officers (IAAO).

1.1.1 Properties to be Appraised

The specific focus of this mass appraisal assignment encompasses the diverse stock of residential properties situated within several key neighbourhoods of Marilu City. These designated areas include Neighbourhoods 713, 715, 716, 731, 732, 734, 736, and 737. The analytical foundation for the valuation models developed rests upon a comprehensive database of property sales transactions occurring within these neighbourhoods, as provided for this examination (Marilu database, Exam Database - January 5, 2021). This dataset contains a wide array of property characteristics deemed pertinent to valuation in the Marilu market. The properties themselves constitute a representative cross-section of the city's single-family housing inventory, exhibiting significant variation in critical attributes such as chronological age, effective age, gross living area, construction quality, physical condition, lot size, location influences, and specific amenities. This heterogeneity reflects the distinct character and evolutionary path of each neighbourhood, aspects explored in detail later in this report. The ultimate goal of the models is to accurately capture and quantify the complex market dynamics influencing the value of these diverse properties as of the specified valuation date.

1.1.2 Property Rights Involved: Fee Simple Estate

The valuations derived and presented throughout this report are specifically intended to estimate the market value of the **Fee Simple** estate for each subject property. Within the framework of real property law, the fee simple estate represents the most complete and absolute form of private ownership recognized. It grants the owner the full 'bundle of rights' associated with the property, subject only to the inherent limitations imposed by standard governmental powers – namely taxation, eminent domain (compulsory purchase), police power (regulation for public welfare, e.g., zoning), and escheat (reversion to the state in absence of heirs) – and potentially subject to private restrictions documented in the title (e.g., easements, covenants). This form of ownership implies absolute title, unencumbered by any other specific interests such as leaseholds, life estates, or liens beyond those typically cleared at the point of sale. Valuing the fee simple interest is the standard and required practice for ad valorem property tax assessment purposes in Marilu City, as it ensures that the assessment reflects the intrinsic value of the property itself, independent of any specific ownership arrangements or encumbrances that might diminish the value of a partial interest. This approach provides a consistent, equitable, and legally sound basis for distributing the property tax burden.

1.1.3 Purpose and Intended Use of the Appraisal

The central purpose of this appraisal assignment is the systematic development, rigorous calibration, and thorough evaluation of mass appraisal models. These models are specifically designed to estimate

the market value of the defined population of residential properties within the designated Marilu City neighbourhoods, serving as the foundation for the upcoming 2018 Assessment Roll. The intended use of these statistically derived models is exclusively by the Marilu City Assessor's office. They are to be employed as the primary tool for establishing equitable, uniform, and market-based property assessments, fulfilling the requirements of the applicable assessment legislation which mandates a full market value standard. This report furnishes the necessary detailed analysis, model specification rationale, calibration procedures and outputs, comprehensive statistical testing results against recognized benchmarks, and clear, actionable recommendations. This complete package is designed to empower the Assessor to select and implement the most suitable valuation model in an effective, efficient, and legally defensible manner. All analytical procedures and model performance evaluations documented herein adhere to the principles and standards for mass appraisal promulgated by the International Association of Assessing Officers (IAAO).

1.1.4 Date of Appraisal (Valuation Date)

The singular, effective date of valuation for this entire appraisal assignment is **July 1, 2017**. This specific date serves as the temporal anchor for all analyses conducted. All market data, particularly sales prices, have been adjusted for time to reflect market conditions prevailing on this date. Consequently, the final value estimates produced by the developed models represent the estimated market value of the subject properties as of July 1, 2017. It is important to note the timeline context: the 2018 Assessment Roll, for which these models provide the valuation basis, is scheduled for presentation to Marilu City on December 31, 2017. This report itself is assumed to be presented to the Assessor on November 30, 2017. This timing provides the Assessor's office with a one-month window for thorough review of the findings, final model selection, system integration, and preparation for the application of the chosen model to produce the final assessment roll.

1.1.5 Definition of Value: Market Value

In strict accordance with the established requirement for a full market value assessment base within Marilu City, the singular value standard employed consistently throughout this report is **Market Value**. While the precise wording of the definition can exhibit minor variations across different jurisdictions, professional organizations, or governing statutes, the underlying concept fundamentally aligns with the internationally recognized definition. Commonly accepted definitions, such as those promulgated by the IAAO or the Appraisal Institute (and reflected in regulations like USPAP or federal banking rules), provide a robust framework. Market Value is typically conceptualized and defined as:

"The most probable price which a property should bring in a competitive and open market under all conditions requisite to a fair sale, the buyer and seller each acting prudently and knowledgeably, and assuming the price is not affected by undue stimulus. Implicit in this definition is the consummation of a sale as of a specified date and the passing of title from seller to buyer under conditions whereby:

- Buyer and seller are typically motivated (i.e., neither under duress);
- Both parties are well informed or well advised, and acting in what they consider their own best interests;
- A reasonable time is allowed for exposure in the open market;
- Payment is made in terms of cash in U.S. dollars or in terms of financial arrangements comparable thereto; and
- The price represents the normal consideration for the property sold unaffected by special or creative financing or sales concessions granted by anyone associated with the sale."

Within the specific context of this report and its application to the Marilu City assessment process, Market Value represents the estimated monetary amount for which a subject property should logically exchange on the effective valuation date (July 1, 2017). This exchange is presumed to occur between a hypothetical willing buyer and a willing seller, engaged in an arm's-length transaction (i.e., unrelated and acting independently), subsequent to adequate marketing efforts, and under conditions where both parties have acted knowledgeably, prudently, and without undue compulsion or external pressure. The mass appraisal models developed and evaluated herein are specifically designed to predict this defined Market Value for the entire population of subject residential properties.

1.1.6 Preliminary Planning and Data Considerations

The analytical core of this mass appraisal exercise is the Marilu database provided for the examination. This database furnishes the essential sales transaction records and associated property characteristics for the neighbourhoods under study. While it is assumed that this dataset has undergone some preliminary screening and validation prior to its provision, a critical step in the model development process involves further in-depth analysis, verification, and potential transformation of the data. This includes identifying and handling outliers, addressing missing data if present, and assessing the suitability of variable formats for statistical modeling. The examination explicitly requires the development and comparison of two distinct mass appraisal models: one predicated on the Cost Approach and the other employing the Direct Sales Comparison Approach, specifically through additive multiple regression analysis.

Comprehensive planning for this task involves several key stages. First, careful selection of relevant variables from the database is required for each model type, guided by appraisal theory and preliminary data analysis. Second, selected variables may need transformation to better conform to model assumptions or to more accurately reflect market behaviour (e.g., creating interaction terms to capture synergistic effects, using non-linear transformations for variables like age or size, converting categorical data into appropriate numerical formats like dummy variables). Third, the models must be calibrated using accepted statistical techniques – deriving coefficients for the Sales Comparison model via regression and estimating depreciation/market adjustments for the Cost model. Fourth, the performance, accuracy, and equity of both calibrated models must be rigorously tested using standard statistical measures and ratio studies, comparing results against IAAO benchmarks. A crucial aspect of the planning for the Sales Comparison model involves leveraging the database's sufficient size to create distinct MODEL and TEST datasets. The model will be initially developed using the MODEL dataset and subsequently validated on the unseen TEST dataset to provide a more robust assessment of its predictive power and generalization capability, before final testing on the full dataset. While the task of primary data collection is outside the scope of this examination (as the dataset is provided), a thorough understanding of the data's characteristics, potential limitations, and inherent quality is paramount for building meaningful and reliable valuation models.

1.2 Preliminary Survey and Analysis

1.2.1 Approaches to Value

Established appraisal theory and professional practice recognize three primary methodologies, or 'approaches,' for estimating the market value of real property: the Cost Approach, the Sales Comparison Approach, and the Income Approach. The relative applicability, reliability, and weight given to each approach are contingent upon several factors, including the specific type of property being appraised (e.g., residential, commercial, industrial, special-purpose), the quantity and quality of available market data pertinent to each approach, and the prevailing characteristics and motivations within the relevant market segment.

1. **Cost Approach:** This approach derives an estimate of value based on the economic principle of substitution. This principle posits that a rational, informed buyer would typically pay no more for an existing property than the cost required to acquire a comparable vacant site and construct new

improvements of equivalent desirability and utility, assuming no undue delay in construction. The methodology involves three main steps: (a) estimating the current cost to construct the existing improvements (either as an exact replica – Reproduction Cost, or as a modern equivalent – Replacement Cost New, RCN); (b) subtracting the total loss in value from RCN due to all forms of accrued depreciation (physical deterioration, functional obsolescence, and external obsolescence); and (c) adding the estimated market value of the land as if vacant and available for its highest and best use. The Cost Approach is often considered most pertinent and reliable for properties that are relatively new (minimal depreciation), unique or special-purpose (lacking comparable sales), or situated in markets where sales data is scarce or unreliable. For this specific assignment, the development of a market-adjusted cost model is an explicit requirement. This necessitates integrating market-derived data, particularly for estimating land value and calibrating depreciation schedules, thereby linking the theoretical cost basis more closely to observed market behaviour.

2. **Sales Comparison Approach:** This approach also fundamentally relies on the principle of substitution, operating under the premise that the market value of a property is directly related to the prices of comparable properties that have recently sold in the same or a similar market. The process involves identifying recent sales of properties that are as similar as possible to the subject property, verifying the details and conditions of these sales to ensure they represent arm's-length market transactions, and then adjusting the sale prices of these comparable properties to account for any significant differences between them and the subject property. Adjustments typically address variations in property rights conveyed, financing terms, conditions of sale, market conditions (time adjustments), location attributes, and physical characteristics (e.g., size, age, quality, condition, features). In mass appraisal, this approach is commonly implemented using statistical techniques, particularly multiple regression analysis (MRA), which allows for the simultaneous analysis of numerous property characteristics across a large dataset of sales to estimate the marginal contribution of each characteristic to the overall property value. Given its direct reflection of market transactions and buyer/seller behaviour, the Sales Comparison Approach is generally considered the most direct, persuasive, and reliable indicator of market value for residential properties like those in Marilu City, provided that a sufficient volume of relevant and reliable comparable sales data is available. The development of a Direct Sales Comparison model using additive MRA is the second explicit modeling requirement of this examination.
3. **Income Approach:** This approach is the primary valuation method for properties that are typically bought and sold based on their ability to generate income, such as apartment buildings, office towers, retail centers, or industrial warehouses. It estimates market value by converting the anticipated future economic benefits (typically the net income stream) generated by the property into a present value. Common techniques include direct capitalization, where a single year's stabilized net operating income (NOI) is divided by a market-derived capitalization rate (cap rate), or discounted cash flow (DCF) analysis, where projected net cash flows over a holding period, plus a reversionary value, are discounted back to present value using an appropriate discount rate. While it is true that some single-family homes in Marilu City are rented (approximately 30% of dwelling units), the primary market driver for these properties is typically owner-occupancy. They are generally not purchased or traded based principally on their income-generating potential in the same manner as purpose-built rental investments. Furthermore, obtaining reliable, verified data on market rents, vacancy rates, and operating expenses for single-family rentals that are truly comparable to the subject properties is often challenging in a mass appraisal context. The motivations and decision criteria of buyers and sellers in the single-family market are less directly tied to quantifiable income streams compared to those in the commercial investment property market. Consequently, the Income Approach is generally considered not applicable or not reliable for valuing the types of single-family residential properties included in this assignment's scope. Therefore, this approach will not be developed or considered further in this report.

Based on the explicit examination requirements and the fundamental characteristics of the properties

being appraised (single-family residential units primarily traded for owner-occupancy), this report will concentrate exclusively on the development, calibration, testing, and evaluation of mass appraisal models derived from the **Cost Approach** and the **Sales Comparison Approach**.

1.2.2 Data Requirements for Selected Approaches

The successful and reliable application of both the Cost and Sales Comparison approaches within a mass appraisal framework hinges critically on the availability, accuracy, and completeness of specific data elements. For this assignment, the primary source is the provided Marilu database, which must be supplemented by external cost data for the Cost Approach.

- **Cost Approach Data:** Effective implementation requires reliable estimates for three core components: land value, improvement costs (specifically, Replacement Cost New - RCN), and total accrued depreciation. As per the requirements, land value is to be estimated using market data, specifically via the land residual technique applied to sales data. This involves subtracting the depreciated cost of improvements from the sale price to isolate the land component. RCN estimates for the improvements will be based on standardized cost data provided in the BUSI 444 Course Workbook. This cost data likely represents benchmark costs per square foot (or metre) based on quality class, potentially requiring further adjustments for specific property features documented in the database (e.g., variations in size, basement type, garage presence, extra fixtures). The estimation of accrued depreciation represents a significant challenge and is critical for model accuracy. It necessitates a thorough analysis of the relationship between property age (both chronological and effective, if available), observed physical condition, and market value trends, typically derived from analyzing the sales data itself. A key task will be the development and calibration of a market-derived Percent Good variable or depreciation schedule. Essential variables from the database for this approach therefore include: identifiers, sale price, sale date, land characteristics (size, possibly location factors influencing land value), and detailed improvement characteristics (living area, quality class, year built/age, condition rating, presence/type of basement, garage details, number of storeys, etc.).
- **Sales Comparison Approach Data:** This approach demands a robust, representative, and sufficiently large dataset of recent, verified, arm's-length sales transactions of properties comparable to those being appraised. The database must contain detailed information on the characteristics thought to influence property values in the Marilu market. Key variables required from the database include: sale price (the dependent variable), sale date (for time adjustments), precise location identifiers (e.g., neighbourhood code, potentially street or map coordinates if available for finer analysis), relevant land characteristics (e.g., lot size, potentially shape, topography, corner influence), and a comprehensive set of improvement characteristics. Improvement data should ideally cover quantitative measures (e.g., gross living area, basement area, garage size), qualitative attributes (e.g., quality class, condition rating, view rating), categorical features (e.g., presence of fireplace, type of heating, exterior finish, number of bathrooms), and age/year built. The primary analytical goal is to use MRA to identify which of these characteristics have a statistically significant and logically sound impact on explaining the variation in sale prices observed in the market. Accurate time adjustments are crucial to trend all sale prices to the single valuation date (July 1, 2017). The database must be large enough not only for initial model calibration but also for rigorous testing and validation, including stratification analysis, as stipulated by the requirement to create separate MODEL and TEST datasets.

1.2.3 Depreciation in the Cost Approach

A cornerstone of the Cost Approach, and often its most challenging component to estimate accurately, is accrued depreciation. Depreciation, in appraisal terms, represents the total loss in value suffered by the improvements (buildings) relative to their Replacement Cost New (RCN), arising from all causes,

measured as of the effective date of appraisal. It is the crucial link between the cost of creating a new asset and the current market value of an existing, older asset. In the context of the Cost Approach formula ($\text{Value} = \text{RCN} - \text{Depreciation} + \text{Land Value}$), depreciation encapsulates the difference between RCN and the market value contribution of the improvements. While sometimes estimated using theoretical age-life methods in single-property appraisal, in mass appraisal, depreciation is ideally estimated through market data analysis, reflecting how the market actually penalizes properties for age, wear and tear, design deficiencies, and adverse external factors. There are three distinct categories of depreciation that must be considered:

1. **Physical Deterioration:** This is the most intuitive form of depreciation, representing the loss in value due to the physical wearing out of the property's components over time. It results from regular wear and tear through use, structural fatigue, decay, insect damage, and the ongoing impact of the elements (weathering). Physical deterioration can be further subdivided into curable items (those repairs and replacements that are economically feasible to undertake, such as repainting, replacing worn flooring, or repairing leaky faucets – the cost to cure is typically less than the anticipated increase in value) and incurable items (defects or wear that are not economically feasible to fix because the cost to cure would exceed the value added, such as major foundation problems or wear on the basic structure – often addressed through 'short-lived' and 'long-lived' component analysis in detailed cost approaches). In mass appraisal, physical deterioration is frequently modeled primarily based on the effective age of the improvements relative to their total expected economic life, often refined by incorporating adjustments based on observed physical condition ratings available in the database.
2. **Functional Obsolescence:** This category represents a loss in value arising from flaws in the design, layout, materials, or features of the property relative to current market standards and preferences. It reflects the property's inability to function as efficiently or desirably as a new property designed for the same purpose would. Functional obsolescence can stem from deficiencies (lacking features that are standard in the current market, e.g., insufficient number of bathrooms, inadequate electrical capacity, poor room layout, outdated kitchen design) or superadequacies (containing features that exceed typical market requirements or desires, where the cost of the feature is greater than its contribution to value, e.g., excessive quality of construction for the neighbourhood, an over-improvement like a swimming pool in a market that doesn't value them). Like physical deterioration, functional obsolescence can be curable (e.g., remodeling an outdated kitchen, adding a needed bathroom if space permits) or incurable (e.g., fundamental poor architectural design, inadequate ceiling heights that cannot be easily changed). In mass appraisal, functional obsolescence is often implicitly captured through adjustments related to quality class, design codes, specific feature variables (or lack thereof), or sometimes through age-related depreciation schedules, as older designs are more likely to be functionally obsolescent.
3. **External (or Economic) Obsolescence:** This type of depreciation signifies a loss in value resulting from adverse factors that are external to the property itself and typically beyond the owner's control. These negative influences originate from the property's surroundings or the broader market environment. Examples include detrimental economic conditions (e.g., high unemployment in the area, closure of a major local industry), negative environmental factors (e.g., proximity to industrial pollution, noise from airports or highways, unpleasant odours), adverse changes in zoning or land use patterns nearby (e.g., encroachment of incompatible uses), lack or inadequacy of public services or utilities, or high crime rates in the vicinity. The detailed descriptions of the Marilu City neighbourhoods provide clear examples: the offensive odour from pulp mills adversely affecting property appeal and value in parts of Neighbourhood 731 and 734, and the potential negative impact of commercial encroachment in Neighbourhood 713. Conversely, proximity to positive external factors like parks, desirable schools, or major employment centers (as noted for Neighbourhood 732) can mitigate or counteract external obsolescence, effectively representing a

positive location influence. External obsolescence is generally considered incurable by the individual property owner. In mass appraisal, its impact is typically quantified through location adjustments derived from analyzing geographic patterns in sale prices (often via the Sales Comparison model) or by applying specific adjustments within the cost model based on delineated neighbourhood boundaries or proximity to known influences.

Accurately estimating the total accrued depreciation, reflecting the combined impact of all three forms, is absolutely essential for developing a reliable and credible Cost Approach valuation model. Within the scope of this assignment, the primary method for achieving this will involve analyzing the market sales data to develop empirical depreciation estimates, most likely through the creation, calibration, and application of a Percent Good variable or a market-derived depreciation schedule, as indicated by the requirement to use PCTGOOD information. This ensures the depreciation applied reflects the actual penalties observed in the Marilu City real estate market.

1.3 Data Collection, Market Analysis, and Highest and Best Use Analysis

1.3.1 Data Collection and Preliminary Market Analysis

While the core transactional and property characteristic data for this examination is provided within the Marilu database, effective model building and subsequent interpretation of results necessitate a thorough understanding of the broader market context from which this data originates. This requires analyzing the interplay of fundamental forces that shape real estate values in Marilu City, as well as delving into the specific attributes and dynamics of the individual neighbourhoods under study. This market analysis provides the essential backdrop against which the statistical models will be developed and evaluated.

1.3.2 Market Influences (The Four Forces)

Real estate values are not determined in isolation; they are the product of a complex and dynamic interplay of four fundamental forces that influence supply and demand within a given market. Understanding these forces provides critical context for interpreting market data, identifying value drivers, and building robust valuation models.

1. **Social Forces:** These forces pertain to the demographic characteristics, trends, and attitudes of the population within the market area. Key aspects include population size and growth rate, age distribution, household size and composition, lifestyle preferences, educational attainment levels, crime rates, and cultural attitudes towards housing and community. In Marilu City, the stable population of approximately 70,563, the high rate of homeownership (70%) compared to renting (30%), and the clear preference for single-detached homes (65% of stock) paint a picture of the prevailing social structure and housing norms. Specific social factors driving localized demand are evident, such as the noted popularity of Neighbourhood 732 among healthcare and education professionals due to its convenient proximity to the regional hospital, college, and high school. Lifestyle considerations associated with the northern climate (e.g., preferences for certain house features, recreational opportunities) also fall under this category, influencing desirability and, consequently, value.
2. **Economic Forces:** These forces relate to the economic health and vitality of the market area, encompassing employment levels and diversity, wage and income rates, the strength of the local industrial and commercial base, availability and cost of financing, construction activity (new supply), inflation rates, and overall market sentiment (e.g., consumer confidence). Marilu's economy shows a significant reliance on the forestry sector, with numerous sawmills and pulp mills being major employers. While this provides a core economic base, it also suggests potential vulnerability to fluctuations in that specific industry. Diversification is provided by significant public sector employment in government, health (regional hospital), and education (regional college).

The crucial observation that the real estate market has been "quite buoyant" in recent years and is expected to remain so indicates generally positive economic influences driving demand and supporting price levels during the period leading up to the July 1, 2017 valuation date. However, negative economic externalities are also present, most notably the adverse impact of pulp mill emissions on the desirability and likely value of properties in adjacent neighbourhoods like 731 and 734.

3. **Governmental Forces:** These forces encompass the wide range of actions and regulations imposed by various levels of government that directly and indirectly affect real property. This includes local zoning ordinances and land use plans, building codes and safety regulations, property taxation policies (including the assessment level and tax rates), the provision and quality of public services (e.g., schools, police and fire protection, parks and recreation, sanitation), infrastructure development (e.g., roads, highways, public transit, utilities), and environmental regulations. The neighbourhood descriptions highlight significant variations in governmental factors across Marilu City: differences in utility provision (overhead versus underground wiring), street layout and access (rear alleys versus direct street access), the presence and quality of parks, and access to major public transit routes. These variations in public services and infrastructure directly impact the utility, desirability, and convenience of properties, thereby influencing their market value differently across various neighbourhoods. The assessment process itself, mandated to achieve full market value, is a critical governmental force shaping the local real estate environment.
4. **Environmental Forces:** These forces relate to the natural aspects and setting of the market area, including climate, topography, soil conditions, natural resources, accessibility, presence of natural amenities (e.g., rivers, lakes, views), and potential environmental hazards or nuisances. Marilu's distinct northern climate, characterized by long, cold winters with significant snowfall and short summers, inevitably influences construction styles (e.g., insulation standards, roof pitches), ongoing maintenance requirements, energy costs, and lifestyle choices, all of which can factor into value. Specific topographical features can create value premiums, such as the river views available from some properties in Neighbourhoods 715 and 716. Conversely, negative environmental factors can detract significantly from value; prominent examples in Marilu include the pervasive pulp mill odours impacting Neighbourhoods 731 and 734 and the potential practical difficulties associated with snow clearing in rear alleys, noted particularly for Neighbourhood 713. Positive environmental amenities, such as the major parks located in or near Neighbourhoods 731 and 732, generally enhance the desirability and value of nearby properties.

Collectively, these four fundamental forces continuously interact to shape the supply and demand dynamics that drive the Marilu City real estate market and create value differentials among its specific neighbourhoods. A clear understanding of these forces provides the essential context for interpreting sales data and constructing valuation models that accurately reflect market realities.

1.3.3 City Overview

Marilu presents as a moderately sized urban centre within Northern Ontario, boasting a 2016 census population of 70,563. Its economic identity is strongly tied to the regional forestry industry, although this is balanced by substantial employment within the public sectors of government, healthcare, and education, anchored by a regional hospital and college. Functioning as a key commercial centre for a broad surrounding region, Marilu features several regional shopping malls and significant commercial floor space. Essential transportation infrastructure includes connectivity via two railway lines and a Class A airport. The city experiences a distinct northern continental climate. Its residential real estate market encompasses approximately 24,000 dwelling units, characterized by a high proportion of owner-occupancy (70%) and a dominance of single-detached housing forms (65%). Notably, the market was described as "quite buoyant" leading up to the valuation period, with expectations of continued strength.

Market activity, as reflected in MLS sales, shows a prevalence of one-storey homes (80%). This composite overview suggests a relatively stable, albeit resource-influenced, regional city experiencing positive momentum in its residential real estate sector around the critical valuation date of July 1, 2017.

1.3.4 Neighbourhood Analysis

The specific neighbourhoods forming the study area for this mass appraisal project exhibit considerable heterogeneity, a factor expected to be a primary driver of property value variation. A summary of their key characteristics, drawn from the provided descriptions, highlights this diversity:

- **Neighbourhood 713:** Situated immediately west of the commercial core. Characterized by some of the city's oldest housing stock, a grid street pattern often incorporating rear alleys (posing potential winter access challenges), and observable commercial encroachment along its eastern edge. Homes are generally of older construction classes (80/90). This area likely represents an older, possibly transitional urban neighbourhood.
- **Neighbourhood 715:** Located west of 713, displaying internal diversity. The northern section features older homes on grid streets, with some properties benefiting from river views. The southern section, generally developed later (1970s), exhibits a more modern layout with curved streets, cul-de-sacs, and underground utility services. Apartment blocks are concentrated along its southern fringe (15th Avenue). This neighbourhood blends older and more contemporary development styles.
- **Neighbourhood 716:** Positioned north and west of 715, developed around the same era (1970s). Features a modern layout with underground services, though some streets retain rear alleys. Properties on northern streets may possess river views. Housing quality is generally considered better than average, with a concentration of higher-quality construction classes (145+). Likely perceived as a desirable, higher-value residential area.
- **Neighbourhood 731:** Located east of the central business district (CBD). Developed mainly in the 1950s with a grid pattern, alleys, and overhead wiring. While initially containing good quality homes near the river, its appeal diminished significantly due to adverse effects (odour) from upstream pulp mills established later. Contains two major parks. Represents a neighbourhood significantly impacted by negative external obsolescence.
- **Neighbourhood 732:** A relatively small neighbourhood situated south of 15th Avenue and adjacent to a major park complex. Features a modern standard layout but predominantly has overhead wiring. Its high desirability stems from its close proximity to the regional hospital, college, and major high school, making it popular among professionals working there and resulting in low market turnover. Housing stock mainly consists of one-storey modern standard homes (classes 90/140). A stable, sought-after neighbourhood due to locational convenience.
- **Neighbourhood 734:** Exhibits mixed quality characteristics. The western portion is low-lying and suffers from the adverse effects of pulp mill emissions. It features a grid pattern, rear alleys, overhead wires, and generally lower-quality housing stock, including properties with basement suites. The eastern section, situated on higher ground, is less affected by emissions, features modern standard layouts with underground services, and generally contains better quality housing. Represents significant internal variation driven by environmental factors and development standards.
- **Neighbourhood 736:** Occupies the southeastern corner of the urban area. Features a mix of areas with underground service and others with overhead wiring. The housing stock generally comprises modern standard one-storey homes (classes 90/140). Represents a typical modern-era suburban neighbourhood.

- **Neighbourhood 737:** Located in the southwestern corner. Primarily developed during the 1970s, with most homes falling into the 90 and 140 quality classes. Features a modern standard layout with underground wiring, numerous cul-de-sacs, and limited through traffic, although, similar to 716, some streets incorporate rear alleys. Another example of a modern-era neighbourhood development with some layout variations.

This detailed neighbourhood analysis underscores the substantial variations in property age, architectural style, layout efficiency, construction quality, utility services, proximity to amenities (positive and negative), and exposure to external influences across the study area. These geographical differences are anticipated to be significant drivers of market value and must be explicitly accounted for within the valuation models, typically through the inclusion of location adjustments, neighbourhood dummy variables, or geographically weighted regression techniques if appropriate and supported by the data. Subsequent data analysis sections will present descriptive statistics (e.g., mean/median size, age, sale price) stratified by neighbourhood to quantitatively illustrate these observed differences.

1.3.5 Highest and Best Use (HBU) Analysis

The concept of Highest and Best Use (HBU) is a cornerstone principle in real property appraisal. It is defined as the reasonably probable and legal use of either vacant land or an improved property, which is physically possible, appropriately supported by market demand, financially feasible, and that ultimately results in the highest possible value for the property as of the effective date of appraisal. The conclusion reached through the HBU analysis dictates the fundamental basis upon which the property will be valued.

In the context of **single-property appraisal**, the HBU analysis is typically performed in two distinct stages: first, analyzing the HBU of the land as if it were vacant, and second, analyzing the HBU of the property as it is currently improved. Comparing these two scenarios helps the appraiser determine whether the existing improvements contribute value to the site or if the land would be more valuable if redeveloped for a different use (implying the existing improvements suffer from obsolescence or represent an 'interim use').

In **mass appraisal**, the HBU analysis is necessarily conducted on a broader scale, often applied to entire neighbourhoods, defined property type groups (e.g., all single-family residences within a certain age range), or market segments. Due to the scale involved, the operational assumption is generally made that the existing use of the vast majority of properties within a well-defined, stable residential area represents the Highest and Best Use. This assumption holds unless there is compelling market evidence indicating a widespread potential for transition or redevelopment to a different, more valuable use (e.g., large-scale rezoning allowing higher density, or significant, observable market trends like the commercial encroachment noted along the edge of Neighbourhood 713). For the specific scope of this assignment, which focuses on existing single-family residential dwellings located within established residential neighbourhoods, the governing presumption is that the HBU for these properties is their continued use as single-family residences. However, this presumption does not negate the need to consider factors identified during the market analysis (like adverse external influences or superior location amenities) that could significantly affect the value *within* that established residential use.

The **four tests** or criteria for determining Highest and Best Use remain relevant and are applied conceptually in the mass appraisal context:

1. **Legally Permissible:** Is the existing residential use consistent with current zoning ordinances, land use regulations, building codes, and any private restrictions (e.g., deed restrictions)? For these established residential neighbourhoods, the answer is assumed to be yes for the vast majority of properties. However, specific local regulations could influence the potential for future renovations, additions, or redevelopment. The noted commercial encroachment in Neighbourhood 713 suggests potential zoning interface complexities or transitional areas where the long-term HBU might eventually shift for properties directly affected.

2. **Physically Possible:** Does the physical nature of the land – its size, shape, topography, soil stability, and the availability of essential utilities – support the existing residential use? Generally, the lots within these neighbourhoods are assumed to be physically suitable for the existing dwellings. However, specific physical attributes can affect the utility and desirability within the residential use category, such as the challenges posed by rear alley access in winter or the potential limitations associated with building on low-lying, potentially emission-affected land as found in parts of Neighbourhood 734.
3. **Financially Feasible:** Considering market conditions, does the existing residential use generate sufficient return (either explicit rental income or implicit amenity value and return for owner-occupants) to support the ongoing use and justify the capital invested? In other words, is the value as improved greater than the value of the land as if vacant? Given the description of a "buoyant" residential market in Marilu City, it is reasonable to conclude that continued residential use is financially feasible for these properties. This test implicitly assumes that the existing improvements contribute positively to the overall property value, exceeding the value of the land alone.
4. **Maximally Productive:** Among all the uses that are legally permissible, physically possible, and financially feasible, does the current residential use yield the highest present value? For the overwhelming majority of properties situated well within these established residential neighbourhoods, continued single-family use is highly likely to be the maximally productive use. Potential exceptions might theoretically exist on the absolute fringes abutting different land use zones (like the commercial edge of 713) or in rare instances where zoning might permit, and market demand might support, redevelopment to a significantly higher density (e.g., replacing a single home with multiple units). However, the provided neighbourhood descriptions do not suggest that such redevelopment pressure is a widespread phenomenon impacting the bulk of the properties under study.

Therefore, based on the application of these four tests within the context of the provided information, the definitive **Highest and Best Use conclusion** for the population of properties subject to this mass appraisal assignment is their **current use as single-family residential dwellings**. The valuation models developed in the subsequent sections will accordingly estimate the market value of these properties based on this concluded use, while fully considering and quantifying the impact of the wide variations in location attributes, physical characteristics, quality, condition, and other value-influencing factors identified during the comprehensive market analysis. Factors such as desirable river views or negative externalities like pulp mill odours will be treated as elements influencing the property's value *within* its established Highest and Best Use, rather than factors fundamentally dictating a change in that use.

This comprehensive introductory framework, encompassing the definition of the appraisal problem, the preliminary survey including the selection of valuation approaches and consideration of depreciation, and the detailed market and highest-and-best-use analyses, establishes the necessary context, scope, definitions, and theoretical underpinnings required before proceeding to the core task of this assignment: the detailed development, calibration, and testing of the Cost and Sales Comparison mass appraisal models in the subsequent sections of this report.

Having established the appraisal problem, the relevant market context, and the theoretical foundations, the report now proceeds to the development and calibration of the first of the two required valuation models. The following section details the specification, data preparation, and calibration process for the market-adjusted Cost Approach model.

2 Valuation using Cost Model

2.1 Introduction

This section details the development, specification, calibration, and initial testing of the market-adjusted Cost Approach model, the first of the two mass appraisal models required by this assignment. The objective is to construct a model that estimates the market value of the subject residential properties in Marilu City as of July 1, 2017, based on the principle of substitution, integrating both standardized cost data and market-derived adjustments. The model follows the fundamental structure: $\text{Market Value} = \text{Land Value} + \text{Replacement Cost New of Improvements} - \text{Accrued Depreciation}$, with components adjusted to reflect local market conditions. The process involves analyzing the provided sales database and cost data (from the Course Workbook) to estimate market-derived land values, develop market-based depreciation schedules (Percent Good), calculate Replacement Cost New Less Depreciation (RCNLD) for improvements, and apply necessary market adjustments or local cost multipliers. The resulting model aims to provide a reliable estimate of market value consistent with the requirements for the 2018 Assessment Roll.

2.2 Preliminary Data Screening

A preliminary review of the provided Marilu database was undertaken to understand its structure, variable types, and general characteristics. The database contains 466 sales records and 57 variables spanning the period from March to October 2017, covering the specified neighbourhoods 713, 715, 716, 731, 732, 734, 736, 737. Key variables relevant to the cost approach include sale price (saleprice), sale date (processed via month column), location (neigh), land characteristics (e.g., lotsize), and various improvement features (e.g., mbmancls as quality, age derived from mbyrblt, flrarea1 as TLA, bsmtarea, gar_area).

A check for completeness revealed no significant missing values for the primary variables required for the cost model. Initial range checks on variables like age (derived from mbyrblt, max 80 years and min 2 years) and flrarea1 indicated generally reasonable distributions. Descriptive analysis for all columns in the dataset is available in Appendix (Section A on page 67). Overall, the database appears suitable for cost model development, subject to the detailed variable preparation and analysis outlined below.

2.3 General Model Specification

The cost approach utilizes a hybrid (market-adjusted) cost model structure that incorporates both additive and multiplicative components. The general form of this model is specified as:

$$MV = \Pi GQ \times [(\Pi BQ \times \Sigma BA) + (\Pi LQ \times \Sigma LA) + \Sigma OA]$$

Where:

- MV = The estimated market value
- ΠGQ = The product of the general qualitative components (e.g., adjustments for time, overall location/market conditions, LCM)
- ΠBQ = The product of the building qualitative components (e.g., adjustments derived from quality class, condition, PCTGOOD factor)
- ΣBA = The sum of the building additive components (e.g., base cost of main structure) - representing the core RCN component.
- ΠLQ = The product of the specific land qualitative adjustment factors (e.g., adjustments for specific site features like conduit, corner, etc.)

- ΣLA = The sum of the land additive components (e.g., base land value derived from size and base unit rate) - representing the core Land Value component.
- ΣOA = The sum of the other additions additive components (e.g., additive values for features like garages, fireplaces, sheds, etc., if handled additively).

This hybrid structure allows for the integration of various property characteristics in a flexible manner. First the focuses will be on developing the land value component ($\Pi LQ \times \Sigma LA$) using a market-derived unit value method, and then other components, including the building component ($\Pi BQ \times \Sigma BA$), other additions (ΣOA), and overall adjustments (ΠGQ) based on market analysis and provided cost data.

2.4 Variable Review and Preparation

The variables available in the Marilu database must be reviewed and prepared to fit into the specified hybrid cost model structure: $MV = \Pi GQ \times [(\Pi BQ \times \Sigma BA) + (\Pi LQ \times \Sigma LA) + \Sigma OA]$. This involves identifying the database fields corresponding to each component and performing necessary transformations.

- **Land Components ($\Pi LQ \times \Sigma LA$):**

- ΣLA (Additive Land Component): The base additive value is primarily driven by ‘lotsize’. The initial analysis used ‘landval’ as a starting point for deriving market adjustments relative to Time Adjusted Sale Price. The final base value incorporates a median unit rate derived from land residual divided by lot size.
- ΠLQ (Multiplicative Land Adjustments): Factors derived from market analysis to adjust the base land value. This includes the size adjustment factor (‘adjsfactpow’) derived from modeling the relationship between Land Value/lotsize and ‘lotsize’ itself using a power curve. It also includes adjustment factors derived from analyzing the ratio Land Value/Predicted Land Value stratified by ‘conduit’ and ‘neigh’. Variables like ‘alley’, ‘sidewalk’, ‘culdesac’, ‘corner’, ‘wooded’, ‘pie_shpe’, ‘slope’, ‘abv_road’ were tested but did not require adjustment factors in this step.

- **Main Building Components ($\Pi BQ \times \Sigma BA$):**

- ΣBA (Additive Building Component): Represents the base Replacement Cost New (RCN) of the main structure, calculated using provided standardized cost data based on variables like ‘flrarea1’ and ‘mbmancls’ (Quality Class), potentially adjusted for ‘nstor’ (Storeys) or ‘bsmtarea’ depending on the cost manual structure.
- ΠBQ (Multiplicative Building Adjustments): Factors adjusting the base RCN. This primarily includes the market-derived depreciation factor (PCTGOOD), which is calculated based on ‘age’ (derived from ‘mbyrblt’). Adjustments for quality (‘mbmancls’) might also be incorporated here or within ΣBA depending on the costing method.

- **Other Additive Components (ΣOA):**

- Represents the additive value contribution of other structures or features not included in the main RCN calculation (ΣBA). This could include the depreciated cost of garages (derived from ‘gar_area’, ‘gar_type’, potentially ‘grmktval’ or cost data), fireplaces (‘fireplcs’), potentially finished basement areas (‘bsmtfin’) if not part of main RCN, or other outbuildings (e.g., ‘ob1mktval’ if applicable). The specific handling (additive vs. part of RCN) depends on the cost manual and modeling choices. For LCM calibration, certain items like garages are typically excluded from the main RCN and handled here.

- **General Qualitative Components (ΠGQ):**

- Represents overall multiplicative adjustments applied to the sum of the land and depreciated improvement components. This includes the Time Adjustment factor derived from ‘saledate’ analysis and the Local Cost Multiplier (LCM) calibrated from the market. Neighbourhood adjustments could be applied here as well, if needed at the end.
- **Dependent Variables:** ‘saleprice’ after time adjustment (if any) is the target variable used for market calibration of ΠLQ , ΠBQ (specifically PCTGOOD), ΣOA contributions, and ΠGQ (specifically LCM).

2.4.1 Time Adjustment Analysis

The Sale Price to Assessed Ratio (SAR) method is used to evaluate if time adjustment is required. Our valuation date is July 1st, 2017, and the sales in this database occurred between March 2017 and October, 2017. Ratio statistics were calculated for the SAR (‘saleprice’ / ‘totalval’) for the 466 valid sales within this period (Table 1).

Table 1: Ratio Statistics for SAR (Sale Price / Total Assessed Value)

Statistic	Value
Mean	1.03
95% CI for Mean (Lower)	1.03
95% CI for Mean (Upper)	1.04
Median	1.04
95% CI for Median (Lower)	1.03
95% CI for Median (Upper)	1.04
Actual Coverage (Median CI)	94.80%
Minimum	0.79
Maximum	1.37
Coefficient of Dispersion (COD)	6.63%
Coefficient of Variation (COV)	8.46%

The 6.63% COD indicates that the previous year’s assessed values (‘totalval’) have been acceptable in relation to the sale prices. That is to say, the Sale Price to Assessed Value ratio (SAR) can be used for time adjustment analysis. The scatter plot between the SAR value and sale month is given below (Figure 1).

There is a non-significant continuous increase in the SAR over the sale period visually. The correlation is not significant as indicated by the value for $R^2 = 0.01$. Therefore, it may be more appropriate to consider adjustments for each month individually, potentially using a non-parametric approach. Here, the Kruskal-Wallis test is used for this purpose: the test determines if there is a significant difference between how SAR ranks are distributed between months. Table 2 shows the mean ranks per month.

Table 3 presents the Kruskal-Wallis test statistics.

The asymptotic significance statistic is approximately 8.3%, which is not small enough (using $\alpha = 0.05$) to reject the null hypothesis that the SAR values are equally distributed among months. That is to say, time adjustment based on median SAR ranks is not statistically necessary. Note that this observation is also aligned with the low value of R^2 derived before.

Although the median of SAR (Sale Price Assessed Value Ratio) does not significantly change month over month, the observed trend in mean SAR values was deemed practically significant, justifying the monthly adjustments. **Therefore, the average value of SAR changes month from month and values should be adjusted to reflect the value as of July 1st, 2017.** The table below (Table 4) shows the variation in the average of SAR in different months. Note that the mean value for SAR is approximately the SAR in the middle of the month and so the target SAR on July 1st is obtained by averaging the mean

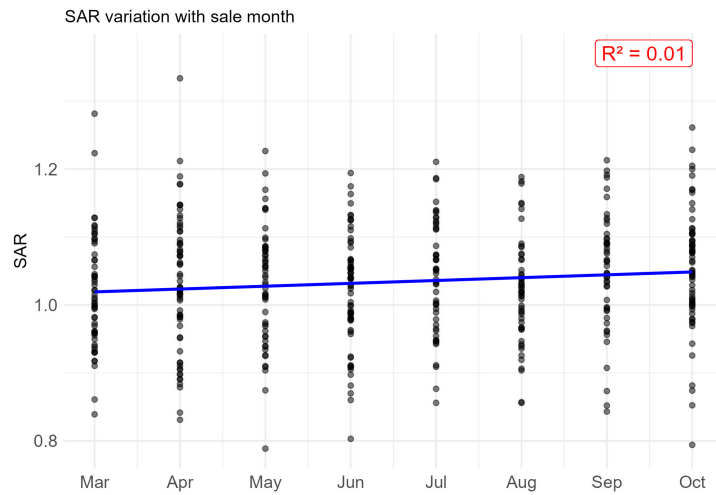


Figure 1: Scatterplot of SAR vs. Sale Month

Table 2: Mean SAR Rank by Sale Month

Sale Month	N	Mean Rank
3	58	204.82
4	66	231.35
5	55	230.13
6	68	217.76
7	52	246.10
8	53	208.68
9	49	265.69
10	65	266.49
Total	466	

Table 3: Kruskal-Wallis Test Statistics: SAR by Sale Month

Statistic	Value
Chi-Square	12.569
df	7
Asymp. Sig. (p-value)	0.083

SAR for the month of June and July. The adjustment factor is calculated as ‘Target SAR (July 1st) / Mean SAR’ for each month.

Therefore, the time adjusted sale price (TASP) will be obtained as follows:

$$\text{time adjusted sale price} = \text{saleprice} \times \text{factor}$$

To check if the time adjustment is done properly, the ratio statistics are derived after adjustment in Table 5. The table shows that the time adjustment is done properly. Also, to ensure that the adjusted sale price is equally distributed over months, a Kruskal-Wallis test was run for the adjusted sale price grouped by month. The p-value for the test is 0.190, which is well above the threshold of 5%, confirming that the adjustment successfully removed significant differences across months (see Appendix Table A9 on page 69). The TASP calculated using these monthly factors will be used in subsequent analyses.

Table 4: Mean SAR by Month and Time Adjustment Factors

Month	Mean SAR	Target SAR (July 1st)	Adjustment Factor
3	1.02	1.03	1.01
4	1.03	1.03	1.00
5	1.03	1.03	1.00
6	1.02	1.03	1.01
7	1.04	1.03	0.99
8	1.02	1.03	1.01
9	1.06	1.03	0.98
10	1.05	1.03	0.98

Table 5: Ratio Statistics for Time-Adjusted Sale Price (TASP) / Total Assessed Value by Sale Month

Month	Mean	95% Confidence Interval for Mean		Median	95% Confidence Interval for Median			Actual Coverage (%)	COD (%)	COV (%)
		Lower Bound	Upper Bound		Lower Bound	Upper Bound				
3	1.03	1.01	1.05	1.02	1.03	1.04		93.26	6.40	8.11
4	1.03	1.01	1.05	1.02	1.03	1.04		95.02	7.69	9.61
5	1.03	1.01	1.05	1.04	1.03	1.04		95.60	6.52	8.32
6	1.03	1.01	1.05	1.05	1.03	1.04		94.67	6.18	7.99
7	1.03	1.01	1.05	1.04	1.03	1.04		94.66	6.45	7.95
8	1.03	1.01	1.05	1.03	1.03	1.04		95.98	5.75	7.58
9	1.03	1.00	1.06	1.03	1.03	1.04		93.51	6.77	9.18
10	1.03	1.01	1.05	1.03	1.03	1.04		93.60	6.34	8.32

2.5 Market-Derived Land Value Adjustments

Estimating land value accurately is crucial for the Cost Approach. To initiate the land valuation process and subsequently isolate the land residual, the time-adjusted contributions of the existing land value ('landval') and improvement value ('imprval') components to TASP are estimated. This is achieved using a linear regression model with TASP as the response variable and 'landval' and 'imprval' as independent variables, with no intercept term. The regression aims to find market-based adjustment factors for the existing value components.

The results of the regression analysis ($TASP \sim \text{landval} + \text{imprval}$ without intercept) are presented in Table 6.

Table 6: Regression Coefficients: TASP vs. Land Value and Improvement Value

Model Component	Coefficient (B)	Std. Error	t	Sig. (p-value)
Land Value ('landval')	1.004	0.042	24.180	<0.001
Improvement Value ('imprval')	1.042	0.023	46.160	<0.001

Dependent Variable: Time-Adjusted Sale Price (TASP)

Linear Regression through the Origin

R-squared: 0.994, Adj. R-squared: 0.994, F-stat: 3.66e+04, p-value: < 2.2e-16

The coefficients indicate the market adjustment factors relative to the existing assessed components. The coefficient for 'landval' is 1.004, suggesting a negligible market adjustment (0.4% increase) is needed for the assessed land values on average. The coefficient for 'imprval' is 1.042, indicating that,

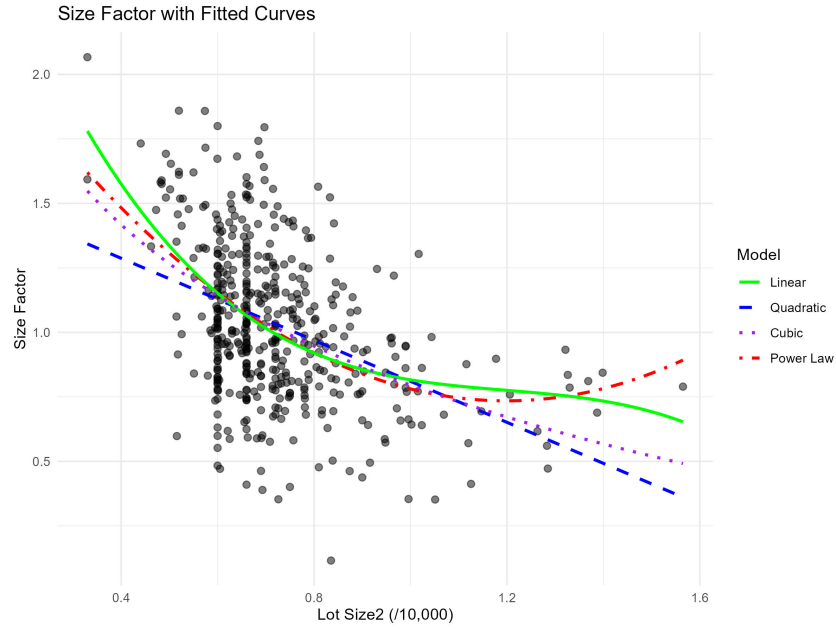


Figure 2: Size Factor vs. Scaled Lot Size (lotsize2)

on average, the market values the improvements at approximately 4.2% higher than their assessed improvement value ('imprval').

Therefore, the time-adjusted market value of the improvements can be estimated as by multiplying 'imprval' by 1.042. Consequently, the Land Residual representing the portion of the TASP attributable to the land after accounting for the adjusted improvement value, is calculated as:

$$\text{Land Residual} = \text{Time Adjusted Sale Price} - \text{Time Adjusted Improvement Value} \quad (1)$$

This 'Land Residual' will serve as the basis for further analysis using the unit value method. First, the relationship between land value and lot size needs to be established. A size factor ('sizefact') is calculated, representing the land residual value per square foot normalized by the median land residual value per square foot:

$$\text{sizefact} = \frac{\text{Land Residual}/\text{lotsize}}{\text{Median}(\text{Land Residual}/\text{lotsize})} \quad (2)$$

A scatter plot of this size factor versus lot size (scaled as 'lotsize2 = lotsize / 10,000') is generated (Figure 2) to visualize the relationship.

To model this relationship, four different functional forms (linear, quadratic, cubic, and power law) were fitted with 'sizefact' as the dependent variable and 'lotsize2' as the independent variable. The summary statistics for these models are presented in Table 7.

Based on the F-statistics and the typical relationship observed between lot size and unit value, the **power law** model is selected for the unit value method, despite the cubic model having a slightly higher R-squared. The power law is selected due to its theoretical soundness in representing diminishing marginal utility with increasing lot size and its common acceptance in appraisal practice for such adjustments. The power law provides a theoretically sound and commonly used functional form for this type of size adjustment. The coefficients for the fitted power law model are shown in Table 8.

The initial general land value estimate based on the power curve ('genlandpow') is then calculated by multiplying the size adjustment factor predicted from the power law ('adjsfactpow'), the 'lotsize', and the median land residual per lot size (which was used initially to normalize the 'sizefact').

$$\text{genlandpow} = \text{adjsfactpow} \times \text{lotsize} \times \text{Median}(\text{land Residual} / \text{lotsize}) \quad (3)$$

Table 7: Model Summary: Size Factor vs. Scaled Lot Size (lotsize2)

Equation	Model Summary				
	R Square	F	df1	df2	Sig.
Linear	0.173	97.121	1	464	<0.001
Quadratic	0.208	60.972	2	463	<0.001
Cubic	0.213	41.730	3	462	<0.001
Power	0.176	99.350	1	464	<0.001

Dependent Variable: sizefact
Independent Variable: lotsize2

Table 8: Power Law Model Coefficients: log(sizefact) vs. log(lotsize2)

Term	Estimate	Std. Error	t value	Sig. (p-value)
(Intercept)	−0.252	0.027	−9.187	<0.001
log(lotsize2)	−0.677	0.068	−9.967	<0.001

Dependent Variable: log(sizefact)
R-squared: 0.176, Adj. R-squared: 0.175

This ‘genlandpow’ (Eq. 4) represents the land value predicted based solely on size, adjusted according to the power curve relationship. This value will then be tested against other land characteristics (like conduit, neighbourhood) using ratio analysis and non-parametric tests to derive further multiplicative adjustment factors (ΠLQ).

Equation for genlandpow

$$\text{genlandpow} = \left(\exp(-0.252) \times \left(\frac{\text{lotsize}}{10000} \right)^{-0.677} \right) \times \text{lotsize} \times 7.2478 \quad (4)$$

2.5.1 Relationship Between Binary Variables

Before applying adjustments based on individual site characteristics, it is prudent to examine the relationships among these variables. Strong correlations or associations between variables can cause an adjustment for one variable to inadvertently impact the apparent effect of a correlated variable.

Reviewing the correlation matrix for the binary site variables (presented previously in Appendix Table A10), a strong correlation is observed between ‘conduit’ and ‘alley’. This suggests that properties with underground wiring are significantly less likely to have alleys, and vice-versa. To further investigate this association, a Chi-squared test of independence was performed between ‘conduit’ and ‘alley’. The contingency table (Table 9) shows the observed counts.

The Pearson’s Chi-squared test results are highly significant (Table 10). The small p-value confirms a strong association between the presence of underground wiring and the absence of an alley. Therefore, when proceeding with the adjustment for one of them, it is important to be mindful of this strong relationship.

2.5.2 Initial Land Value Adjustments (Conduit)

First, the Land Ratio (genlandpow / land residual) is examined based on the ‘conduit’ variable. A Kruskal-Wallis test (equivalent to Mann-Whitney for two groups) was performed. The results (Table 11)

Table 9: Contingency Table: Conduit vs Alley

Conduit	Alley		Row Total
	0 (No)	1 (Yes)	
0 (Overhead)	29	137	166
1 (Underground)	289	11	300
Column Total	318	148	466

Table 10: Chi-Squared Test: Conduit vs Alley

Statistic	Value
X-squared	303.050
df	1
p-value	< 2.2e-16

Note: Yates' continuity correction applied.

show a statistically significant difference ($p < 2.2e-16$) between properties with and without underground wiring.

Table 11: Kruskal-Wallis Test: Land Ratio by Conduit (Before Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	75.582
df	1
p-value	< 2.2e-16

Ratio statistics (Table 12) confirm this difference, with properties lacking conduit (0) having a lower median ratio (0.89) than those with conduit (1) (median ratio 1.13). This indicates that the initial 'genlandpow' estimate is relatively too high for properties without conduit and too low for properties with conduit, requiring adjustment.

Table 12: Ratio Statistics for Land Ratio by Conduit (Before Adjustment)

Conduit	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
0	0.89	0.90	0.87	0.93	0.37	1.55	19.10	24.20	0.84	0.93
1	1.13	1.12	1.09	1.15	0.13	1.81	18.60	24.00	1.11	1.16

Adjustments are based on the median ratios. Factors are calculated to bring the median ratio for each group towards the overall target median (ideally 1.0). The adjustment factor for conduit=0 is 0.89 and for conduit=1 is 1.13. These factors are applied multiplicatively to 'genlandpow' based on the 'conduit' value.

After applying the adjustment, the Kruskal-Wallis test was re-run on the adjusted land ratio. The result ($p = 0.483$, see Appendix Table A12) on page 70 is no longer statistically significant, indicating the adjustment successfully addressed the bias related to the 'conduit' variable. The ratio statistics for the adjusted ratio also show median values close to 1.0 for both groups (see Appendix Table A11 on page 70).

The adjusted land value estimate, incorporating both size and conduit adjustments, will now be tested against other variable, starting from Neighbourhood.

2.5.3 Adjustments for Other Binary Variables

The conduit-adjusted land ratio is then tested against other binary site characteristics using the Kruskal-Wallis (Mann-Whitney) test ($\alpha = 0.05$) to determine if further adjustments were needed. The variables tested included: ‘alley’, ‘sidewalk’, ‘culdesac’, ‘corner’, ‘wooded’, ‘pie_shpe’, ‘slope’, and ‘abv_road’.

The results of the Kruskal-Wallis tests are summarized in Table 13. None of the p-values were below the 0.05 significance threshold. Specifically, the results for ‘alley’ remain biased despite the adjustment on ‘conduit’, particularly in cases of strong correlation.

Table 13: Kruskal-Wallis Test Results: Conduit-Adjusted Land Ratio by Binary Site Variables

Variable Tested	p-value
alley	0.199
sidewalk	0.189
culdesac	0.285
corner	0.084
wooded	0.067
pie_shpe	0.189
slope	0.688
abv_road	0.228

Ratio statistics stratified by these variables (Tables 14 and 15) also showed median ratios generally close to 1.0 for both levels of each variable.

Table 14: Ratio Statistics for Conduit-Adjusted Land Ratio by Binary Site Variables (Part 1)

Variable	Level	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
alley	0	1.00	0.98	0.96	1.01	0.12	1.60	18.30	23.70	0.97	1.02
	1	1.01	1.02	0.98	1.06	0.38	1.75	19.50	24.70	0.95	1.06
sidewalk	0	1.00	0.99	0.97	1.01	0.12	1.66	18.40	23.70	0.97	1.01
	1	1.08	1.04	0.95	1.13	0.41	1.75	20.00	27.70	0.91	1.13
culdesac	0	1.00	1.00	0.97	1.02	0.35	1.75	18.70	23.90	0.97	1.02
	1	1.08	1.01	0.90	1.11	0.12	1.45	17.40	27.80	0.97	1.11
corner	0	0.99	0.99	0.96	1.01	0.12	1.72	18.50	24.00	0.97	1.01
	1	1.07	1.04	0.98	1.10	0.41	1.75	18.60	24.40	0.98	1.13

Table 15: Ratio Statistics for Conduit-Adjusted Land Ratio by Binary Site Variables (Part 2)

Variable	Level	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
wooded	0	1.01	1.00	0.98	1.03	0.12	1.75	18.60	24.00	0.98	1.03
	1	0.92	0.92	0.85	1.00	0.35	1.43	19.50	24.60	0.81	1.01
pie_shpe	0	1.00	0.99	0.97	1.02	0.35	1.75	18.60	23.80	0.97	1.02
	1	1.06	1.04	0.92	1.15	0.12	1.48	20.10	28.50	0.91	1.15
slope	0	1.00	1.00	0.97	1.02	0.12	1.75	18.90	24.40	0.97	1.03
	1	1.00	0.98	0.92	1.04	0.51	1.45	17.60	22.10	0.92	1.08
abv_road	0	1.00	1.00	0.98	1.03	0.35	1.75	18.60	24.10	0.98	1.03
	1	0.96	0.96	0.91	1.01	0.12	1.45	19.10	24.00	0.91	1.01

Therefore, based on these statistical tests and ratio analyses, no further adjustments for these binary site characteristics were deemed necessary. The adjusted land value estimate, incorporating size (via power curve) and conduit adjustments, will now be tested against the Neighbourhood variable.

2.5.4 Adjustment for Neighbourhood

The land value estimate, now adjusted for size (via power curve) and conduit (‘genlandpow_adj_conduit’), is tested against the ‘Neighbourhood’ variable. The ratio used for testing is the conduit-adjusted land ratio = Land Residual / genlandpow_adj_conduit. A Kruskal-Wallis test was performed to determine if the

median of this adjusted land ratio differs significantly across the various neighbourhoods. The results (Table 16) show a highly significant difference ($p < 1.24\text{e-}12$), indicating that location adjustments are likely necessary.

Table 16: Kruskal-Wallis Test: Conduit-Adjusted Land Ratio by Neighbourhood

Statistic	Value
Kruskal-Wallis chi-squared	70.377
df	7
p-value	$< 1.24\text{e-}12$

Ratio statistics stratified by neighbourhood (Table 17) provide the median ratios and their confidence intervals, which are used to determine the specific adjustments. An adjustment is applied only if a neighbourhood's median ratio falls outside the acceptable range of [0.95, 1.05] or its confidence interval for the median does not contain 1.0. Based on these criteria and the results in Table 17, adjustments are required for Neighbourhoods 715, 731, 734, 736, and 737. Neighbourhoods 713, 716, and 732 do not require adjustment. The adjustments are based on the median ratios. Factors are calculated for the qualifying neighbourhoods to bring their median ratio towards the overall target median (ideally 1.0). The adjustment factor for each neighbourhood 'i' is calculated as $\text{Factor}_i = \text{Median Ratio}_i$. These factors represent the neighbourhood component of ILQ .

These neighbourhood-specific factors are then applied multiplicatively to the previously adjusted land value estimate to arrive at the final Base Land Value. This value now incorporates adjustments for size (via power curve), conduit, and neighbourhood, providing the land value input for the final cost model calculation. Subsequent testing confirms that the ratios based on this final Land Value are uniform across neighbourhoods (see Tables 18 and Appendix A13 on page 70). The Kruskal-Wallis test on the final adjusted ratio yields a p-value of 0.673 (Appendix Table A13 on page 70), indicating no remaining significant differences attributable to neighbourhood.

Table 17: Ratio Statistics for Conduit-Adjusted Land Ratio by Neighbourhood

Neigh.	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
713	1.01	0.99	0.94	1.04	0.38	1.53	17.20	22.70	0.93	1.06
715	0.83	0.85	0.79	0.92	0.41	1.54	22.20	28.20	0.78	0.90
716	1.04	1.04	1.01	1.08	0.51	1.60	15.00	19.00	1.00	1.07
731	1.13	1.20	1.09	1.30	0.68	1.75	18.50	22.50	1.04	1.31
732	1.02	1.03	0.81	1.26	0.65	1.44	22.00	26.20	0.65	1.33
734	0.85	0.87	0.77	0.98	0.62	1.24	14.10	18.60	0.79	0.95
736	0.81	0.79	0.70	0.88	0.12	1.24	18.90	27.90	0.73	0.87
737	1.05	1.03	0.98	1.08	0.35	1.53	16.80	22.90	1.01	1.09

Table 18: Ratio Statistics for Land Ratio by Neighbourhood (After Adjustment)

Neigh.	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
713	1.01	0.99	0.94	1.04	0.38	1.53	17.20	22.70	0.93	1.06
715	1.00	1.03	0.95	1.10	0.49	1.85	22.20	28.20	0.93	1.09
716	1.04	1.04	1.01	1.08	0.51	1.60	15.00	19.00	1.00	1.07
731	1.00	1.06	0.97	1.15	0.60	1.55	18.50	22.50	0.92	1.16
732	1.02	1.03	0.81	1.26	0.65	1.44	22.00	26.20	0.65	1.33
734	1.00	1.03	0.91	1.15	0.73	1.46	14.10	18.60	0.93	1.12
736	0.99	0.98	0.87	1.08	0.14	1.53	18.90	27.90	0.90	1.07
737	1.00	0.98	0.93	1.03	0.34	1.46	16.80	22.90	0.96	1.04

2.5.5 Final Land Value Model

The final base land value incorporates adjustments derived from the preceding analyses for lot size, conduit, and neighbourhood. The model uses a unit value approach, starting with a base rate per square foot, adjusting for size effects using a power law, and then applying multiplicative factors for specific site and location characteristics. The components are:

- **Base Rate:** The median land residual value per square foot, calculated as \$7.2478.
- **Lot Size:** The size of the lot in square feet ('lotsize').
- **Size Adjustment Factor ('adjsfactpow'):** Derived from the power law regression, adjusting the base rate for the non-linear effect of size. It is calculated as

$$\text{adjsfactpow} = \exp(-0.252) \times \left(\frac{\text{lotsize}}{10000} \right)^{-0.677}$$

- **Conduit Adjustment Factor ('factor_conduit'):** A multiplicative factor based on the median land ratio for properties with ('conduit'=1) or without ('conduit'=0) underground wiring. The factors are 0.89 for 'conduit'=0 and 1.13 for 'conduit'=1.
- **Neighbourhood Adjustment Factor ('factor_neighbourhood'):** A multiplicative factor applied based on the median land ratio for specific neighbourhoods identified as needing adjustment. The factors are 0.83 (Neigh 715), 1.13 (Neigh 731), 0.85 (Neigh 734), 0.81 (Neigh 736), 1.05 (Neigh 737), and 1.0 otherwise.

Combining these elements, the final equation for the Base Land Value (Land Value) expresses the adjustments using indicator variables in the exponents:

Final Land Value Model Equation

$$\begin{aligned} \text{Land Value} = & \underbrace{\left(\exp(-0.252) \times \left(\frac{\text{lotsize}}{10000} \right)^{-0.677} \right)}_{\text{adjsfactpow}} \times \text{lotsize} \times 7.2478 \\ & \times (0.89)^{I_{\text{conduit}=0}} \times (1.13)^{I_{\text{conduit}=1}} \\ & \times (0.83)^{I_{\text{neigh}=715}} \times (1.13)^{I_{\text{neigh}=731}} \times (0.85)^{I_{\text{neigh}=734}} \\ & \times (0.81)^{I_{\text{neigh}=736}} \times (1.05)^{I_{\text{neigh}=737}} \quad (5) \end{aligned}$$

Where:

- $I_{\text{condition}}$ is an indicator variable that equals 1 if the condition is true for the property, and 0 otherwise. For example, $I_{\text{conduit}=0}$ is 1 if 'conduit'=0, and 0 if 'conduit'=1. Similarly, $I_{\text{neigh}=715}$ is 1 only if 'neigh'=715, and 0 for all other neighbourhoods.
- Neighbourhoods 713, 716, and 732 effectively have a factor of $1.0^1 = 1.0$ as they did not require adjustment.

This Land Value represents the final land value estimate used as input for the cost model.

2.6 Replacement Cost New Less Depreciation (RCNLD) Calculation

This step focuses on estimating the value contribution of the improvements, represented by Replacement Cost New Less Depreciation (RCNLD). This involves calculating the base RCN based on the provided cost data and property characteristics, deriving market-based depreciation (PCTGOOD), and applying a local cost multiplier (LCM).

2.6.1 Replacement Cost New (RCN) Calculation

The base RCN for each property's improvements (including main building, garages, and carports) is calculated using the standardized cost data provided for this assignment. The calculation follows a component-based approach, summing the costs of different parts of the property and applying quality adjustments. The specific steps implemented are:

1. **Determine Building Structure:** Identify the number of stories (1 or 2 based on 'flrarea2' > 0) and the basement type ("Full" if 'bsmtarea' > 0, "Crawl" if 'crlwarea' > 0, "None" otherwise).
2. **Calculate Base Constant Cost:** Apply a fixed base cost depending on the structure:
 - 1-storey with Full Basement: \$51,500
 - 1-storey with Crawl Space: \$45,500
 - 2-storey with Full Basement: \$51,500 + \$16,300 = \$67,800
 - 2-storey with Crawl Space: \$45,500 + \$16,300 = \$61,800
3. **Calculate Area-Based Costs (Main Floors):** Add costs based on floor area:
 - 1-storey: 'flrarea1' * \$42.50
 - 2-storey: ('flrarea1' * \$42.50) + ('flrarea2' * \$33.40)
4. **Add Basement/Crawl/Finished Basement Area Costs:** Add costs based on the area of these components:
 - Full Basement: 'bsmtarea' * \$21.25 (if 'basement_type' is "Full")
 - Crawl Space: 'crlwarea' * \$14.90 (if 'basement_type' is "Crawl")
 - Finished Basement: 'bsmtfin' * \$15.60 (if 'bsmtfin' > 0)
5. **Add Hot Water Heat Adjustment:** For properties with 'heattype' coded as 7 or 8, add an additional cost calculated as ('flrarea1' * \$4.40) + ('flrarea2' * \$3.30).
6. **Add Fireplace and Bath Costs:** Add costs for specific features:
 - Fireplaces ('fireplcs'): Number of fireplaces * \$3,850
 - Baths ('baths'): Number of baths * \$4,000 (Note: This calculation uses the value in the 'baths' field directly, assuming it represents the total equivalent full baths, including fractions).
7. **Sum Main Building Cost ('mb_cost'):** Sum the costs from steps 2 through 6 to get the total cost for the main structure before quality adjustment.
8. **Apply Main Building Quality Factor:** Multiply the 'mb_cost' by the appropriate Main Building Manual Class factor ('mb_factor') based on the 'mbmancls' variable (e.g., 0.945 for class 90, 1.000 for class 140, 1.220 for class 145, etc.) to get the adjusted main building RCN.
9. **Calculate Garage RCN:**
 - Determine the raw garage cost based on 'gar_type':
 - Type 1 (Attached): \$11,600 + ('gar_area' * \$30.00)
 - Type 2 (Detached): \$16,300 + ('gar_area' * \$30.00)
 - Type 3 (Basement): \$7,100
 - Other types/No garage: \$0

- Determine the Garage Quality Factor ('gr_factor') based on 'gr_manc1' (e.g., 1.000 for class 920/921, 1.200 for class 930/931, etc.).
- Multiply the raw garage cost by the 'gr_factor' to get 'gr_rcn'.

10. Calculate Carport RCN ('cp_rcn'):

- Determine the raw carport cost: If 'cp_area' > 0, Cost = \$5,850 + ('cp_area' * \$22.00), otherwise \$0.
- Determine the Carport Quality Factor ('CP_factor') based on 'cp_manc1' (using the same factors as garages).
- Multiply the raw carport cost by the 'CP_factor' to get 'cp_rcn'.

11. Calculate Total RCN ('rcn'): Sum the adjusted RCN values for the main building, garage, and carport:

$$rcn = mb_rcn + gr_rcn + cp_rcn \quad (6)$$

This calculated 'rcn' represents the total estimated Replacement Cost New for the property's improvements based on the provided cost data and characteristics, before considering depreciation or local market adjustments (LCM).

2.6.2 Depreciation Analysis (Percent Good - PCTGOOD)

Market-derived depreciation is estimated by analyzing the relationship between the Improvement Residual and the RCN. Observed Depreciation ('obsdeprn') is calculated as the proportion of RCN lost:

$$obsdeprn = 1 - \frac{\text{Time Adjusted Sale Price} - \text{Land Value}}{\text{Total RCN} + \text{Outbuilding Market Value}} \quad (7)$$

This observed depreciation is then modeled as a function of the property's effective age ('effage').

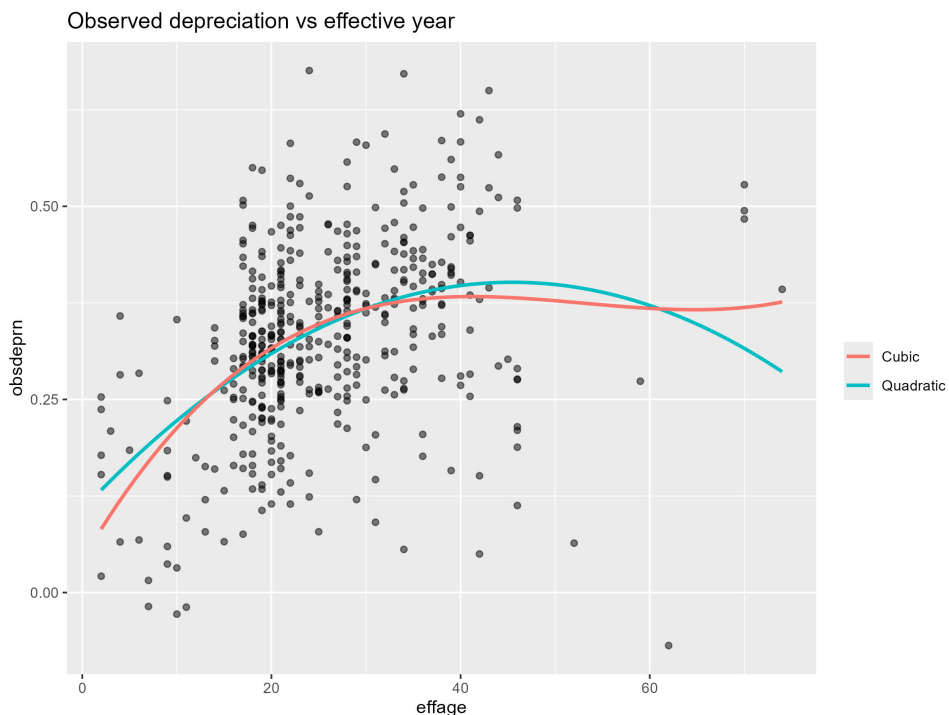


Figure 3: Observed Depreciation vs. Effective Age

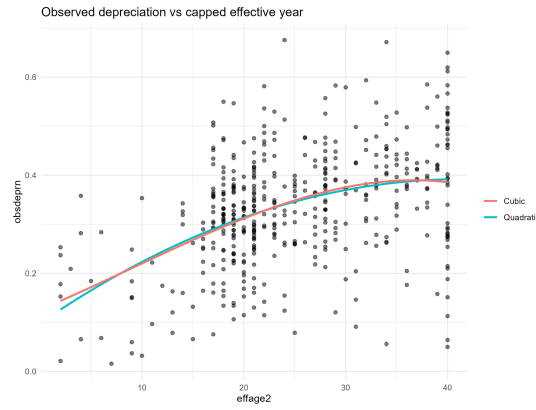


Figure 4: Observed Depreciation vs. Capped Effective Age After Removing Negative Depreciation

An initial plot of ‘obsdeprn’ versus effective age (‘effage’) (Figure 3) suggested potential issues with fitting a smooth curve, particularly for older properties. Note that ‘effage’ is derived as follows:

$$\text{effage} = 2017 - \text{mbeffyr}$$

The observed depreciation appeared to potentially increase again after a certain age, around 40 years old, which contradicts typical depreciation patterns where the rate slows down over time. This might be due to data variability or the influence of renovations not fully captured by ‘effage’. To address this and reflect typical depreciation patterns, the effective age was capped at 40 years. The relationship between ‘obsdeprn’ and this capped effective age (‘effage2’) was then re-examined. For this analysis, properties where the calculated ‘obsdeprn’ was non-positive were dismissed. Figure 4 plots this relationship for the filtered data and overlays fits from both quadratic and cubic polynomial regression models.

Both quadratic and cubic models were fitted to the positive ‘obsdeprn’ values against the capped effective age (‘effage2’). Comparing the model summaries (see Appendix Tables A14 on page 71 and A15 on page 71), the cubic model (Adj. R-squared = 0.204) provides only a marginal improvement in explanatory power over the quadratic model (Adj. R-squared = 0.203). Furthermore, examining the coefficients of the cubic model reveals that none of its terms reach statistical significance. In contrast, all terms in the quadratic model are statistically significant ($p < 0.01$), as shown in Table 19.

Table 19: Quadratic Depreciation Model Coefficients (Dependent: obsdeprn)

Term	Estimate	Std. Error	t value	Sig. (p-value)
(Intercept)	8.71200×10^{-2}	3.20900×10^{-2}	2.71400	0.007
effage2	1.42000×10^{-2}	2.62400×10^{-3}	5.41200	<0.001
effage2^2	-1.65500×10^{-4}	5.07700×10^{-5}	-3.26000	0.001

Based on model: obsdeprn ~ effage2 + effage2^2, filtered for obsdeprn > 0

R-squared: 0.207, Adj. R-squared: 0.203

Given the lack of significance for the terms in the cubic model and the preference in appraisal practice for simpler, stable depreciation curves, the quadratic model is selected as the more appropriate representation of the depreciation trend.

Percent Good is the complement of depreciation. Therefore, the market-derived Percent Good for each property was calculated using the coefficients from the selected quadratic model based on its capped effective age (‘effage2’):

$$\text{predicted obsdeprn} = 0.08712 + (0.01420 \times \text{effage2}) - (0.0001655 \times \text{effage2}^2) \quad (8)$$

$$\text{PCTGOOD} = 1 - \text{predicted obsdeprn} \quad (9)$$

The resulting relationship between the calculated ‘pctgood’ and the capped effective age (‘effage2’) is shown in Figure 5. This curve represents the estimated percentage of RCN remaining as a function of age, reflecting market-observed depreciation.

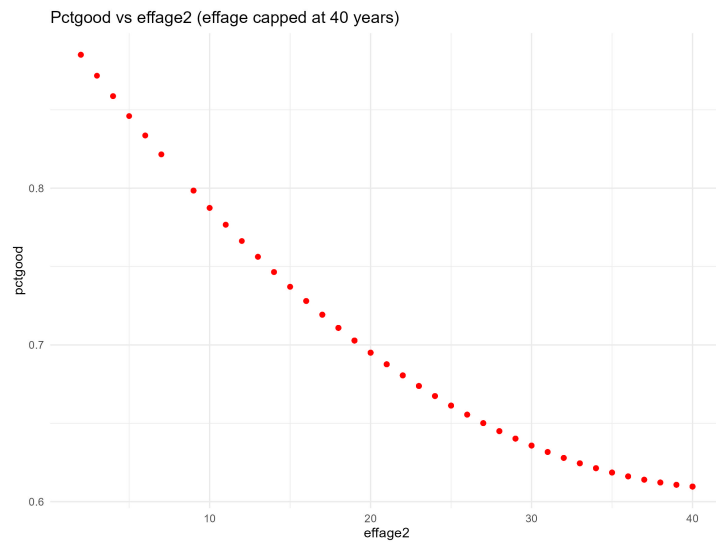


Figure 5: Percent Good vs. Capped Effective Age

The final step before calibrating the Local Cost Multiplier (LCM) is to calculate the preliminary Replacement Cost New Less Depreciation (RCNLD) based on the previously calculated RCN components and the market-derived Percent Good. The equation structure below mirrors the logic used in the analysis code, applying the depreciation factor to the primary structures (main building, garage, carport) and adding separately valued components like other buildings (‘ob1mktvl’).

The preliminary RCNLD (‘imprcnld’) is calculated as follows, applying the Percent Good only to the primary structure components (Main Building, Garage, Carport) and adding the value of other buildings:

$$\begin{aligned}
 \text{imprcnld} = & \text{PCTGOOD} \times (\text{linmcdv} \times (\text{genconstant} + (42.50 \times \text{flarea1}) + \\
 & (33.40 \times \text{flarea2}) + 21.25 \times \text{bsmtarea} + 14.90 \times \text{crlwarea} + 15.60 \times \text{bsmtfin} + \\
 & \text{heatwater} + 4000 \times \text{baths} + 3850 \times \text{firepls}) + \\
 & \text{cp_qfac} \times \text{carpfl} \times (5850 + 22.00 \times \text{cp_area}) + \\
 & \text{gr_qfac} \times \text{garagefl} \times (\text{attgar} \times (11600 + 30.00 \times \text{gar_area}) + \\
 & \text{detgar} \times (16300 + 30.00 \times \text{gar_area}) + (\text{bsmtgar} \times 7100))) + \text{ob1mktvl}
 \end{aligned} \tag{10}$$

This value represents the estimated depreciated value of the improvements before applying the final Local Cost Multiplier (LCM). The next step involves comparing this value to the Improvement Residual to calibrate the LCM.

2.6.3 Market-Derived Improvement Value Adjustments

After calculating the ‘imprcnld’, further market adjustments might be necessary. This is evaluated by calculating an Improvement Value Ratio, termed Market Factor (‘market_factor’), which compares the market-derived value of the improvements (‘imprrsid’) to ‘imprcnld’ calculated before LCM).

$$\text{imprrsid} = \text{Time Adjusted Sale Price} - \text{Land Value (Eq. 5)} \tag{11}$$

$$\text{market_factor} = \frac{\text{imprrsid (Eq. 11)}}{\text{imprcnld (Eq. 10)}} \tag{12}$$

Ideally, this ratio should center around 1.0 if the RCNLD estimate accurately reflects the market.

Overall ratio statistics for the ‘market_factor’ are presented in Table 20.

Table 20: Overall Ratio Statistics for Market Factor (Improvement Residual / Preliminary RCNLD)

Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
0.99	1.00	0.99	1.02	0.48	1.70	12.37	16.26	0.98	1.01

95% CI Actual Coverage: 94.8%

The overall statistics are very good, with a median of 0.99 (within the [0.95, 1.05] range), a tight confidence interval for the median [0.98, 1.01] which includes 1.0, and a COD well below 15%. This suggests that, overall, the preliminary RCNLD estimate is well-calibrated to the market residuals and major adjustments might not be necessary.

However, before concluding, potential biases related to specific improvement characteristics are examined. Similar to the land analysis, relationships between improvement-related variables are checked first. A high correlation was noted between ‘garagefl’ (presence of a garage) and ‘gar_type’. A Chi-squared test confirmed this strong association (see contingency table A17 on page 71 and Chi-squared test on A18 on page 72) for more details. This implies that any adjustment considered for ‘gar_type’ should be evaluated carefully for its potential impact on ‘garagefl’, due to their high correlation.

Next, Kruskal-Wallis tests were performed on the ‘market_factor’ stratified by key binary improvement-related variables. The results are summarized in Table 21. Ratio studies for Improvement Ratio in Table A19 on 72 also shows that the only variable that needs adjustment is ‘gar_type’.

Table 21: Kruskal-Wallis Test Results: Market Factor by Binary Improvement Variables

Variable Tested	p-value
heattype	0.149
baseste	0.063
garagefl	0.527
carptfl	0.095
gar_type	<0.001
splitlvl	0.075

At the $\alpha = 0.05$ significance level, only ‘gar_type’ shows a statistically significant difference in the distribution of the Market Factor ($p < 0.001$). This suggests an adjustment based on garage type may be warranted.

Ratio statistics stratified by ‘gar_type’ (Table 22) provide further detail.

Table 22: Ratio Statistics for Market Factor by Garage Type

gar_type	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
0 (None)	1.00	1.00	0.99	1.02	0.48	1.70	12.20	16.40	0.99	1.01
1 (Attached)	0.97	1.00	0.96	1.04	0.78	1.25	11.00	13.20	0.95	1.05
2 (Detached)	0.95	0.93	0.89	0.97	0.62	1.39	13.10	17.50	0.89	0.97
3 (Basement)	1.09	1.08	1.04	1.13	0.86	1.32	9.61	12.00	1.02	1.13

Examining the median ratios and confidence intervals in Table 22, Type 0 (No Garage) has a median of 1.00 with a CI [0.99, 1.01], and Type 1 (Attached) has a median of 0.97 with a CI [0.95, 1.05]; both meet the criteria and require no adjustment. Type 2 (Detached) has a median of 0.95 with a CI [0.89, 0.97]; although the median is borderline, the confidence interval does not contain 1.0, indicating an adjustment is needed. Type 3 (Basement) has a median of 1.09, which is outside the [0.95, 1.05] range, and its CI [1.02, 1.13] also does not contain 1.0, confirming an adjustment is required. Therefore, adjustments based on the median ratios will be applied for garage types 2 (Detached) and 3 (Basement),

using factors of 0.95 and 1.09 respectively. These factors will be applied multiplicatively to the ‘imprc-nld’ estimate. It will be important to re-check the Kruskal-Wallis test results for ‘garagefl’ and other variables after applying the ‘gar_type’ adjustment to ensure no adverse effects due to the correlation. Tables A20 on page 72 and A21 on page 72 present the ratio statistics and Kruskal-Wallis test results for after applying the adjustments based on the garage type. These results are used to confirm that the adjustment successfully removed the statistically significant bias related to garage type.

After applying the adjustments for garage types 2 and 3, the Kruskal-Wallis test was re-run on the adjusted market factor stratified by the binary improvement variables to check if the initial adjustment caused any unintended bias or if other variables now show significant differences. The results are summarized in Table 23.

Table 23: Kruskal-Wallis Test Results: Garage-Adjusted Market Factor by Binary Variables

Variable Tested	p-value
heattype	0.029
baseste	0.139
garagefl	0.480
carptfl	0.192
gar_type	0.849
splitlvl	0.141

The test results indicate that after adjusting for ‘gar_type’, only ‘heattype’ shows a statistically significant difference ($p = 0.029$) in the market factor distribution. The test for ‘gar_type’ itself is now non-significant ($p = 0.849$), confirming the effectiveness of the previous adjustment. The test for ‘garagefl’ remains non-significant ($p=0.480$), indicating no adverse effect from the ‘gar_type’ adjustment despite their correlation.

Further investigation into the ‘heattype’ variable is warranted. Ratio statistics for the garage-adjusted market factor stratified by ‘heattype’ are presented in Appendix Table A22. Examining the median ratios and their confidence intervals reveals potential issues primarily with ‘heattype’ codes 1 (Median 1.14), 3 (Median 1.11), and 7 (Median 0.94). However, ‘heattype’ = 1 (Gas Wall Furnace) has only 3 observations, making its statistics unreliable and its confidence interval incalculable. Given its similarity to ‘heattype’ = 2 (Gas Fired Hot Air), these two categories were merged for a more robust analysis.

The ratio statistics for the merged ‘heattype’ categories are shown in Appendix Table A23. Based on the criteria (Median outside $[0.95, 1.05]$ and CI not containing 1.0), adjustments are needed for ‘heattype’ = 3 (Median 1.11, CI $[1.01, 1.33]$) and potentially ‘heattype’ = 7 (Median 0.94, CI $[0.78, 1.04]$). Although the confidence interval for ‘heattype’ = 7 contains 1.0, its median is below the 0.95 threshold, suggesting an adjustment based on the median may improve equity.

Therefore, multiplicative adjustment factors based on the median ratios are applied for ‘heattype’ = 3 (Factor = 1.11) and ‘heattype’ = 7 (Factor = 0.94). For all other ‘heattype’ categories (including the merged 1 2), the factor is 1.0.

After applying these ‘heattype’ adjustments, the Kruskal-Wallis tests were re-run for all binary variables against the fully adjusted market factor. The results (Appendix Table A24) show that no variables exhibit statistically significant differences at the $\alpha = 0.05$ level. Ratio statistics also confirm that median values are now generally centered around 1.0 across ‘heattype’ categories (Appendix Table A25). This indicates that the necessary market-based adjustments for the preliminary RCNLD estimate for binary variables have been completed.

The final adjusted RCNLD, incorporating factors for ‘gar_type’ and ‘heattype’, is now ready for LCM calibration.

2.6.4 Adjustments for Other Building Characteristics

With adjustments applied for 'gar_type' and 'heattype', the analysis proceeds to test the now fully adjusted improvement ratio against other non-binary building characteristics.

First, correlations among these characteristics were examined (see Appendix Table A26). The only very high correlation observed was between 'carptfl' and 'cp_manc1' (0.99999), which is expected as carport quality ('cp_manc1') is only relevant if a carport exists ('carptfl'=1). Other correlations are generally low to moderate.

Kruskal-Wallis tests were performed on improvement ratio stratified by 'stories', grouped 'mbmancls', 'familyrm', and 'bedrooms'.

- **Stories:** The test showed no significant difference between 1-storey and 2-storey homes ($p = 0.152$). Ratio statistics also indicated acceptable levels and uniformity (see Appendix Tables A27 and A28). No adjustment was applied.
- **Main Building Manual Class ('mbmancls'):** Due to low frequencies in some original classes (see Appendix Table A29), classes were grouped: 40-81 into Group 8; 90 & 92 into Group 9; 140 & 141 into Group 14; and 142-150 into Group 15. The Kruskal-Wallis test on the grouped classes was significant ($p = 0.0017$, Appendix Table A30). Ratio statistics (Appendix Table A31) showed that Group 8 (lower quality classes) had a median ratio of 1.08, which was outside the acceptable range [0.95, 1.05] and its CI [1.01, 1.16] did not contain 1.0. An adjustment factor based on the median (1.08) was applied to properties in Group 8. After adjustment, the Kruskal-Wallis test became non-significant ($p = 0.078$), indicating the adjustment was successful (see Appendix Tables A32).
- **Family Room ('familyrm'):** The Kruskal-Wallis test on the improvement ratio (now adjusted for 'gar_type', 'heattype', and grouped 'mbmancls') stratified by 'familyrm' count (0, 1, or 2) was significant ($p = 0.013$, Appendix Table A33). Ratio statistics (Appendix Table A34) showed that properties with 0 family rooms had a median ratio of 0.95, meeting the criteria for adjustment. A factor based on this median (0.95) was applied. The post-adjustment Kruskal-Wallis test was non-significant ($p = 0.669$), and ratio statistics showed improved uniformity (see Appendix Tables A35 and A36).
- **Bedrooms ('bedrooms'):** The Kruskal-Wallis test on the improvement ratio (adjusted for previous factors) stratified by 'bedrooms' (2, 3, or 4) was significant ($p = 0.025$, Appendix Table A37). However, examining the ratio statistics (Appendix Table A38), the median ratios for all groups (2 beds: 0.98, 3 beds: 1.01, 4 beds: 0.97) were within the acceptable [0.95, 1.05] range. Despite the KW test result, the ratio statistics suggest reasonable equity. Nevertheless, following the principle of addressing statistical significance where practical, an adjustment based on the median ratio was applied to properties with 2 bedrooms (Factor = 0.98). The post-adjustment Kruskal-Wallis test became non-significant ($p = 0.265$), and ratio statistics remained acceptable (see Appendix Tables A39 and A40).

Finally, Kruskal-Wallis tests were run on the fully adjusted improvement ratio against all remaining building characteristic variables ('carptfl', 'fireplcs', 'baseste', 'gr_manc1', 'cp_manc1'). All p -values were well above the 0.05 threshold (see Table 24), indicating that no further adjustments are required for the improvement ratio based on these characteristics.

The final adjusted improvement ratio now incorporates adjustments for 'gar_type', 'heattype', grouped 'mbmancls', 'familyrm', and 'bedrooms'. This adjusted factor represents the market-derived multiplier to be potentially applied as the Local Cost Multiplier (LCM).

2.6.5 Replacement Cost New Less Depreciation (RCNLD) Model

The final market-adjusted cost model ($MV_{CostModel}$) combines the fully adjusted Base Land Value (LV_{Base}) as specified in Equation 5, and the final adjusted Replacement Cost New Less Depreciation

Table 24: Kruskal-Wallis Test Results: Final Adjusted Improvement Ratio by Other Building Variables

Variable Tested	p-value
carptfl	0.310
fireplcs	0.605
bedrooms	0.265
familyrm	0.668
baseste	0.259
gr_mancl	0.132
cp_mancl	0.469

($RCNLD_{Final}$).

The $imprcnld_{final}$ is calculated by taking the preliminary ‘imprcnld’ from Equation 10) and applying the Local Cost Multiplier (LCM) along with the multiplicative adjustment factors derived from the improvement ratio analysis. The LCM is taken as the overall median of the initial market factor (Improvement Residual / imprcnld), which was 0.99 (Table 20). The specific adjustment factors are applied using indicator variables:

Final RCNLD Equation

$$\begin{aligned}
 imprcnld_{final} = & \text{imprcnld}_{prelim} \times \underbrace{0.99}_{\text{LCM}} \\
 & \times (0.95)^{I_{gar_type=2}} \times (1.09)^{I_{gar_type=3}} \\
 & \times (1.11)^{I_{heattype=3}} \times (0.94)^{I_{heattype=7}} \\
 & \times (1.08)^{I_{mbmancls_grp=8}} \\
 & \times (0.95)^{I_{familyrm=0}} \times (0.98)^{I_{bedrooms=2}} \quad (13)
 \end{aligned}$$

Where:

- $imprcnld$ is the preliminary $imprcnld$ calculated before adjustments (Eq. 10).
- $I_{condition}$ is an indicator variable (1 if true, 0 otherwise). For factors not listed (e.g., ‘gar_type’ 0 or 1, ‘heattype’ 2,4,5,6,8, ‘mbmancls’ 9,14,15, ‘familyrm’ 1 or 2, ‘bedrooms’ 3 or 4), the effective factor is $Factor^0 = 1$.

The complete final market-adjusted cost model is therefore the sum of the final land value and the final RCNLD:

Final Market-Adjusted Cost Model Equation

$$MV = \text{Land Value} + imprcnld_{final} \quad (14)$$

Where Land Value is defined in Equation 5 and $imprcnld_{final}$ is defined in Equation 13. This equation represents the final calibrated cost model estimate for market value as of July 1, 2017. The performance of this final model will be evaluated in the Model Testing section.

2.7 Cost Model Testing

The final step in evaluating the cost approach is to test the performance of the fully calibrated model. The final Cost Model Value ('costval') is calculated as the sum of the final adjusted land value (Land Value) and the final adjusted RCNLD (imprcnld_{final}), as defined in Equation 2.6.5:

$$\text{costval} = \text{Land Value} + \text{imprcnld}_{final} \quad (15)$$

The primary metric for evaluating model performance is the Cost Assessment Ratio (Cost ASR), defined as the ratio of the model's predicted value to the time-adjusted sale price:

$$\text{costasr} = \frac{\text{costval}}{\text{Time Adjusted Sale Price}} \quad (16)$$

2.7.1 Overall Performance and Scatter Plot

A scatter plot comparing the calculated 'costval' against the Time-Adjusted Sale Price provides a visual assessment of the model's predictive accuracy (Figure 6). Ideally, points should cluster closely around the 45-degree line.

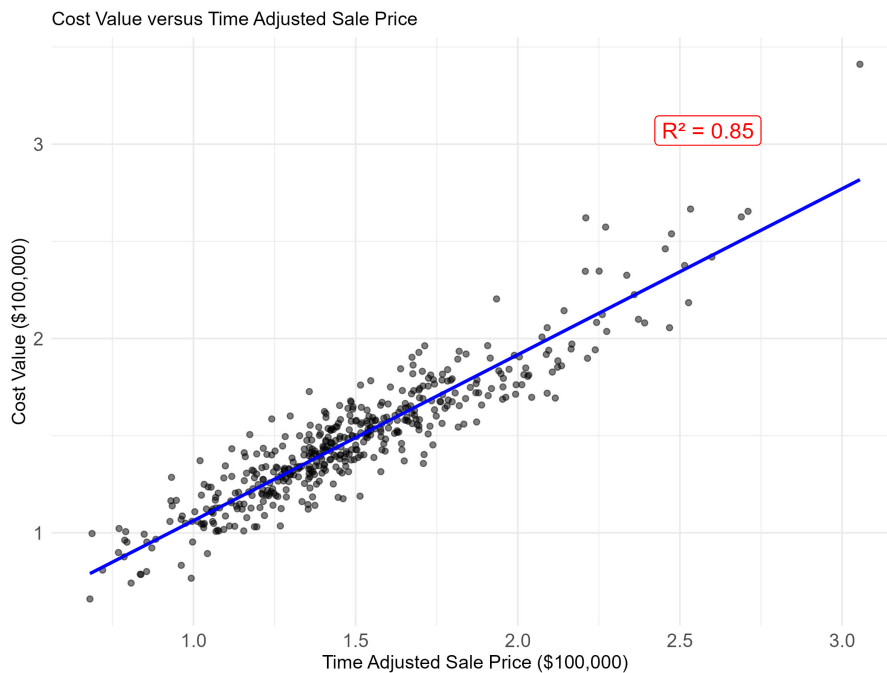


Figure 6: Cost Model Value vs. Time Adjusted Sale Price

Overall ratio statistics were calculated for the 'costasr' across all valid sales (Table 25).

Table 25: Overall Ratio Statistics for Cost ASR ('costval' / TASP)

Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
1.00	1.01	1.00	1.01	0.69	1.30	7.36	9.50	0.99	1.01

95% CI Actual Coverage: 94.8%

The overall results indicate excellent performance. The median ASR is 1.00, perfectly centered within the IAAO target range of 0.90-1.10. The Coefficient of Dispersion (COD) is 7.36%, which is well below the IAAO standard of 15% (and even meets stricter standards often aiming for <10%), indicating very good uniformity across the entire dataset.

2.7.2 Neighbourhood Stratification Analysis

To check for potential bias related to location, the ‘costasr’ was analyzed by neighbourhood. A Kruskal-Wallis test was performed to assess if the median ASR differs significantly across neighbourhoods. The results (Table 26) show a p-value of 0.491, which is well above the 0.05 significance level. This indicates no statistically significant difference in the median ‘costasr’ across the different neighbourhoods.

Table 26: Kruskal-Wallis Test: Cost ASR by Neighbourhood

Statistic	Value
Kruskal-Wallis chi-squared	6.448
df	7
p-value	0.491

Ratio statistics stratified by neighbourhood are presented in Table 27.

Table 27: Ratio Statistics for Cost ASR by Neighbourhood

Neigh.	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
713	0.98	0.97	0.95	0.99	0.74	1.30	7.41	9.94	0.96	1.00
715	0.99	1.00	0.98	1.02	0.78	1.22	6.89	8.77	0.97	1.02
716	1.01	1.01	1.00	1.03	0.83	1.22	6.36	7.99	1.00	1.03
731	1.09	1.09	1.05	1.13	0.85	1.27	7.29	9.53	1.06	1.15
732	1.01	1.05	0.96	1.14	0.92	1.25	6.65	10.18	0.92	1.18
734	0.82	0.84	0.78	0.90	0.69	1.03	7.69	11.59	0.80	0.85
736	1.01	1.02	0.99	1.05	0.89	1.14	5.70	6.80	0.98	1.07
737	1.02	1.01	0.99	1.03	0.78	1.20	6.95	8.76	0.98	1.05

Although the Kruskal-Wallis test indicated no overall significant difference, examining the median ratios reveals potential issues. Neighbourhood 731 (Median 1.09) and Neighbourhood 734 (Median 0.82) have median ASRs outside the acceptable [0.90, 1.10] range established by IAAO standards. The confidence intervals for the median in these neighbourhoods ([1.06, 1.15] for 731 and [0.80, 0.85] for 734) also do not contain 1.0. This suggests that despite the overall non-significance in the rank test, there are practical equity concerns regarding these two specific neighbourhoods, which appear to be relatively over-assessed (731) and under-assessed (734) by the current cost model.

A Kruskal-Wallis test was performed to assess if the median ASR differs significantly across neighbourhoods after incorporating the neighbourhood adjustments derived during the land value analysis. The results (Table 28) show a p-value of 0.491, which is well above the 0.05 significance level. This indicates no remaining statistically significant difference in the median ‘costasr’ across the different neighbourhoods.

Table 28: Kruskal-Wallis Test: Cost ASR by Neighbourhood (After Land Value Adjustments)

Statistic	Value
Kruskal-Wallis chi-squared	6.448
df	7
p-value	0.491

Ratio statistics stratified by neighbourhood, reflecting the model after land value adjustments were incorporated, are presented in Table 29.

The ratio statistics confirm the effectiveness of the adjustments applied during the land valuation stage. All neighbourhood median ASRs now fall comfortably within the [0.90, 1.10] IAAO range.

Table 29: Ratio Statistics for Cost ASR by Neighbourhood (After Land Value Adjustments)

Neigh.	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
713	0.98	0.97	0.95	0.99	0.74	1.30	7.41	9.94	0.96	1.00
715	0.99	1.00	0.98	1.02	0.78	1.22	6.89	8.77	0.97	1.02
716	1.01	1.01	1.00	1.03	0.83	1.22	6.36	7.99	1.00	1.03
731	1.00	1.00	0.96	1.04	0.78	1.17	7.29	9.53	0.97	1.05
732	1.01	1.05	0.96	1.14	0.92	1.25	6.65	10.18	0.92	1.18
734	1.00	1.02	0.95	1.10	0.84	1.26	7.69	11.59	0.97	1.04
736	1.01	1.02	0.99	1.05	0.89	1.14	5.70	6.80	0.98	1.07
737	1.02	1.01	0.99	1.03	0.78	1.20	6.95	8.76	0.98	1.05

The previously problematic neighbourhoods 731 (Median 1.00) and 734 (Median 1.00) are now well-aligned with the overall level of assessment. Combined with the non-significant Kruskal-Wallis test result ($p=0.491$), this indicates that the cost model exhibits good equity across neighbourhoods, and no further location-based adjustments to the final Cost ASR are required.

To further assess model balance and equity, the neighborhood adjusted Cost ASR ('ncostasr') was examined against other key property characteristics. The overall ratio statistics after adjustments (Table 30) confirm the model's strong performance continues.

Table 30: Overall Ratio Statistics for ncostasr

Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
1.00	1.00	1.00	1.01	0.74	1.30	6.89	8.89	0.99	1.01

95% CI Actual Coverage: 94.8%

The performance remains excellent, with a median of 1.00 and a COD of 6.89%, well within IAAO standards.

Scatter plots of the final Cost ASR versus basement area ('bsmtarea') and lot size ('lotsize') were generated (Figures 8 and 7). Visual inspection revealed no apparent trend in the ratios relative to these size measures. Corresponding R-squared values from simple linear regressions were near zero, confirming a very low correlation and indicating no significant size-related bias in the final model values.

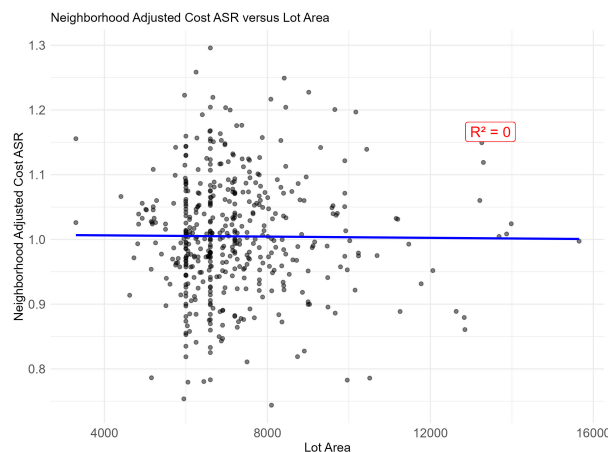


Figure 7: Final Cost ASR vs. Lot Size

Additionally, a box plot of the final Cost ASR stratified by the number of fireplaces ('fireplcs') was created (Figure 9). This plot showed similar median levels and dispersion across groups (0, 1, or 2 fireplaces), suggesting the model adequately accounts for this feature without introducing bias.

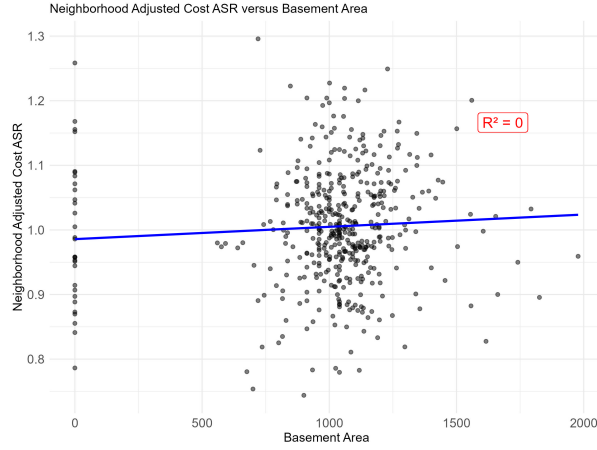


Figure 8: Final Cost ASR vs. Basement Area

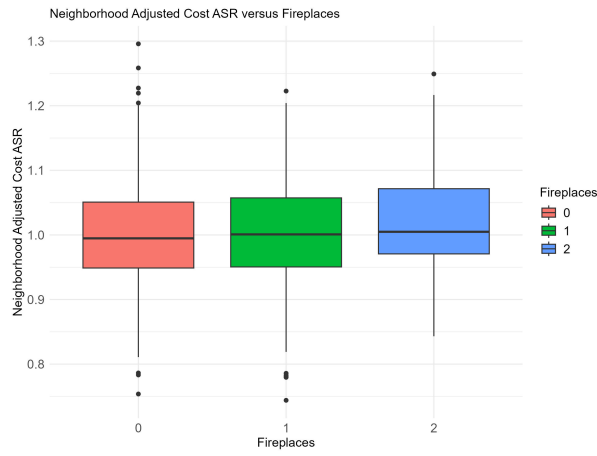


Figure 9: Final Cost ASR vs Fireplaces

These final checks indicate that the cost model is well-balanced and does not exhibit significant bias related to the tested property characteristics.

2.8 Final Cost Model Summary and Equation

This section summarizes the final calibrated market-adjusted cost model derived from the preceding analyses. The model estimates Market Value (MV) as the sum of the final adjusted Base Land Value (Land Value) and the final adjusted Replacement Cost New Less Depreciation (imprcnld_final).

The Land Value incorporates adjustments for size, conduit, and neighbourhood (Eq. 5) and ‘ imprcnld ’ incorporates building characteristics factors (Eq. 13)

Combining these, the final operational equation for the Cost Model Value is presented below:

Final Market-Adjusted Cost Model Equation

$$MV_{final} = (\text{Land Value} + \text{imprcnld}_{final}) \times (0.917)^{I_{neigh=731}} \times (1.220)^{I_{neigh=734}} \quad (17)$$

Where:

- Land Value is the fully adjusted land value (Eq. 5).
- imprcnld_{final} is the fully adjusted depreciated improvement value (Eq. 13).

- $I_{condition}$ is an indicator variable (1 if true, 0 otherwise). The factors $0.917 \approx 1/1.09$ and $1.220 \approx 1/0.82$ are applied only to properties in neighbourhoods 731 and 734 respectively. For all other neighbourhoods, this final adjustment factor is effectively 1.

This equation provides the final estimate of market value based on the fully calibrated and adjusted cost approach.

2.9 Cost Model Conclusions

The development of the market-adjusted cost model involved several key steps, using sales data from March 2017 to October 2017: derivation of land value using a residual technique with adjustments for size (power curve), conduit, and neighbourhood; calculation of RCN based on standardized cost schedules; estimation of market depreciation via a quadratic model applied to capped effective age (40 years); and calibration of improvement value adjustments, including an LCM of 0.99 and multiplicative factors for garage type, heat type, grouped building class, family room count, and bedroom count. Final assessment equity adjustments based on neighbourhood ASR analysis were applied to the overall model value for Neighbourhoods 731 and 734.

The resulting cost model (Eq. 17) demonstrates excellent overall performance, as summarized by the final Cost ASR statistics in Table 31.

Table 31: Overall Ratio Statistics for Final Cost ASR (After All Adjustments)

Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
1.00	1.00	1.00	1.01	0.74	1.30	6.89	8.89	0.99	1.01

The final median Cost ASR is 1.00, indicating accurate valuation at the market level and falling precisely in the middle of the IAAO target range of 0.90-1.10. Uniformity is exceptionally strong, with an overall COD of 6.89%, significantly better than the IAAO standard of 15% for single-family residential properties. Final testing confirmed that assessment ratios are equitable across neighbourhoods and various property characteristics, with no remaining statistically significant biases detected by Kruskal-Wallis tests at the $\alpha = 0.05$ level. This fully calibrated cost model provides a statistically sound, accurate, and equitable basis for estimating market value for the subject properties as of July 1, 2017.

3 Valuation using Sales Comparison Model

3.1 Introduction

This section details the development of the second required valuation model, based on the Direct Sales Comparison Approach using Multiple Regression Analysis (MRA). The objective is to create a statistically sound model that predicts the market value of the subject residential properties in Marilu City as of July 1, 2017. The analysis utilizes the provided database containing 466 sales records with 44 variables, spanning a period relevant to the valuation date (sales occurred between March 2017 and October 2017). By analyzing the relationship between sale prices and various property characteristics observed in this market data, the MRA model aims to directly capture market behaviour. This approach is often considered the primary method for valuing residential properties when sufficient sales data is available. The following subsections outline the data preparation, model specification, variable analysis, calibration, and testing procedures employed for this MRA model.

3.2 Preliminary Data Screening

The initial data review and preparation steps are largely consistent with those performed for the cost model. A general overview of the database characteristics, including record counts, date ranges, and initial variable checks, can be found in the Preliminary Data Screening subsection of the Cost Model section (refer to page Section 2.2 on page 17).

For the Sales Comparison model, the dependent variable will be the Time-Adjusted Sale Price (TASP). The methodology and results for the time adjustment calculation were detailed previously in Section 2.4.1 on page 19. No obvious data entry errors warranting immediate removal were found, and so consistent with the analysis for the cost model's land value component, no records are filtered out based on preliminary screening.

3.3 General Model Specification

The Direct Sales Comparison model will be developed using an additive Multiple Regression Analysis (MRA) framework. This approach estimates market value by summing the contributions of various property characteristics, each weighted by a coefficient derived from the regression calibration. The general form of the additive MRA model can be conceptualized, following standard mass appraisal structures, as:

$$MV = \Sigma GBQ + (\Sigma BQB + \Sigma (BQM \times BA)) + (\Sigma LQB + \Sigma (LQM \times LA)) + \Sigma (OQM \times OA)$$

Where:

- MV = Estimated Market Value (predicted Time Adjusted Sale Price).
- ΣGBQ = Sum of General Base Qualitative components (e.g., Intercept term, potentially time adjustment if not applied directly to TASP). Note: As per instructions, **Neighbourhood effects will NOT be included here** and will be adjusted later, if needed. Therefore, no General Base Qualitative component is considered ($\Sigma GBQ = 0$)
- ΣBQB = Sum of Building Qualitative Base components (e.g., base value contribution associated with specific building types or styles, often captured by intercepts or base levels of categorical variables).
- $\Sigma (BQM \times BA)$ = Sum of Building Quantitative components, where BA represents additive building characteristics (like square footage of living area, basement area) and BQM represents their corresponding regression coefficients (marginal values per unit). Qualitative aspects like quality or condition might be incorporated as multiplicative adjustments (BQM) to these additive terms or handled via separate categorical variables.

- ΣLQB = Sum of Land Qualitative Base components (e.g., base value associated with a typical lot in a reference neighbourhood).
- $\Sigma(LQM \times LA)$ = Sum of Land Quantitative components, where LA represents additive land characteristics (like lot size) and LQM represents their coefficients. Qualitative land factors (e.g., neighbourhood, corner influence, conduit) are typically incorporated as additive adjustments (LQM) via dummy variables or interaction terms.
- $\Sigma(OQM \times OA)$ = Sum of Other Quantitative components, representing additive values (OA) for features like garages, fireplaces, extra baths, etc., multiplied by their respective regression coefficients (OQM).

In practice, the MRA calibration directly estimates the coefficients (β_i) for each independent variable (X_i) included in the model, resulting in the familiar linear equation: $MV = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$. The selection and transformation of database variables (X_i) to represent the land, building, and other components within this additive structure is detailed in the following sections.

3.4 Variable Review

The variables available in the Marilu database must be reviewed and prepared specifically for use within the additive MRA framework. Unlike the cost model where variables directly feed into pre-defined cost components, in MRA, variables are selected based on their expected theoretical relationship with sale price and their statistical significance in explaining price variation. The goal is to identify and appropriately code variables representing land, improvement, and location characteristics that significantly influence market value.

- **Dependent Variable:** Time-Adjusted Sale Price (TASP).
- **Land Characteristics:** Variables like 'lotsize', 'neigh', 'conduit', 'corner', 'alley', 'wooded', 'pieshpe', 'slope', 'culdesac', 'abv_road' will be considered. 'lotsize' might require transformation (e.g., logarithm, polynomial) to capture non-linear effects. Categorical variables like 'neigh' and binary flags (e.g., 'conduit', 'corner') will likely be converted into dummy variables.
- **Improvement Characteristics:**
 - *Size/Area:* 'flrarea1', 'flrarea2', 'bsmtarea', 'bsmtfin', 'gar_area', 'cp_area'. These are primary drivers of value and might enter the model directly or require transformations.
 - *Quality/Condition:* 'mbmancls', 'gr_mancl', 'cp_mancl' (representing quality classes) and potentially condition ratings if available. These will likely be represented by dummy variables.
 - *Age:* using *PCTGOOD* derived for cost model in Section 2.6.2 to capture depreciation effects within the MRA framework.
 - *Features:* 'bedrooms', 'baths' (potentially split into full/partial components), 'fireplcs', 'familyrm', 'heattype', 'stories', 'baseste', 'gar_type', 'splitlvl'.

The process involves analyzing the relationship of each potential independent variable with the dependent variable (TASP) using graphical analysis, correlation, and preliminary statistical tests to guide the selection and transformation process for the final MRA model specification. This differs significantly from the cost model, where variables were mapped to pre-defined cost components rather than being selected based on their direct statistical relationship with sale price.

3.5 Variable Examination

3.5.1 Descriptive Analysis

This section provides a more detailed examination of the key variables identified for potential inclusion in the MRA model, based on the descriptive statistics presented in Appendix Tables A1 and A2.

- **Dependent Variable (TASP):** The Time-Adjusted Sale Price ranges from approximately \$68,100 to \$305,500, with a median value of \$144,440 and a mean of \$148,954. The distribution appears reasonably symmetric, although the maximum value is significantly higher than the 3rd quartile (\$167,858), suggesting some positive skewness or potential high-value outliers.
- **Key Predictors (Continuous/Numeric):**
 - *Floor Area ('flrarea1')*: Represents the main floor area, ranging from 552 sq ft to 1,978 sq ft, with a median of 1,058 sq ft. This is expected to be a primary driver of value.
 - *Lot Size ('lotsize')*: Varies considerably, from 3,300 sq ft to 15,650 sq ft. The median lot size is 6,839 sq ft. Given the wide range and potential non-linear relationship with price, transformations (e.g., logarithm) may be necessary.
 - *Effective Age ('effage')*: Ranges from 2 years to 74 years, with a median of 23 years. This variable will be crucial for capturing depreciation effects. Capping might be considered.
 - *Bedrooms ('bedrooms')*: Most properties have 3 bedrooms (71.9%), with a smaller group having 2 (23.8%) and very few having 4 (4.3%). This will likely be treated as a categorical variable or dummy variables.
 - *Bathrooms ('baths')*: The count ranges from 1 to 2.75, indicating the presence of partial baths. The most common counts are 1.0 (34.1%) and 1.5 (24.2%). This variable may need careful handling, potentially splitting into full and partial bath counts or using dummy variables for common configurations.
 - *Fireplaces ('fireplcs')*: Properties typically have 0 (30.7%), 1 (46.1%), or 2 (23.2%) fireplaces. This can be used as a count or dummy variables.
 - *Garage Area ('gar_area')*: The median is 0, indicating most properties in the dataset lack a garage area value (or have no garage). For those with garages, the area goes up to 756 sq ft. This variable is linked to 'gar_type' and 'garagefl' and requires careful consideration, likely using dummy variables for presence/type rather than area directly.
 - *Basement Area ('bsmtarea')*: Ranges from 0 to 1,978 sq ft, with a median of 1,040 sq ft. This suggests most properties have basements, and the area varies significantly.
- **Key Predictors (Categorical/Factor):** Variables like 'neigh', 'mbmancls', 'heattype', 'gar_type', 'conduit', 'alley', 'sidewalk', 'corner', and 'pie_shpe' represent important qualitative differences. Their frequency distributions (summarized in Appendix Table A3 on page 67, Table A29 on page 74, Table A4 on page 68, Table A5 on page 68, Table A6 on page 68, Table A7 on page 69, and Table A8 on page 69) show varying levels of prevalence. For instance, Neighbourhood 716 is the most common (33.5%), while 732 is the least (1.7%). Gas forced air heating ('heattype'=2) is dominant (86.5%). Most properties lack a garage ('gar_type'=0, 71.5%), but underground wiring ('conduit'=1, 64.4%) is more common than overhead. Features like sidewalks, cul-de-sacs, wooded lots, and pie-shaped lots are relatively infrequent (<10% each). For the MRA model, these categorical variables will typically be converted into sets of dummy (indicator) variables, with one category chosen as the reference level.

3.5.2 Graphical Analysis

To visually assess the relationship between the dependent variable (TASP) and key continuous predictors, scatter plots were generated. These plots help identify the direction and strength of the relationships, guiding potential variable transformations or inclusion decisions for the MRA model.

Figure 10 presents scatter plots for TASP against Floor Area ('flrarea1'), Effective Age ('effage'), Basement Area ('bsmtarea'), Lot Size ('lotsize'), and Garage Area ('gar_area'). Each plot includes a linear trend line and the corresponding R-squared value from a simple linear regression.

Observations from the plots:

- **Floor Area ('flrarea1'):** Shows a strong positive linear relationship with TASP (Figure 10a), as expected. The R-squared value of approximately 0.53 indicates that floor area alone explains a significant portion of the price variation.
- **Effective Age ('effage'):** Exhibits a clear negative relationship with TASP (Figure 10b), reflecting depreciation. The relationship appears reasonably linear, with an R-squared of 0.44.
- **Basement Area ('bsmtarea'):** Displays a positive relationship with TASP (Figure 10c), although with considerable scatter (R-squared $\approx$ 0.39). The cluster of points at 'bsmtarea' = 0 represents properties without basements.
- **Lot Size ('lotsize'):** Shows a positive but weaker relationship with TASP compared to floor area or age (Figure 10f), with an R-squared of only 0.13. There is significant variation in price for similar lot sizes, and the relationship might be non-linear, potentially warranting transformation.
- **Garage Area ('gar_area'):** The relationship appears very weak (Figure 10e), with an R-squared near zero (0.03). The large cluster at 'gar_area' = 0 dominates. This suggests that simply using garage area might not be effective; using garage type or presence as a categorical variable might be more appropriate.

These visualizations suggest that 'flrarea1' and 'effage' are strong linear predictors. 'bsmtarea' and 'lotsize' show expected positive relationships but with more variance or potential non-linearity. 'gar_area' appears to have little direct linear relationship with TASP. These insights will inform the variable selection and transformation process for the MRA model.

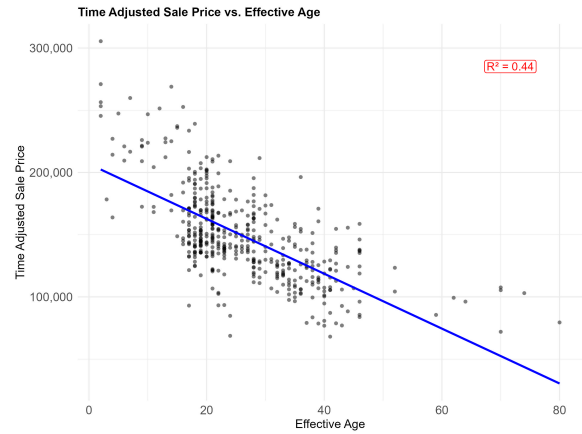
To understand how TASP varies across different categories of key discrete or categorical variables, box plots were generated (Figures 11 and 12).

Observations from the box plots:

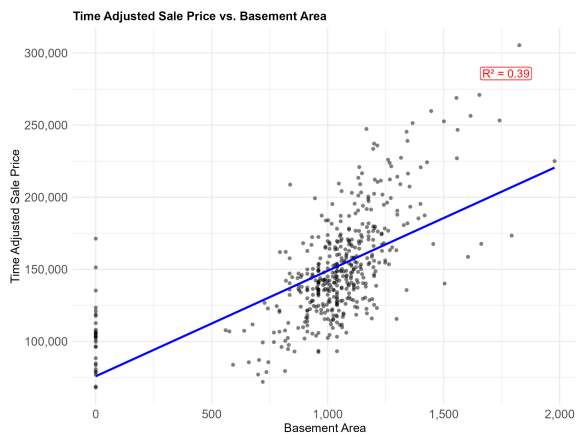
- **Heating Type ('heattype'):** While Type 2 (Gas Forced Air) dominates, Figure 11b indicates potential price differences. Notably, Type 1 (Gas Wall Furnace, N=3) appears associated with much lower prices, while Type 8 (Electric Baseboard?) might be linked to higher prices, although sample sizes for types other than 2 are small. This variable likely requires dummy coding.
- **Garage Type ('gar_type'):** Figure 11c suggests differences in median price based on garage type. Type 1 (Attached) and Type 3 (Basement) appear associated with higher median prices than Type 0 (None) or Type 2 (Detached). This supports using dummy variables for garage types.
- **Conduit ('conduit'):** Properties with underground wiring ('conduit'=1) have a distinctly higher median TASP than those without (Figure 11d), indicating this is likely a valuable predictor.
- **Fireplaces ('fireplcs'):** Figure 12a shows a positive relationship between the number of fireplaces and median TASP, as expected. The increase appears somewhat non-linear (larger jump from 0 to 1 than from 1 to 2). This could be included as a count or dummy variables.
- **Bedrooms ('bedrooms'):** Median TASP increases with the number of bedrooms (Figure 12b), although the difference between 3 and 4 bedrooms might be less pronounced than between 2 and 3, possibly due to the small sample size for 4-bedroom homes.



(a) TASP vs. Floor Area 1 ('flrarea1') ($R^2 \approx 0.53$)



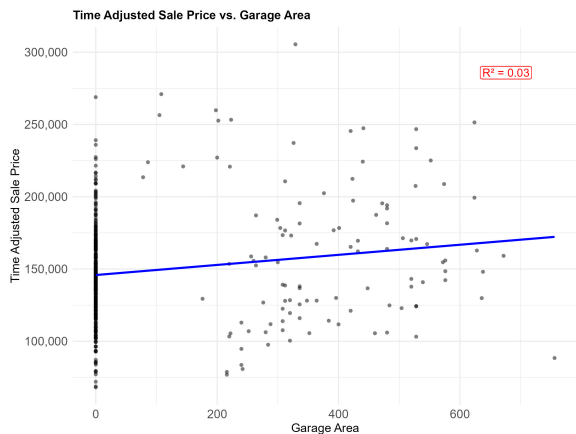
(b) TASP vs. Effective Age ('effage') ($R^2 \approx 0.44$)



(c) TASP vs. Basement Area ('bsmtarea') ($R^2 \approx 0.39$)



(d) TASP vs. Lot Size ('lotsize') ($R^2 \approx 0.13$)



(e) TASP vs. Garage Area ('gararea') ($R^2 \approx 0.03$)

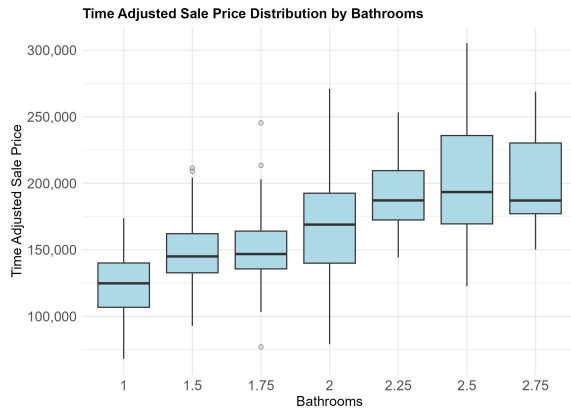


(f) TASP vs. Floor Area 2 ('flrarea2') ($R^2 \approx 0.05$)

Figure 10: Scatter plots of Time-Adjusted Sale Price (TASP) against selected predictors.

- **Bathrooms ('baths'):** Figure 11a shows a generally positive trend between the number of bathrooms (including partials) and median TASP. The steps are not perfectly uniform, suggesting dummy variables for common configurations (e.g., 1.0, 1.5, 2.0, 2.5+) might be more effective than treating it as a purely continuous variable.

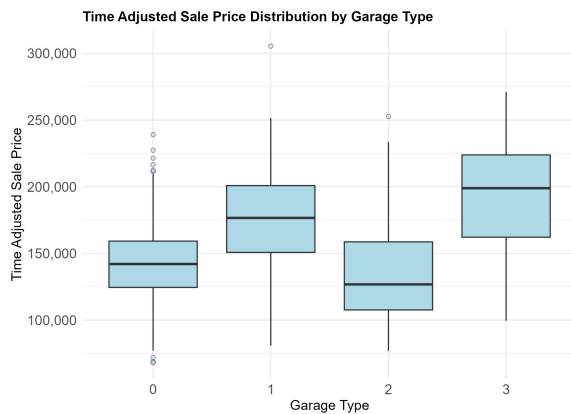
These graphical analyses reinforce the importance of key size, age, location, and feature variables. They also highlight potential non-linearities (e.g., 'lotsize') and suggest appropriate ways to code categorical



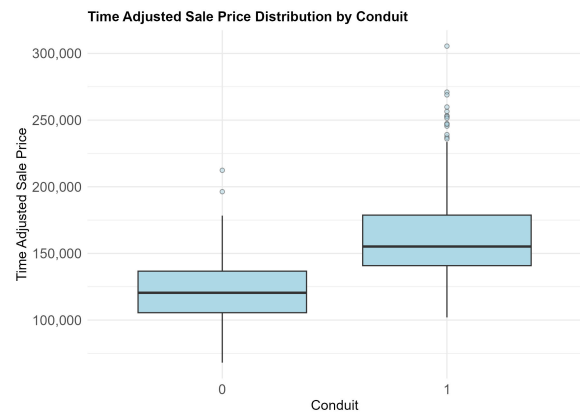
(a) TASP vs. Bathrooms ('baths')



(b) TASP vs. Heating Type ('heattype')



(c) TASP vs. Garage Type ('gar_type')



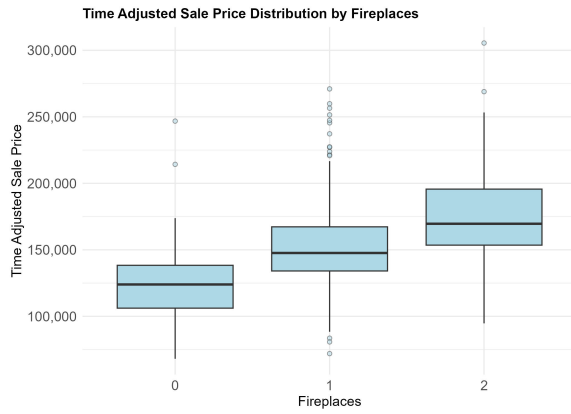
(d) TASP vs. Conduit ('conduit')

Figure 11: Box Plots of TASP vs. Key Categorical Predictors (Part 1)

or discrete variables (e.g., dummy variables for 'neigh', 'gar_type', 'baths') for the MRA model.

3.5.3 Correlation Matrix

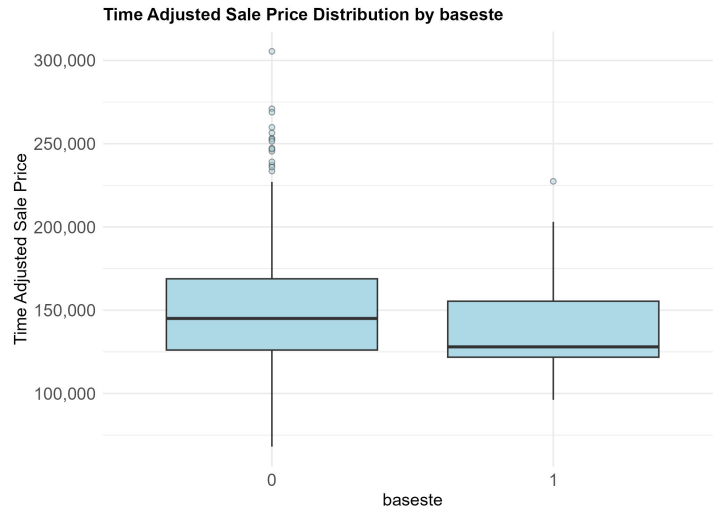
To assess potential multicollinearity issues among the continuous predictor variables considered for the MRA model, a correlation matrix was generated (Tables 32 and 33). Multicollinearity occurs when predictor variables are highly correlated with each other, which can inflate the variance of regression coefficient estimates and make it difficult to interpret their individual effects. Generally, absolute correlation coefficients above 0.7 or 0.8 warrant closer inspection.



(a) TASP vs. Fireplaces ('fireplcs')



(b) TASP vs. Bedrooms ('bedrooms')



(c) TASP vs. Basement Presence ('baseste')

Figure 12: Box Plots of TASP vs. Key Discrete Predictors (Part 2)

Table 32: Correlation Matrix for Selected Variables (Part 1)

Variable	<i>flrarea1</i>	<i>flrarea2</i>	<i>lot_size</i>	<i>bedrooms</i>	<i>baths</i>	<i>fireplcs</i>	<i>gar_area</i>
<i>flrarea1</i>	1.000	0.021	0.331	0.401	0.546	0.484	0.128
<i>flrarea2</i>	0.021	1.000	0.027	0.365	0.165	0.030	0.149
<i>lot_size</i>	0.331	0.027	1.000	0.098	0.229	0.214	0.103
<i>bedrooms</i>	0.401	0.365	0.098	1.000	0.228	0.180	0.093
<i>baths</i>	0.546	0.165	0.229	0.228	1.000	0.484	0.124
<i>fireplcs</i>	0.484	0.030	0.214	0.180	0.484	1.000	0.051
<i>gar_area</i>	0.128	0.149	0.103	0.093	0.124	0.051	1.000
<i>bsmtarea</i>	0.715	-0.049	0.262	0.265	0.469	0.452	0.018
<i>crwlaarea</i>	-0.194	0.081	-0.101	-0.043	-0.226	-0.250	0.091
<i>conduit</i>	0.338	-0.059	0.160	0.169	0.360	0.316	-0.074
<i>alley</i>	-0.326	0.059	-0.189	-0.169	-0.298	-0.340	0.109
<i>sidewalk</i>	-0.035	0.041	-0.027	-0.024	-0.108	-0.028	0.081
<i>corner</i>	0.058	-0.033	0.021	-0.060	-0.002	0.052	0.198
<i>pie_shpe</i>	0.147	0.035	0.304	0.098	0.133	0.076	0.011

Table 33: Correlation Matrix for Selected Variables (Part 2)

Variable	<i>bsmtarea</i>	<i>crwlnarea</i>	<i>conduit</i>	<i>alley</i>	<i>sidewalk</i>	<i>corner</i>	<i>pie_shpe</i>
flrarea1	0.715	-0.194	0.338	-0.326	-0.035	0.058	0.147
flrarea2	-0.049	0.081	-0.059	0.059	0.041	-0.033	0.035
lot_size	0.262	-0.101	0.160	-0.189	-0.027	0.021	0.304
bedrooms	0.265	-0.043	0.169	-0.169	-0.024	-0.060	0.098
baths	0.469	-0.226	0.360	-0.298	-0.108	-0.002	0.133
fireplcs	0.452	-0.250	0.316	-0.340	-0.028	0.052	0.076
gar_area	0.018	0.091	-0.074	0.109	0.081	0.198	0.011
bsmtarea	1.000	-0.820	0.391	-0.382	-0.050	0.089	0.123
crwlnarea	-0.820	1.000	-0.301	0.295	0.054	-0.085	-0.064
conduit	0.391	-0.301	1.000	-0.811	-0.289	-0.017	0.165
alley	-0.382	0.295	-0.811	1.000	0.228	-0.049	-0.149
sidewalk	-0.050	0.054	-0.289	0.228	1.000	0.007	-0.047
corner	0.089	-0.085	-0.017	-0.049	0.007	1.000	-0.030
pie_shpe	0.123	-0.064	0.165	-0.149	-0.047	-0.030	1.000

Note: Absolute correlation values > 0.7 are highlighted in red and bold.

The correlation matrix reveals several high correlations (absolute value > 0.7):

- ‘flrarea1’ and ‘bsmtarea’ (0.715): Larger homes tend to have larger basements.
- ‘bsmtarea’ and ‘crwlnarea’ (-0.820): Mutually exclusive basement types (a property has one or the other, or neither).
- ‘conduit’ and ‘alley’ (-0.811): Properties with underground wiring are much less likely to have alleys.

These high correlations, particularly between ‘flrarea1’ and ‘bsmtarea’, and between ‘conduit’ and ‘alley’, suggest potential multicollinearity if these pairs are included together in the MRA model without transformation or careful consideration. The strong negative correlation between ‘bsmtarea’ and ‘crwlnarea’ is expected due to their definition. Other correlations are generally moderate or low.

Based on the variable examination, several variable transformations are recommended to prepare the data for the MRA model calibration. Binary variables (‘conduit’, ‘alley’, ‘sidewalk’, ‘corner’, ‘pie_shpe’, ‘wooded’, ‘slope’, ‘abv_road’, ‘splitlvl’, ‘baseste’, ‘garagefl’, ‘carptfl’) will be included directly as dummy variables (0/1 coding). Beside these transformations, more transformations would be needed which will be discussed in the later section.

3.6 Variable Transformations and Examination

3.6.1 Variable Transformations

Based on the preceding variable examination and the requirements for MRA, several variables were transformed or created to better capture market effects and prepare predictors for the regression model. The following transformations were implemented:

- **Adjusted Land Size (‘adjusted_land_size’)**: To incorporate the comprehensive land value adjustments derived in the cost model analysis (including size, conduit, and neighbourhood factors), a variable representing the adjusted land value per unit of size was created. This was calculated by dividing the final adjusted land value from the cost model by the median land residual value per square foot (7.2478). This variable aims to represent the adjusted land contribution in a single term.

$$\text{adjusted_land_size} = \frac{\text{land_value_final}}{7.2478}$$

- **Depreciated Linear Building Areas ('dlin...')**: Several variables were created to represent the depreciated contribution of various area components, simultaneously accounting for quality (via `mb_factor` or `cp_factor`) and market depreciation (via `pctgood` derived from the quadratic model on capped effective age). `mb_factor` and `cp_factor` are derived from Appendix tables A41 on page 76 and A42 on page 76. This transformation combines size, quality, and age effects into single terms:
 - First Floor Area: $dlin_flrarea1 = mb_factor \times flrarea1 \times pctgood$
 - Second Floor Area: $dlin_flrarea2 = mb_factor \times flrarea2 \times pctgood$
 - Finished Basement Area: $dlin_bsmtfin = mb_factor \times bsmtfin \times pctgood$
 - Carport Area: $dlin_cp_area = cp_factor \times cp_area \times pctgood$
- **Depreciated Linear Garage/Carport Areas ('dlin_gar_area...')**: Similar variables were created for different garage types, applying the appropriate quality factor (`cp_factor` as per R code) and depreciation (`pctgood`). Indicator variables (`gar_type == x`) ensure the calculation applies only to the correct garage type:
 - Attached Garage: $dlin_gar_area_attached = cp_factor \times gar_area \times (gar_type == 1) \times pctgood$
 - Detached Garage: $dlin_gar_area_detached = cp_factor \times gar_area \times (gar_type == 2) \times pctgood$
 - Basement Garage: $dlin_gar_area_basement = cp_factor \times flrarea1 \times (gar_type == 3) \times pctgood$

(Note: As approximately 70% of properties with basement garages (`gar_type == 3`) have zero `gar_area` values, `flrarea1` was used as a proxy for the area component in this transformation. For R Syntax, see Appendix Section F.1 on page 84)
- **Heating Type Dummies ('gas_heattype', 'oil_heattype', 'electric_heattype')**: The categorical `heattype` variable was converted into three binary dummy variables based on the fuel source, allowing for separate coefficients for each fuel type relative to a reference category (which will be omitted in the regression):
 - `gas_heattype` = 1 if `heattype` $\in \{1, 2, 7\}$, 0 otherwise.
 - `oil_heattype` = 1 if `heattype` $\in \{3, 8\}$, 0 otherwise.
 - `electric_heattype` = 1 if `heattype` $\in \{4, 5, 6\}$, 0 otherwise.

Note that the transformations above remove the need to effective age to be included in the model as the values are after depreciation. Also, since all the properties have one type of heat type (no properties come with heat type of zero), I would only consider oil and electric heat type in the modeling and consider gas heat type as the baseline due to its more appearance in the data.

3.6.2 Variable Examination after Transformation

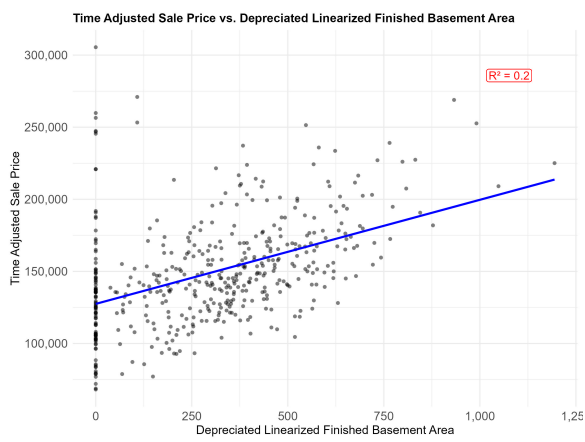
Following the variable transformations, scatter plots were generated to visualize the relationship between the dependent variable (TASP) and the key transformed "depreciated linear" area variables (Figures 13 and 14). These plots help assess if the transformations, which combine size, quality, and age effects, exhibit stronger linear relationships with TASP compared to the raw area variables.



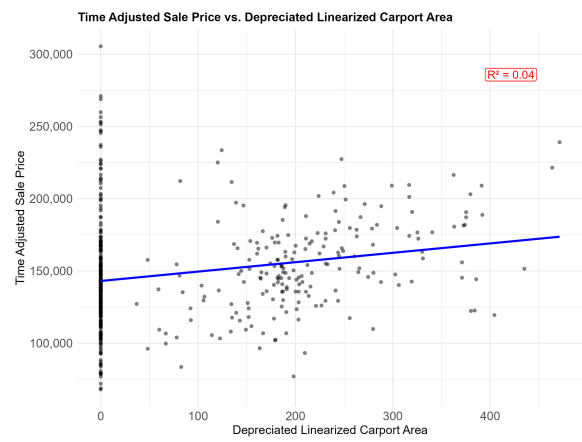
(a) TASP vs. Dep. Lin. Floor Area 1 ($R^2 \approx 0.69$)



(b) TASP vs. Dep. Lin. Floor Area 2 ($R^2 \approx 0.08$)

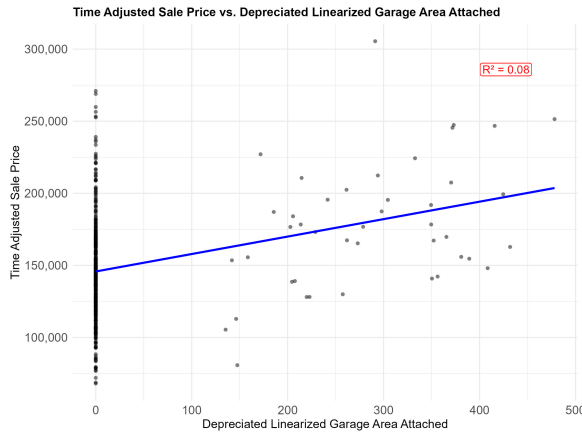


(c) TASP vs. Dep. Lin. Finished Bsmt Area ($R^2 \approx 0.20$)

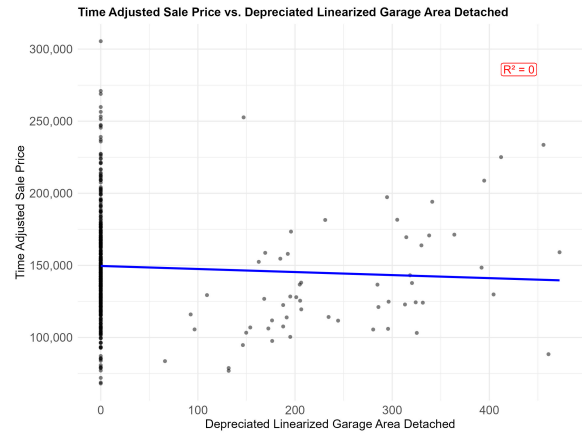


(d) TASP vs. Dep. Lin. Carport Area ($R^2 \approx 0.04$)

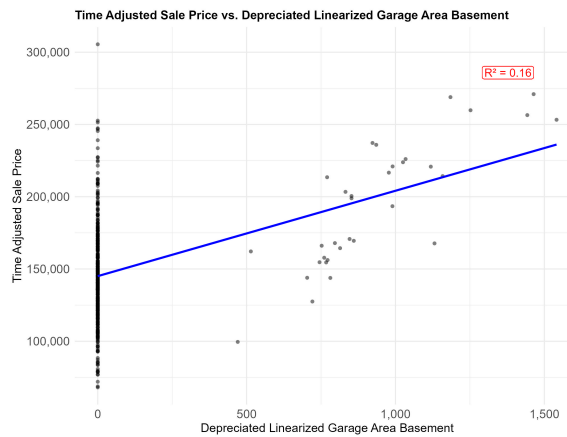
Figure 13: Scatter Plots of TASP vs. Transformed Depreciated Linear Area Variables (Part 1)



(a) TASP vs. Dep. Lin. Attached Garage Area ($R^2 \approx 0.08$)



(b) TASP vs. Dep. Lin. Detached Garage Area ($R^2 \approx 0.00$)



(c) TASP vs. Dep. Lin. Basement Garage Area ($R^2 \approx 0.16$)

Figure 14: Scatter Plots of TASP vs. Transformed Depreciated Linear Area Variables (Part 2)

Observations from the transformed variable plots:

- **Dep. Lin. Floor Area 1 ('dlin_flarea1'):** Figure 13a shows a very strong positive linear relationship ($R^2 \approx 0.69$), significantly improved compared to the raw 'flarea1'. This transformation successfully combines size, quality, and depreciation effects into a potent predictor.
- **Dep. Lin. Floor Area 2 ('dlin_flarea2'):** The relationship remains weak ($R^2 \approx 0.08$, Figure 13b), dominated by the large number of 1-storey homes where this value is zero.
- **Dep. Lin. Finished Basement Area ('dlin_bsmtfin'):** Shows a positive relationship ($R^2 \approx 0.20$, Figure 13c), somewhat stronger than raw 'bsmtarea' but still with considerable scatter.
- **Dep. Lin. Carport Area ('dlin_cp_area'):** Exhibits a weak positive relationship ($R^2 \approx 0.04$, Figure 13d).
- **Dep. Lin. Garage Areas:** The relationships for attached ($R^2 \approx 0.08$, Figure 14a), detached ($R^2 \approx 0.00$, Figure 14b), and basement (proxy) ($R^2 \approx 0.16$, Figure 14c) garages remain relatively weak when plotted individually against TASP. This reinforces the idea that using dummy variables for garage presence/type might be more effective than using these transformed area variables directly in the MRA.

The transformation creating ‘`dlin_flrarea1`’ appears particularly successful in capturing combined size, quality, and age effects. The other transformed area variables show weaker individual linear relationships with TASP.

Box plots were generated to visualize the relationship between TASP and the created dummy variables for heating fuel type (Figure 15).

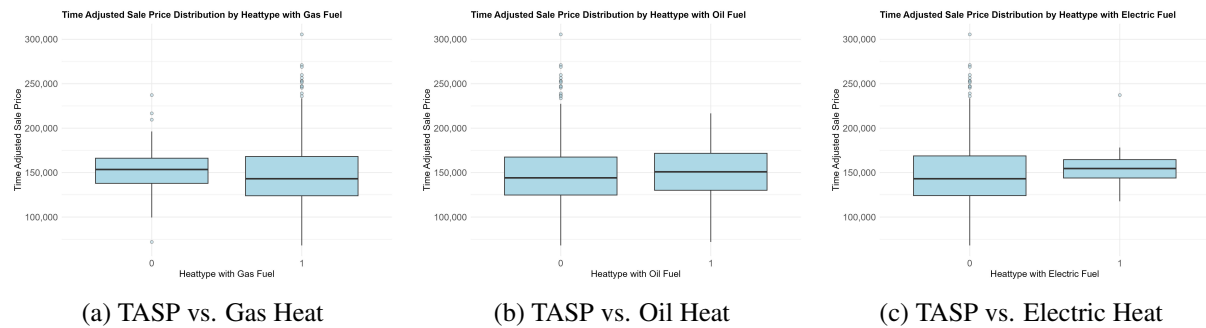


Figure 15: Box Plots of TASP vs. Transformed Heating Type Dummy Variables

Observations from the heat type dummy variable plots:

- **Gas Heat (‘`gas_heattype`’):** Figure 15a shows that properties with gas heat (Value=1, combining types 1, 2, and 7) have a similar median TASP to those without gas heat (Value=0), although the distribution for non-gas properties is wider due to the mix of other types.
- **Oil Heat (‘`oil_heattype`’):** Figure 15b suggests that properties with oil heat (Value=1, types 3 and 8) might have a slightly higher median TASP compared to properties without oil heat, but the sample size for oil heat is small.
- **Electric Heat (‘`electric_heattype`’):** Figure 15c indicates that properties with electric heat (Value=1, types 4, 5, and 6) may have a slightly higher median TASP than non-electric properties.

These plots suggest potential differences based on fuel type, validating the use of these dummy variables in the MRA model, where their statistical significance and contribution can be formally tested.

3.7 Model Calibration

The calibration phase involves building and evaluating several Multiple Regression Analysis (MRA) models to predict the dependent variable (‘`log(TASP)`’ or ‘`TASP`’) using different combinations of the transformed predictor variables identified previously. The goal is to select a final model that demonstrates good statistical fit, logical coefficient signs and magnitudes, predictive accuracy, and minimal multicollinearity issues.

Four distinct models will be specified and tested. These models share a core set of common predictor variables, deemed fundamentally important based on theory and initial analysis. Each model then includes a different set of specific variables, primarily varying how potentially correlated variables (like ‘`conduit`’/‘`alley`’ and area measures) are handled or which additional features are included.

The common variables included in all four models are listed in Table 34. Note that the gas heat type serves as the reference category in the model and hence is not included in the table (all properties have one of the types of heat type).

Table 34: Common Predictor Variables for MRA Model Calibration

Variable Name	Description
adjusted_land_size	Land value component adjusted for lot size, conduit, and neighborhood
dlin_flrarea1	Depreciated linear first floor area (includes quality)
dlin_flrarea2	Depreciated linear second floor area (includes quality)
dlin_bsmtfin	Depreciated linear finished basement area (includes quality)
dlin_gar_area_attached	Depreciated linear area of attached garage (includes quality)
dlin_gar_area_detached	Depreciated linear area of detached garage (includes quality)
dlin_gar_area_basement	Depreciated linear area of basement garage (proxy; includes quality)
dlin_cp_area	Depreciated linear carport area (includes quality)
oil_heattype	Dummy variable: 1 if oil heating is used
electric_heattype	Dummy variable: 1 if electric heating is used
baths	Number of bathrooms
bedrooms	Number of bedrooms
fireplcs	Number of fireplaces
baseste	Dummy variable: 1 if basement or crawl space exists
crwlarea	Area of crawl space

Building upon this common set, four models are specified with the following distinct variable additions, primarily focusing on alternative ways to handle site features or area components (Table 35).

Table 35: Specific Variable Sets for MRA Model Variations

Model	Specific Variables Added to Common Set
Model 1	'conduit' dummy variable.
Model 2	'alley' dummy variable.
Model 3	'bsmtarea' (raw basement area); 'conduit' dummy variable.
Model 4	'bsmtarea' (raw basement area); 'alley' dummy variable.

Note: All models implicitly include an intercept.

Each of these four models was calibrated using Ordinary Least Squares (OLS) regression. A summary of the overall model fit statistics is presented in Table 36.

Table 36: MRA Model Summary Statistics Comparison

Model	Model Summary				
	R Square	Adj. R Sq.	F-statistic	df1	df2
precision Model 1 (conduit)	0.856	0.851	166.903	16	449
Model 2 (alley)	0.856	0.851	167.246	16	449
Model 3 (bsmtarea, conduit)	0.856	0.851	157.094	17	448
Model 4 (bsmtarea, alley)	0.857	0.851	157.567	17	448

Dependent variable: 'saleprice_time_adjusted'.

Table 36 provides a high-level comparison of the models based on overall fit statistics. All four models exhibit very similar and high R-squared and Adjusted R-squared values (approx. 0.856 and 0.851 respectively), indicating that all specified sets of predictors explain roughly 85% of the variation in the time-adjusted sale price. The F-statistics for all models are highly significant ($p < 2.2e-16$), confirming the overall validity of each specification. Notably, the inclusion of raw 'bsmtarea' in Models 3 and 4 does not substantially improve the overall model fit compared to Models 1 and 2.

Selecting the best model requires further analysis beyond overall fit. The detailed coefficient estimates for each model, including their statistical significance (p-values) and logical signs, must be examined. Crucially, multicollinearity diagnostics, specifically Variance Inflation Factors (VIF) and Tolerance values for each predictor, need to be assessed to ensure the stability and reliability of the coefficient estimates. High VIF values (> 3.33) or low Tolerance values (< 0.33) would indicate problematic multicollinearity.

The detailed coefficient tables and collinearity statistics for each of the four models are presented in Appendix Tables A43 through A46. A review of these tables indicates that Models 1 and 2 generally exhibit acceptable collinearity statistics, with VIF values well below common thresholds. However, Models 3 and 4 show high VIF values (and correspondingly low Tolerance) for several variables, particularly 'dlin_flarea1' (VIF > 8), 'crwlarea' (VIF > 10), and 'bsmtarea' (VIF > 19). VIFs exceeding 3.33 suggest potential multicollinearity issues; the values observed for these variables in Models 3 and 4 strongly indicate that including raw 'bsmtarea' alongside the depreciated linear area variables introduces significant multicollinearity, making the coefficient estimates for these variables unreliable. In contrast, no such problematic VIF values are observed for Models 1 and 2. Considering the similar overall fit statistics (Adjusted R-squared, F-statistic) and the superior collinearity diagnostics, Model 2 appears to be the most promising specification among the four tested. These tables will be used to evaluate the individual predictor performance and collinearity within each model to inform the final model selection.

3.8 MODEL/TEST Data Split

A crucial step in developing a robust predictive model like MRA, especially for assessment purposes, is to evaluate its performance on data not used during calibration. This helps ensure the model generalizes well to new data and avoids overfitting to the specific characteristics of the calibration sample. As required by the examination instructions for the Sales Comparison model, the dataset is split into distinct MODEL and TEST subsets.

Following the specified proportions, the available sales data (466 records) will be randomly partitioned:

- **MODEL Set:** Approximately 2/3 of the records (approx. 310 records) will be assigned to the MODEL set. This subset will be used exclusively for calibrating the MRA models (i.e., estimating the regression coefficients β_i).
- **TEST Set:** Approximately 1/3 of the records (approx. 156 records) will be assigned to the TEST set. This subset will be held out during calibration and used for independent validation of the selected model's predictive accuracy and performance based on standard ratio statistics.

The random assignment ensures that both the MODEL and TEST sets are likely representative of the overall dataset. The subsequent MRA calibration will be performed using only the MODEL data, and the resulting preferred model will then be applied to the TEST data to assess its out-of-sample performance.

3.9 Stepwise Regression

To arrive at a statistically sound MRA model, a stepwise regression procedure was employed using the MODEL dataset (approximately 2/3 of the total sales data). Stepwise regression automatically adds or removes predictor variables based on predefined statistical criteria (typically p-values for entry and removal) to find a subset of variables that best explains the variation in the dependent variable ('saleprice_time_adjusted'). This analysis started with the common variables and the specific variable set identified as most promising from the initial calibration comparison (Model 2, using 'alley' instead of 'conduit' due to lower collinearity concerns).

The stepwise procedure resulted in a final model after 4 steps. Table 37 summarizes the overall fit statistics at each step.

Table 37: MRA Stepwise Regression Summary Statistics

Step	Model Summary				
	R Square	Adj. R Sq.	F-statistic	df1	df2
1	0.837	0.828	93.694	16.000	293.000
2	0.836	0.828	100.273	15.000	294.000
3	0.836	0.829	107.693	14.000	295.000
4	0.836	0.828	115.743	13.000	296.000

Dependent variable: 'saleprice_time_adjusted'. Based on MODEL dataset.

The variable selection process across the steps was as follows:

- **Step 1:** Started with the initial set of 16 predictors from Model 2.
- **Step 2:** Removed 'electric_heattype' (due to non-significance).
- **Step 3:** Removed 'oil_heattype' (due to non-significance after removing electric).
- **Step 4:** Removed 'baseste' (due to non-significance).

The final model (Step 4) retained 13 predictor variables plus the intercept. Notably, the Adjusted R-squared remained stable at approximately 0.828 throughout the removal steps, suggesting the removed variables contributed little unique explanatory power. The F-statistic increased at each step, indicating improved model efficiency (explaining variance with fewer variables).

The detailed coefficient estimates and collinearity statistics for the final selected model (Step 4) are presented in Table 38. The coefficients generally have logical signs (e.g., positive for size variables, negative for 'crlwarea' relative to basement). Most predictors are statistically significant at conventional levels ($p < 0.05$ or $p < 0.10$). Interestingly, all variables retained are statistically significant based on a 10% significance threshold ($p < 0.10$). Note that the variables with the lowest statistical significance, 'bedrooms' ($p=0.085$) and 'fireplcs' ($p=0.090$), are still considered important based on the earlier variable examination (see Figure 12). Collinearity diagnostics (VIF and Tolerance) are all within acceptable limits ($VIF < 3.33$, $Tolerance > 0.3$), indicating that multicollinearity is not a concern in this final specification. Based on these results, the model from Step 4 is selected as the preferred MRA specification. (Detailed results for Steps 1-3 are provided in Appendix Tables A47 through A49).

Table 38: Final MRA Model (Step 4) Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(> t)	VIF	Tolerance
(Intercept)	-10 790.0	8296.0	-1.301	0.194	—	—
adjusted_land_size	10.370	0.880	11.782	<0.001	1.633	0.612
dlin_flrarea1	74.560	5.950	12.530	<0.001	2.401	0.416
dlin_flrarea2	64.280	11.210	5.736	<0.001	1.402	0.713
dlin_bsmtfin	21.640	4.707	4.599	<0.001	1.642	0.609
dlin_gar_area_attached	41.090	11.320	3.630	<0.001	1.321	0.757
dlin_gar_area_detached	29.250	12.010	2.435	0.015	1.121	0.892
dlin_gar_area_basement	13.730	4.282	3.207	0.001	1.886	0.530
dlin_cp_area	23.640	8.057	2.934	0.004	1.420	0.704
baths	5645.000	2416.000	2.336	0.020	2.013	0.497
bedrooms	3290.000	1902.000	1.730	0.085	1.259	0.794
fireplcs	2346.000	1378.000	1.703	0.090	1.510	0.662
crlwarea	-13.610	4.226	-3.220	0.001	1.196	0.836
alley1	4726.000	2383.000	1.983	0.048	1.780	0.562

Note: Tolerance = 1/VIF. Dependent Variable: 'saleprice_time_adjusted'. Based on MODEL dataset.

Based on the stepwise regression analysis performed on the MODEL dataset, the final preferred MRA model (Step 4) includes the intercept and the 13 predictor variables listed in Table 38. This model demonstrated strong overall fit (Adjusted R-squared ≈ 0.828) while maintaining acceptable collinearity diagnostics (all VIFs < 3.33).

The final equation for the estimated market value (MV_{MRA}) based on this model is:

MRA Model Equation

$$\begin{aligned}
 MV = & -10790 \\
 & + (10.370 \times \text{adjusted_land_size}) \\
 & + (74.560 \times \text{dlin_flrarea1}) \\
 & + (64.280 \times \text{dlin_flrarea2}) \\
 & + (21.640 \times \text{dlin_bsmtfin}) \\
 & + (41.090 \times \text{dlin_gar_area_attached}) \\
 & + (29.250 \times \text{dlin_gar_area_detached}) \\
 & + (13.730 \times \text{dlin_gar_area_basement}) \\
 & + (23.640 \times \text{dlin_cp_area}) \\
 & + (5645.000 \times \text{baths}) \\
 & + (3290.000 \times \text{bedrooms}) \\
 & + (2346.000 \times \text{fireplcs}) \\
 & - (13.610 \times \text{crlwarea}) \\
 & + (4726.000 \times \text{alley1})
 \end{aligned} \tag{18}$$

Where the variable names correspond to the transformed predictors defined previously (e.g., ‘dlin_...’ variables incorporate quality and depreciation, ‘**alley1**’ is the dummy variable for alley presence). This equation represents the calibrated Sales Comparison model that will be used for prediction and testing on the held-out TEST dataset.

3.10 Casewise Diagnostics

As a final check on the model calibrated using the MODEL dataset, casewise diagnostics were performed to identify influential outliers. Standardized residuals were examined, and cases with absolute values greater than 3 were flagged.

This analysis identified one outlier (Row 249) with a standardized residual of -3.276. This indicates the model significantly over-predicted the sale price for this particular property compared to the general pattern observed in the MODEL dataset.

To assess the impact of this outlier, it was removed from the MODEL dataset, and the final Step 4 model specification was refit to the cleaned data. The detailed coefficients and collinearity statistics for this refitted model are presented in Table 39. The overall fit remained very similar (Adjusted R-squared = 0.829), and the coefficients retained similar signs and magnitudes, although some p-values shifted slightly (e.g., ‘fireplcs’ became non-significant at $p=0.151$). Importantly, collinearity remained low (all VIFs < 3.33).

A check of the standardized residuals for the refitted model confirmed that no further observations had absolute standardized residuals greater than 3. Therefore, the model refitted after removing the single outlier (Row 249) is considered the final calibrated MRA model based on the MODEL dataset. This refitted model will be used for subsequent testing on the TEST dataset.

Table 39: Final MRA Model Coefficients and Collinearity Statistics (Cleaned MODEL Data)

Variable	Estimate	Std. Error	t value	Pr(> t)	VIF	Tolerance
(Intercept)	−13 240.0	8191.0	−1.616	0.107	—	—
adjusted_land_size	10.470	0.866	12.094	<0.001	1.629	0.614
dlin_flrarea1	78.970	5.999	13.164	<0.001	2.316	0.432
dlin_flrarea2	77.100	11.670	6.605	<0.001	1.328	0.753
dlin_bsmtfin	21.030	4.632	4.540	<0.001	1.636	0.611
dlin_gar_area_attached	39.600	11.140	3.555	<0.001	1.322	0.756
dlin_gar_area_detached	30.600	11.820	2.590	0.010	1.122	0.892
dlin_gar_area_basement	14.430	4.216	3.422	<0.001	1.719	0.582
dlin_cp_area	23.460	7.923	2.961	0.003	1.417	0.706
baths	5363.000	2378.000	2.256	0.025	1.989	0.503
bedrooms	3094.000	1871.000	1.654	0.099	1.235	0.809
fireplcs	1958.000	1360.000	1.440	0.151	1.521	0.658
crwlarea	−13.610	4.156	−3.275	0.001	1.196	0.836
alley1	4878.000	2344.000	2.081	0.038	1.778	0.562

Note: Tolerance = 1/VIF. Dependent Variable: 'saleprice_time_adjusted'. R-squared: 0.836, Adj. R-squared: 0.829, F-stat: 115.9, p-value: < 2.2e-16.

3.11 Model Testing

Having calibrated the final MRA model using the MODEL dataset and removing influential outliers (resulting in the model specified in Eq. 18 based on the coefficients in Table 39), the next critical step is to evaluate its predictive performance.

3.11.1 Mean Prediction Accuracy Test

A fundamental check is to compare the mean of the predicted values generated by the MRA model for the MODEL set against the mean of the actual TASP values in the MODEL set. If the model is unbiased on average, these two means should be very close. The analysis confirms that the mean predicted sale price for the MODEL set is \$147,718.1, which is identical to the mean actual TASP for the MODEL set (\$147,718.1). This indicates that the model predicts the correct average value for the holdout sample, passing this initial basic accuracy test.

3.11.2 ASR Ratio for MODEL and TEST set

Having calibrated the final MRA model using the cleaned MODEL dataset (Equation 18), the next step is to evaluate its performance. This involves applying the calibrated model to predict values (MV_{MRA}) and then calculating the Assessment Sales Ratio (ASR) using the Time-Adjusted Sale Price (TASP):

$$ASR = \frac{MV}{TASP} \quad (19)$$

The performance is first evaluated on the MODEL dataset itself to confirm internal validity and check for biases introduced or remaining after calibration. Subsequently, the model will be applied to the held-out TEST dataset for independent validation.

The overall ratio statistics for the ASR calculated on the cleaned MODEL dataset (N=309 after removing one outlier) are presented in Table 40.

Table 40: Overall Ratio Statistics for MRA ASR (MODEL Dataset)

Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
1.00	1.01	1.00	1.02	0.77	1.33	7.34	9.58	1.00	1.02

95% CI Actual Coverage: 95.3%

The overall performance on the MODEL set is excellent. The median ASR is 1.00, indicating accurate prediction at the central tendency. The COD of 7.34% demonstrates very strong uniformity, well below the IAAO standard of 15%. To ensure equity, A Kruskal-Wallis test was performed on the ASR stratified by neighbourhood ('neigh'). The initial test result was statistically significant ($p = 0.0019$), indicating differences in median ASR across neighbourhoods before any final neighbourhood-specific adjustments to the MRA model value.

Table 41: Kruskal-Wallis Test: MRA ASR by Neighbourhood (MODEL Set - Before Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	22.704
df	7
p-value	0.002

Ratio statistics by neighbourhood (Table 42) revealed that Neighbourhood 734 had a median ASR of 1.16, which is outside the acceptable [0.90, 1.10] range, and its confidence interval [1.05, 1.19] did not contain 1.0. All other neighbourhoods had median ASRs within the acceptable range, although Neighbourhood 731 was borderline at 0.95.

Table 42: Ratio Statistics for MRA ASR by Neighbourhood (MODEL Set - Before Adjustment)

Neigh.	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
713	1.01	1.02	0.99	1.06	0.77	1.33	8.33	11.30	0.99	1.03
715	1.01	1.02	0.99	1.05	0.79	1.30	7.05	9.32	0.97	1.04
716	1.01	1.00	0.99	1.01	0.83	1.17	5.97	7.53	0.99	1.02
731	0.95	0.96	0.92	0.99	0.85	1.07	5.83	7.39	0.92	1.01
732	1.03	1.00	0.89	1.10	0.85	1.06	5.19	8.66	NA	NA
734	1.16	1.14	1.06	1.23	0.88	1.30	7.14	10.60	1.05	1.19
736	0.98	0.97	0.93	1.01	0.80	1.12	6.55	8.22	0.91	1.01
737	1.00	1.01	0.98	1.04	0.82	1.29	8.07	10.25	0.96	1.03

Based on these results, a final multiplicative adjustment factor was applied to the MRA model predictions (MV) specifically for Neighbourhood 734 (Factor = $1.0 / 1.16 \approx 0.862$). After applying this adjustment, the Kruskal-Wallis test for the adjusted ASR ('ASR_neigh') by neighbourhood became non-significant ($p = 0.209$), and the ratio statistics (Appendix Table A52) confirmed that all neighbourhood medians were within acceptable ranges.

To ensure the final model (including the neighbourhood adjustment) is balanced across other key characteristics, Kruskal-Wallis tests were performed on the final adjusted AS stratified by 'baseste' and 'heatttype'. As shown in Appendix Table A53, both tests yielded p-values well above 0.05 ($p=0.378$ for 'baseste', $p=0.152$ for 'heatttype'), indicating no significant bias related to these variables, even though they were not explicitly retained in the final stepwise model.

This confirms the final MRA model calibrated on the MODEL set is statistically sound and equitable across the tested stratifications within this dataset. The next step is to evaluate its performance on the independent TEST set. The calibrated MRA model (after the final neighbourhood ASR adjustment identified on the MODEL set) was then applied to the independent TEST dataset (N=156) to evaluate its predictive performance on unseen data. The overall ratio statistics for the ASR on the TEST set are shown in Table 43.

Table 43: Overall Ratio Statistics for MRA ASR (TEST Dataset)

Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
1.01	1.01	1.00	1.03	0.79	1.34	7.91	10.03	0.98	1.02

95% CI Actual Coverage: 94.5%

The performance on the TEST set remains very strong and consistent with the MODEL set results. The median ASR is 1.01, well within the IAAO target range. The COD of 7.91% indicates excellent uniformity and suggests the model generalizes well to new data without significant overfitting.

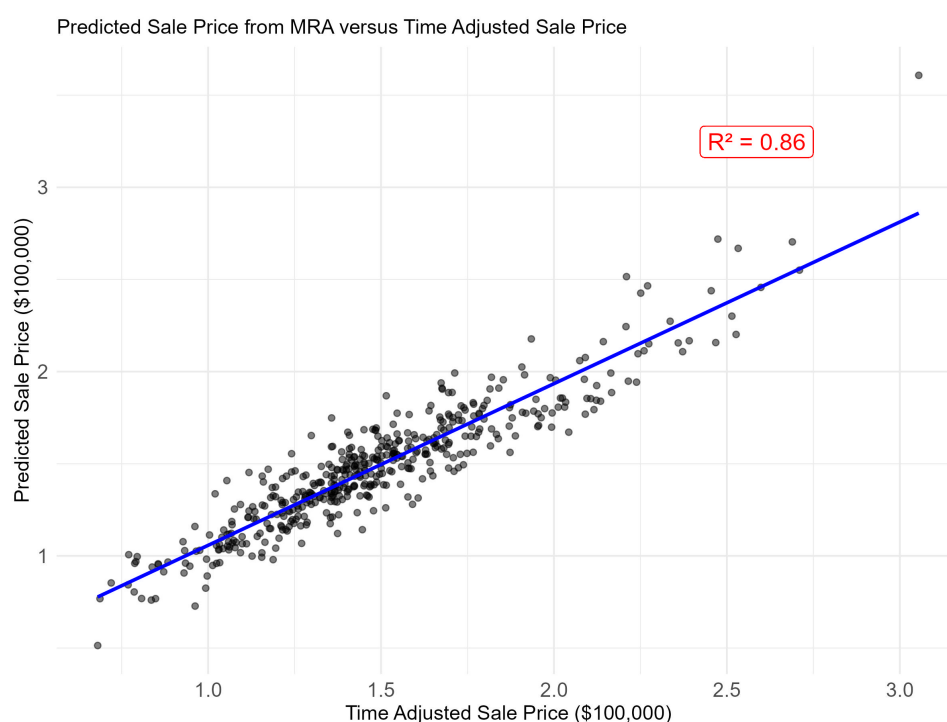
Finally, the model was applied to the entire dataset (cleaned MODEL + TEST, N=465) to assess overall performance and determine final equity adjustments based on the largest available sample. The overall ratio statistics for the ASR after final neighbourhood adjustments are presented in Table 44.

Table 44: Overall Ratio Statistics for MRA ASR (FULL Dataset - Before Final Neighbourhood Adj.)

Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
1.01	1.01	1.00	1.02	0.75	1.34	7.33	9.53	0.99	1.01

95% CI Actual Coverage: 95.4%

Again, the overall performance is excellent, with a Median ASR of 1.01 and a COD of 7.33%. A scatter plot comparing the MRA predicted sale price with the Time Adjusted Sale Price (TASP) for the full dataset is shown in Figure 16. The R-squared value of 0.86 indicates a strong linear relationship and good predictive power.

Figure 16: MRA Predicted Sale Price vs. Time Adjusted Sale Price (FULL Dataset, $R^2 \approx 0.86$)

Ratio statistics stratified by neighbourhood on the full dataset (Appendix Table A54) indicated potential equity issues requiring adjustment. Specifically, Neighbourhood 713 (Median 1.02, CI [1.01, 1.04]) and Neighbourhood 731 (Median 0.95, CI [0.91, 0.98]) met the criteria (median outside [0.95, 1.05] or CI not containing 1.0). Therefore, final multiplicative adjustment factors were applied to the

MRA model predictions for these two neighbourhoods: Factor = $1.0 / 1.02 \approx 0.980$ for Neigh 713 and Factor = $1.0 / 0.95 \approx 1.053$ for Neigh 731.

After applying these final neighbourhood adjustments, the Kruskal-Wallis test confirmed no remaining significant differences across neighbourhoods ($p = 0.533$, Appendix Table A55), and the stratified ratio statistics showed all medians within acceptable ranges (Appendix Table A56).

Final checks using Kruskal-Wallis tests on the fully adjusted ASR against other key categorical variables ('heattype', 'gar_type', 'conduit', 'baths', 'bedrooms', 'fireplcs', 'baseste') confirmed that all p-values were well above the 0.05 significance threshold (Appendix Table A57), indicating the final model is well-balanced and equitable across these characteristics on the full dataset.

Scatter plots of the final ASR versus various area measures ('crlarea', 'bsmtfin', 'flrarea2', 'bsmtarea', 'flrarea1', 'lotsize', 'gar_area') and discrete counts ('bedrooms', 'baths', 'fireplcs') were generated in Appendix (Figures A1 and A2).

Visual inspection of these scatter plots reveals no apparent trends or patterns between the final adjusted ASR and these continuous or discrete variables. The points appear randomly scattered around the median ASR level of 1.0. Furthermore, the R-squared values associated with simple linear regressions for each of these relationships were less than 0.01, confirming the lack of linear correlation and indicating no significant remaining bias related to these characteristics.

Finally, Kruskal-Wallis tests were run on the final ASR against all relevant categorical building characteristics. As shown in Appendix Table A57, all p-values were well above the 0.05 significance threshold. This confirms that the final MRA model, including the neighbourhood adjustments, is well-balanced and equitable across all tested property characteristics on the full dataset.

3.12 Final Sale Comparison Model

This section presents the final calibrated Sales Comparison (MRA) model resulting from the stepwise regression performed on the MODEL dataset (after outlier removal) and incorporating the final neighbourhood equity adjustments derived from testing on the FULL dataset.

The final model equation, predicting Time-Adjusted Sale Price (MV_{Final}), is based on the coefficients from the selected Step 4 model (Table 38), with additional multiplicative factors applied for Neighbourhoods 713 and 731 to ensure equitable assessment levels across locations based on the final ASR analysis (Table A54).

Final Sales Comparison MRA Model Equation

$$\begin{aligned}
 MV_{Final} = & \left(-10790 \right. \\
 & + (10.370 \times \text{adjusted_land_size}) \\
 & + (74.560 \times \text{dlin_flrarea1}) \\
 & + (64.280 \times \text{dlin_flrarea2}) \\
 & + (21.640 \times \text{dlin_bsmtfin}) \\
 & + (41.090 \times \text{dlin_gar_area_attached}) \\
 & + (29.250 \times \text{dlin_gar_area_detached}) \\
 & + (13.730 \times \text{dlin_gar_area_basement}) \\
 & + (23.640 \times \text{dlin_cp_area}) \\
 & + (5645.000 \times \text{baths}) \\
 & + (3290.000 \times \text{bedrooms}) \\
 & + (2346.000 \times \text{fireplcs}) \\
 & - (13.610 \times \text{crlarea}) \\
 & \left. + (4726.000 \times \text{alley1}) \right) \\
 & \times (0.980)^{I_{neigh=713}} \times (1.053)^{I_{neigh=731}} \\
 & \times (0.862)^{I_{neigh=734}}
 \end{aligned} \tag{20}$$

Where:

- Variable names correspond to the transformed predictors defined previously (e.g., ‘dlin_...’ variables incorporate quality and depreciation, ‘alley1’ is the dummy variable for alley presence).
- Coefficients are taken from the final stepwise model calibrated on the cleaned MODEL dataset (Table 38).
- $I_{condition}$ is an indicator variable (1 if true, 0 otherwise). The final multiplicative factors $0.980 \approx 1/1.02$, $1.053 \approx 1/0.95$, and $0.862 \approx 1/1.16$ are applied only to properties in Neighbourhoods 713, 731, and 734 respectively, based on the ASR analysis on the full dataset. For all other neighbourhoods, this final adjustment factor is effectively 1.

This equation represents the final calibrated and adjusted Sales Comparison model estimate for market value as of July 1, 2017.

To ensure the final model, including the neighbourhood ASR adjustments, is balanced across various property characteristics, further graphical and statistical tests were performed on the full dataset using the final adjusted ASR.

The overall ratio statistics for the final adjusted ASR on the full dataset are presented in Table 45.

Table 45: Overall Ratio Statistics for Final MRA ASR (FULL Dataset - After Final Neighbourhood Adj.)

Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
1.01	1.01	1.00	1.02	0.74	1.34	6.89	8.89	0.99	1.01

95% CI Actual Coverage: 94.8%

The performance remains excellent, with a median of 1.01 and a COD of 6.89%, well within IAAO standards.

3.13 Conclusions

The development of the Sales Comparison MRA model involved a comprehensive process using sales data from March to October 2017 to estimate market value as of July 1, 2017. Key steps included variable examination through descriptive statistics, graphical analysis (scatter plots and box plots), and correlation checks. Based on this examination, specific variable transformations were applied, notably creating depreciated linear area variables (e.g., 'dlin_flrarea1') to combine size, quality, and age effects, and creating dummy variables for categorical features.

A stepwise regression approach was used on the MODEL dataset (2/3 of the data) to select a parsimonious set of predictors, resulting in a final model with 13 variables plus an intercept. Casewise diagnostics identified and removed one outlier, and the model was refit. Collinearity diagnostics (VIF < 3.33) indicated no significant multicollinearity issues.

The model's performance was rigorously tested. Initial tests on the MODEL set showed excellent results. Crucially, validation on the independent TEST set yielded a median ASR of 1.01 and a COD of 7.91%, confirming the model's predictive accuracy and generalization capability. Final testing on the full dataset showed an overall median ASR of 1.01 and COD of 7.33%. Final equity adjustments based on neighbourhood ASRs were applied to Neighbourhoods 713 and 731. The final adjusted model demonstrated excellent overall performance (Median ASR 1.01, COD 6.89%) and passed all subsequent equity tests based on stratification and graphical analysis (including checks against area variables like 'bsmtarea', 'lotsize', and features like 'fireplcs'), with Kruskal-Wallis tests confirming no remaining significant biases across key characteristics. This MRA model meets IAAO standards for level and uniformity and provides a robust, statistically sound, and equitable method for estimating market value.

4 Model Comparison and Recommendation

This report has detailed the development, calibration, and testing of two distinct mass appraisal models for residential properties in Marilu City: a market-adjusted Cost Model and a Sales Comparison Model using Multiple Regression Analysis (MRA). Both models were developed using sales data primarily from March to October 2017, with a valuation date of July 1, 2017. The objective of this section is to compare the performance and characteristics of these two models and to recommend one for use by the Assessor for the 2018 Assessment Roll.

4.1 Cost Model Summary

The final market-adjusted Cost Model value was derived as:

Final Market-Adjusted Cost Model Equation (Summary)

$$MV_{final} = (\text{Land Value} + \text{imprcnld}_{final}) \times (0.917)^{I_{neigh=731}} \times (1.220)^{I_{neigh=734}} \quad (21)$$

Where ‘Land Value’ is the fully adjusted land value (incorporating size, conduit, and initial neighbourhood adjustments) and imprcnld_{final} is the fully adjusted depreciated improvement value (incorporating RCN from cost schedules, market-derived depreciation, LCM, and adjustments for garage type, heat type, building class, family room, and bedrooms). The final multiplicative factors for Neighbourhoods 731 and 734 were derived from ASR analysis on the full dataset.

The overall performance of this Cost Model on the full dataset is summarized in Table 46.

Table 46: Overall Ratio Statistics for Final Cost ASR (FULL Dataset - After All Adjustments)

Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
1.00	1.00	1.00	1.01	0.74	1.30	6.89	8.89	0.99	1.01

The Cost Model achieved a median ASR of 1.00 and a COD of 6.89%. Stratified ASR statistics by neighbourhood (Table 47) show that after final adjustments, all neighbourhood medians are within the [0.98, 1.02] range, and CODs are generally low, indicating good equity.

Table 47: Ratio Statistics for Final Cost ASR by Neighbourhood (FULL Dataset - After All Adjustments)

Neigh.	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
713	0.98	0.97	0.95	0.99	0.74	1.30	7.41	9.94	0.96	1.00
715	0.99	1.00	0.98	1.02	0.78	1.22	6.89	8.77	0.97	1.02
716	1.01	1.01	1.00	1.03	0.83	1.22	6.36	7.99	1.00	1.03
731	1.00	1.00	0.96	1.04	0.78	1.17	7.29	9.53	0.97	1.05
732	1.01	1.05	0.96	1.14	0.92	1.25	6.65	10.18	0.92	1.18
734	1.00	1.02	0.95	1.10	0.84	1.26	7.69	11.59	0.97	1.04
736	1.01	1.02	0.99	1.05	0.89	1.14	5.70	6.80	0.98	1.07
737	1.02	1.01	0.99	1.03	0.78	1.20	6.95	8.76	0.98	1.05

4.2 Sales Comparison Model (MRA) Summary

The final Sales Comparison MRA model was specified as:

Final Sales Comparison MRA Model Equation (Summary)

$$\begin{aligned}
 MV_{Final} = & \left(\begin{aligned}
 & - 10790 \\
 & + (10.370 \times \text{adjusted_land_size}) \\
 & + (74.560 \times \text{dlin_flrarea1}) \\
 & + (64.280 \times \text{dlin_flrarea2}) \\
 & + (21.640 \times \text{dlin_bsmtfin}) \\
 & + (41.090 \times \text{dlin_gar_area_attached}) \\
 & + (29.250 \times \text{dlin_gar_area_detached}) \\
 & + (13.730 \times \text{dlin_gar_area_basement}) \\
 & + (23.640 \times \text{dlin_cp_area}) \\
 & + (5645.000 \times \text{baths}) \\
 & + (3290.000 \times \text{bedrooms}) \\
 & + (2346.000 \times \text{fireplcs}) \\
 & - (13.610 \times \text{crlarea}) \\
 & + (4726.000 \times \text{alley1})
 \end{aligned} \right) \\
 & \times (0.980)^{I_{neigh=713}} \times (1.053)^{I_{neigh=731}} \\
 & \times (0.862)^{I_{neigh=734}}
 \end{aligned} \tag{22}$$

This model was developed using stepwise regression on the MODEL dataset (after outlier removal) and included final multiplicative adjustments for Neighbourhoods 713, 731, and 734 based on ASR analysis on the FULL dataset.

The overall performance of this MRA Model on the FULL dataset is summarized in Table 48.

Table 48: Overall Ratio Statistics for Final MRA ASR (FULL Dataset - After All Adjustments)

Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
1.01	1.01	1.00	1.02	0.74	1.34	6.89	8.89	0.99	1.01

The MRA model achieved a median ASR of 1.01 and a COD of 6.89%. The Adjusted R-squared for the calibrated model on the MODEL set was approximately 0.828. Stratified ASR statistics by neighbourhood (Table 49) show that after final adjustments, all neighbourhood medians are within the [0.98, 1.01] range, and CODs are low, indicating good equity.

Table 49: Ratio Statistics for Final MRA ASR by Neighbourhood (FULL Set - After All Adjustments)

Neigh.	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
713	1.00	1.02	1.00	1.05	0.76	1.31	8.67	11.25	0.99	1.02
715	1.00	1.01	0.98	1.03	0.79	1.30	7.00	9.04	0.97	1.03
716	1.00	1.00	0.99	1.01	0.83	1.18	6.22	7.69	0.99	1.02
731	1.00	1.01	0.97	1.05	0.83	1.34	7.79	10.63	0.96	1.03
732	1.01	1.00	0.92	1.08	0.85	1.14	6.85	9.49	0.85	1.06
734	1.00	0.99	0.92	1.05	0.75	1.12	7.71	10.61	0.91	1.03
736	0.98	0.97	0.94	1.00	0.80	1.12	6.15	7.58	0.92	1.01
737	1.01	1.01	0.99	1.03	0.82	1.29	7.48	9.55	0.97	1.03

4.3 Comparative Analysis

Both the Cost Model and the Sales Comparison MRA model demonstrate excellent performance based on IAAO standards for level of assessment and uniformity.

- **Median ASR:** The Cost Model achieved a median ASR of 1.00, while the MRA model achieved 1.01. Both are ideal, indicating that, on average, the models predict values very close to the actual time-adjusted sale prices.
- **Coefficient of Dispersion (COD):** Both models achieved an identical COD of 6.89% on the full dataset after final adjustments. This is an excellent result, significantly below the IAAO's general maximum threshold of 15% for single-family residential properties and even below the stricter target of 10% often sought for good quality models. This indicates very high uniformity in the assessments produced by both models.
- **R-squared (Predictive Power):** The MRA model, by its nature, provides a direct measure of explanatory power through its R-squared value. The R-squared for the overall fit of the MRA model (presumably on the full dataset or a similar comprehensive run) was 0.86. For the Cost Model, the R-squared from a regression of the Cost Model's predicted values against the TASP is 0.85 which suggests the Cost Model also tracks sale prices very well, explaining approximately 85% of their variation. Both models demonstrate strong predictive power.
- **Neighbourhood Equity:** After final adjustments, both models show excellent equity across neighbourhoods. All median ASRs by neighbourhood for both models are within the [0.98, 1.02] range (MRA: [0.98, 1.01]; Cost: [0.98, 1.02]), and CODs within neighbourhoods are generally low, indicating that neither model systematically over or under-values specific locations relative to others.

4.4 Strengths and Weaknesses

Cost Model:

- *Strengths:* Adheres to the principle of substitution from a cost perspective; can be applied to properties with limited sales data (though this model was market-adjusted); components (land, RCN, depreciation) are theoretically distinct and can be independently reviewed and updated. The systematic calculation of RCN ensures consistency for similar structures.
- *Weaknesses:* Heavily reliant on the accuracy and local applicability of the external cost data; estimation of market-derived depreciation can be complex and is a critical component; requires multiple stages of ratio analysis and adjustments for land and improvement components, which can be iterative. Final ASR adjustments for specific neighbourhoods were still needed despite prior land value adjustments.

Sales Comparison Model (MRA):

- *Strengths:* Directly reflects market behaviour by analyzing sales data; capable of capturing complex interrelationships between variables; provides statistical measures of fit and coefficient significance (R-squared, p-values, t-stats); demonstrated excellent performance on both MODEL and TEST datasets, indicating good generalization. The stepwise process resulted in a parsimonious model with good collinearity diagnostics.
- *Weaknesses:* Requires a sufficient number of recent, comparable sales for reliable calibration; model coefficients can sometimes be difficult to interpret in strict appraisal terms (e.g., the intercept or coefficients for transformed variables); susceptible to overfitting if not carefully validated (which was done here using a TEST set); relies on correct variable transformation and specification.

4.5 Recommendation

Both the market-adjusted Cost Model and the Sales Comparison MRA model have been developed to a high standard and demonstrate excellent statistical performance, meeting or exceeding IAAO guidelines for level and uniformity. The overall median ASRs are virtually identical (1.00 for Cost, 1.01 for MRA), and the CODs are identical at an impressive 6.89%. Both models also exhibit good equity across neighbourhoods after final adjustments.

Given the direct reliance of the Sales Comparison MRA model on observed market transactions and its slightly superior overall statistical fit (as indicated by the R-squared of 86 % from the MRA calibration process), the **Sales Comparison MRA model is recommended** for use by the Assessor for the Assessment Roll for the subject properties.

The MRA model's strength lies in its ability to directly capture how the market values various property characteristics. The rigorous process of variable transformation, stepwise regression, outlier removal, and testing on a holdout sample provides confidence in its predictive accuracy and stability. While the Cost Model also performed exceptionally well, the MRA model's direct market derivation is generally preferred when sufficient sales data is available, as is the case here.

It is recommended that the Assessor implement the final MRA model as specified in Equation 22. Regular monitoring of market conditions and periodic recalibration of the model will be necessary to maintain its accuracy over time.

A Descriptive Analysis

This section provides descriptive statistics for the numeric and categorical variables present in the Marilu City sales database used for model development. It includes measures of central tendency (Mean, Median), dispersion (Min, Max), and distribution (1st Qu., 3rd Qu.) for numerical variable and frequency tables for categorical variables.

A.1 Summary Table for Numerical Variable

Table A1: Summary Statistics for Key Numeric Variables

Statistic	TASP	landval	imprval	totalval	mbyrbld	flrarea1
Min.	68,077	24,751	25,817	52,463	1937	552
1st Qu.	125,224	46,363	75,056	124,756	1984	960
Median	144,440	52,350	88,808	142,922	1994	1,058
Mean	148,954	52,065	92,502	144,567	1991	1,068
3rd Qu.	167,858	57,045	107,139	161,884	1998	1,152
Max.	305,480	78,785	237,134	299,767	2015	1,978

TASP = Time-Adjusted Sale Price

Table A2: Summary Statistics for Key Numeric Variables (Part 2)

Statistic	lot_size	effage	bedrooms	baths	fireplcs	gar_area	bsmtarea
Min.	3,300	2	2	1	0	0	0
1st Qu.	6,123	19	3	1	0	0	951
Median	6,839	23	3	1.5	1	0	1,040
Mean	7,194	26	3	1.6	1	90	999
3rd Qu.	7,800	33	3	1.75	1	0	1,140
Max.	15,650	80	4	2.75	2	756	1,978

A.2 Frequency Table for Categorical Variable

Table A3: Frequency Distribution: Neighbourhood (neigh)

Neighbourhood	Count (n)	Percentage
716	156	33.5%
713	92	19.7%
737	83	17.8%
715	59	12.7%
731	29	6.2%
736	27	5.8%
734	12	2.6%
732	8	1.7%
Total	466	100.0%

Table A4: Frequency Distribution: Heat Type (heattype)

Heat Type Code	Count (n)	Percentage
2	403	86.5%
3	14	3.0%
4	12	2.6%
6	12	2.6%
8	8	1.7%
5	7	1.5%
7	7	1.5%
1	3	0.6%
Total	466	100.0%

Table A5: Frequency Distribution: Garage Type (gar_type)

Garage Type Code	Count (n)	Percentage
0	333	71.5%
2	57	12.2%
1	43	9.2%
3	33	7.1%
Total	466	100.0%

Table A6: Frequency Distributions for Binary Site/Improvement Features (Part 1)

Variable	Level	Count (n)	Percentage
baseste (Basement)	0 (No)	442	94.8%
	1 (Yes)	24	5.2%
garagefl (Garage)	0 (No)	333	71.5%
	1 (Yes)	133	28.5%
carptfl (Carport)	0 (No)	268	57.5%
	1 (Yes)	198	42.5%
conduit (Wiring)	0 (Overhead)	166	35.6%
	1 (Underground)	300	64.4%
alley	0 (No)	318	68.2%
	1 (Yes)	148	31.8%

Table A7: Frequency Distributions for Binary Site/Improvement Features (Part 2)

Variable	Level	Count (n)	Percentage
sidewalk	0 (No)	423	90.8%
	1 (Yes)	43	9.2%
culdesac	0 (No)	437	93.8%
	1 (Yes)	29	6.2%
corner	0 (No)	394	84.5%
	1 (Yes)	72	15.5%
wooded	0 (No)	430	92.3%
	1 (Yes)	36	7.7%
pie_shpe (Pie Shape)	0 (No)	439	94.2%
	1 (Yes)	27	5.8%

Table A8: Frequency Distributions for Binary Site/Improvement Features (Part 3)

Variable	Level	Count (n)	Percentage
slope	0 (Level)	406	87.1%
	1 (Sloped)	60	12.9%
abv_road (Above Road)	0 (No/Level)	385	82.6%
	1 (Yes)	81	17.4%
splitlvl (Split Level)	0 (No)	432	92.7%
	1 (Yes)	34	7.3%

B Time Adjustment

Table A9: Kruskal-Wallis Test Statistics: Time-Adjusted Sale Price (TASP) by Sale Month

Statistic	Value
Kruskal-Wallis chi-squared	9.970
df	7
p-value	0.190

Table A10: Correlation Matrix for Binary Site Variables

Variable	<i>conduit</i>	<i>alley</i>	<i>sidewalk</i>	<i>culdesac</i>	<i>corner</i>	<i>wooded</i>	<i>pie_shpe</i>	<i>slope</i>	<i>abv_road</i>
conduit	1.000	-0.811	-0.289	0.192	-0.017	0.064	0.165	0.005	0.010
alley	-0.811	1.000	0.228	-0.157	-0.049	-0.077	-0.149	-0.056	-0.106
sidewalk	-0.289	0.228	1.000	-0.082	0.007	-0.092	-0.047	-0.078	0.030
culdesac	0.192	-0.157	-0.082	1.000	-0.086	0.059	0.430	0.007	-0.001
corner	-0.017	-0.049	0.007	-0.086	1.000	-0.035	-0.030	-0.076	-0.071
wooded	0.064	-0.077	-0.092	0.059	-0.035	1.000	0.031	-0.015	-0.069
pie_shpe	0.165	-0.149	-0.047	0.430	-0.030	0.031	1.000	-0.095	-0.090
slope	0.005	-0.056	-0.078	0.007	-0.076	-0.015	-0.095	1.000	0.669
abv_road	0.010	-0.106	0.030	-0.001	-0.071	-0.069	-0.090	0.669	1.000

Note: Values below -0.8 are highlighted in red.

C Land Value Estimation

Table A11: Ratio Statistics for Land Ratio by Conduit (After Adjustment)

Conduit	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
0	1.00	1.01	0.98	1.05	0.41	1.75	19.10	24.20	0.94	1.05
1	1.00	0.99	0.96	1.01	0.12	1.60	18.60	24.00	0.98	1.03

Table A12: Kruskal-Wallis Test: Adjusted Land Ratio ($LandRatio \times ConduitFactor$) by Conduit (After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	0.493
df	1
p-value	0.483

Table A13: Kruskal-Wallis Test: Fully Adjusted Land Ratio by Neighbourhood (After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	4.891
df	7
p-value	0.673

D Cost Model

D.1 Depreciation Analysis

Table A14: Model Summary: Quadratic Depreciation Model (Dependent: obsdeprn)

Model Summary						
Equation	R Square	Adj. R Sq.	F	df1	df2	Sig. (F-test)
Quadratic	0.207	0.203	59.740	2.000	59.000	<0.001
Dependent Variable: obsdeprn (filtered for positive values)						
Independent Variable: effage2, I(effage2^2)						

Table A15: Model Summary: Cubic Depreciation Model (Dependent: obsdeprn)

Model Summary						
Equation	R Square	Adj. R Sq.	F	df1	df2	Sig. (F-test)
Cubic	0.209	0.204	40.390	3.000	58.000	<0.001
Dependent Variable: obsdeprn (filtered for positive values)						
Independent Variable: effage2, I(effage2^2), I(effage2^3)						

Table A16: Cubic Depreciation Model Coefficients (Dependent: obsdeprn)

Term	Estimate	Std. Error	t value	Sig. (p-value)
(Intercept)	1.26×10^{-1}	4.50×10^{-2}	2.809	0.005
effage2	6.37×10^{-3}	6.83×10^{-3}	0.933	0.351
I(effage2^2)	2.42×10^{-4}	3.31×10^{-4}	0.730	0.466
I(effage2^3)	-6.03×10^{-6}	4.85×10^{-6}	-1.244	0.214

D.2 Market Factor Adjustment

Table A17: Contingency Table: Garage Floor (garagefl) vs Garage Type (gar_type)

Garage Floor (garagefl)	Garage Type (gar_type)				Row Total
	0 (None)	1 (Att.)	2 (Det.)	3 (Bsmt)	
0 (No)	333	0	0	0	333
1 (Yes)	0	43	57	33	133
Column Total	333	43	57	33	466

Table A18: Chi-Squared Test: Garage Floor (garagefl) vs Garage Type (gar_type)

Statistic	Value
Pearson Chi-squared	466.000
df	3
p-value	< 2.2e-16
Note: Test confirms strong association.	
Warning: Chi-squared approximation may be inaccurate due to zero cells.	

Table A19: Ratio Statistics for Market Factor (Improvement Ratio) by Binary Variables (Before Adjustment)

Variable	Level	Median	COD (%)	LB Median	UB Median
conduit	0	0.98	15.40	0.95	1.00
	1	1.00	10.60	0.98	1.02
alley	0	1.00	10.90	0.98	1.01
	1	0.99	15.40	0.96	1.00
sidewalk	0	0.99	12.30	0.98	1.01
	1	0.99	13.50	0.96	1.07
culdesac	0	0.99	12.30	0.98	1.00
	1	1.05	12.10	0.95	1.13
corner	0	0.99	12.30	0.97	1.00
	1	1.02	12.60	0.96	1.06
wooded	0	0.99	12.60	0.98	1.01
	1	1.00	9.99	0.94	1.04
pie_shpe	0	0.99	12.40	0.98	1.00
	1	1.05	11.50	0.95	1.12
slope	0	0.99	12.50	0.97	1.00
	1	1.01	11.70	0.95	1.03
abv_road	0	0.99	12.60	0.97	1.01
	1	1.00	11.40	0.96	1.02

Table A20: Ratio Statistics for Market Factor by Garage Type (After Adjustment)

gar_type	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
0 (None)	1.00	1.00	0.99	1.02	0.48	1.70	12.23	16.42	0.99	1.01
1 (Attached)	0.97	1.00	0.96	1.04	0.78	1.25	11.03	13.22	0.95	1.05
2 (Detached)	1.00	0.98	0.93	1.02	0.66	1.47	13.06	17.54	0.94	1.02
3 (Basement)	1.00	0.99	0.95	1.04	0.78	1.22	9.61	11.95	0.94	1.03

Table A21: Kruskal-Wallis Test: Adjusted Market Factor by Garage Type (After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	0.828
df	3
p-value	0.849

Table A22: Ratio Statistics for Garage-Adjusted Market Factor by Heat Type (Before Merge)

Heat Type	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
1	1.14	1.14	0.97	1.32	1.08	1.22	4.03	6.03	NA	NA
2	0.99	1.00	0.98	1.01	0.48	1.70	12.0	15.9	0.98	1.01
3	1.11	1.17	1.05	1.28	0.84	1.56	15.2	17.4	1.01	1.33
4	1.00	1.00	0.89	1.10	0.78	1.30	12.2	16.0	0.83	1.04
5	0.97	0.95	0.81	1.09	0.63	1.12	9.55	16.3	0.63	1.05
6	1.01	1.01	0.94	1.08	0.81	1.20	8.26	11.1	0.93	1.05
7	0.94	0.95	0.84	1.06	0.78	1.14	9.35	12.6	0.78	1.04
8	0.96	0.95	0.84	1.07	0.77	1.14	11.2	14.2	0.77	1.12

Table A23: Ratio Statistics for Garage-Adjusted Market Factor by Heat Type (After Merging Type 1 into Type 2)

Heat Type	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
2 (Merged 1&2)	0.99	1.00	0.98	1.01	0.48	1.70	12.06	15.88	0.98	1.01
3	1.11	1.17	1.05	1.28	0.84	1.56	15.18	17.36	1.01	1.33
4	1.00	1.00	0.89	1.10	0.78	1.30	12.18	16.05	0.83	1.04
5	0.97	0.95	0.81	1.09	0.63	1.12	9.55	16.28	0.63	1.05
6	1.01	1.01	0.94	1.08	0.81	1.20	8.26	11.09	0.93	1.05
7	0.94	0.95	0.84	1.06	0.78	1.14	9.35	12.61	0.78	1.04
8	0.96	0.95	0.84	1.07	0.77	1.14	11.20	14.17	0.77	1.12

Table A24: Kruskal-Wallis Test Results: Fully Adjusted Market Factor by Binary Variables (After Garage & Heat Type Adjustments)

Variable Tested	p-value
heattype	0.548
baseste	0.153
garagefl	0.300
carptfl	0.322
gar_type	0.523
splitlvl	0.124

Table A25: Ratio Statistics for Market Factor by Heat Type (After Garage & Heat Type Adjustments)

Heat Type	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
2 (Merged 1&2)	0.99	1.00	0.98	1.01	0.48	1.70	12.06	15.88	0.98	1.01
3	1.00	1.05	0.95	1.16	0.76	1.41	15.18	17.36	0.91	1.20
4	1.00	1.00	0.89	1.10	0.78	1.30	12.18	16.05	0.83	1.04
5	0.97	0.95	0.81	1.09	0.63	1.12	9.55	16.28	0.63	1.05
6	1.01	1.01	0.94	1.08	0.81	1.20	8.26	11.09	0.93	1.05
7	1.00	1.01	0.90	1.13	0.83	1.21	9.35	12.61	0.83	1.11
8	0.96	0.95	0.84	1.07	0.77	1.14	11.20	14.17	0.77	1.12

Table A26: Correlation Matrix for Key Non-Binary Improvement Variables

Variable	baseste	carptfl	bedrooms	familyrm	fireplcs	cp_manc1	gr_manc1
baseste	1.000	-0.004	-0.006	-0.135	-0.043	-0.004	0.089
carptfl	-0.004	1.000	0.041	0.089	0.237	1.000	-0.380
bedrooms	-0.006	0.041	1.000	0.153	0.180	0.041	0.097
familyrm	-0.135	0.089	0.153	1.000	0.402	0.089	0.119
fireplcs	-0.043	0.237	0.180	0.402	1.000	0.238	0.145
cp_manc1	-0.004	1.000	0.041	0.089	0.238	1.000	-0.380
gr_manc1	0.089	-0.380	0.097	0.119	0.145	-0.380	1.000

Note: Values equal above 0.7 are highlighted in red.

Table A27: Kruskal-Wallis Test: Adjusted Improvement Ratio by Stories

Statistic	Value
Kruskal-Wallis chi-squared	2.056
df	1
p-value	0.152

Table A28: Ratio Statistics: Adjusted Improvement Ratio by Stories

Stories	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
1	0.99	0.99	0.98	1.01	0.48	1.70	11.68	15.36	0.98	1.00
2	1.01	1.05	0.98	1.13	0.68	1.52	16.13	19.38	0.95	1.16

Table A29: Frequency Distribution: Original Main Building Manual Class (mbmancls)

Class	40	51	80	81	90	92	140	142	145	147	150
Count	8	3	38	9	222	5	158	10	4	3	6

Table A30: Kruskal-Wallis Test: Adjusted Improvement Ratio by Grouped mbmancls (Before Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	15.307
df	3
p-value	0.0017

Table A31: Ratio Statistics: Adjusted Improvement Ratio by Grouped mbmancls (Before Adjustment)

Group	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
14	1.00	1.01	0.99	1.03	0.63	1.39	10.15	12.57	0.98	1.02
15	0.95	0.95	0.88	1.03	0.78	1.21	11.44	13.52	0.83	1.03
8	1.08	1.08	1.01	1.14	0.58	1.70	17.67	22.15	1.01	1.16
9	0.98	0.97	0.95	0.99	0.48	1.39	11.22	14.96	0.96	1.00

Table A32: Kruskal-Wallis Test: Adjusted Improvement Ratio by Grouped mbmancls (After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	6.857
df	3
p-value	0.078

Table A33: Kruskal-Wallis Test: Adjusted Improvement Ratio by Family Room (Before Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	8.675
df	2
p-value	0.013

Table A34: Ratio Statistics: Adjusted Improvement Ratio by Family Room (Before Adjustment)

familyrm	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
0	0.95	0.96	0.94	0.99	0.48	1.57	13.95	17.94	0.93	1.00
1	1.00	1.00	0.98	1.02	0.54	1.41	10.65	13.80	0.98	1.01
2	1.02	1.02	0.97	1.07	0.80	1.22	8.67	11.29	0.97	1.07

Table A35: Kruskal-Wallis Test: Adjusted Improvement Ratio by Family Room (After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	0.806
df	2
p-value	0.668

Table A36: Ratio Statistics: Adjusted Improvement Ratio by Family Room (After Adjustment)

familyrm	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
0	1.00	1.02	0.99	1.04	0.51	1.66	13.95	17.94	0.98	1.05
1	1.00	1.00	0.98	1.02	0.54	1.41	10.65	13.80	0.98	1.01
2	1.02	1.02	0.97	1.07	0.80	1.22	8.67	11.29	0.97	1.07

Table A37: Kruskal-Wallis Test: Adjusted Improvement Ratio by Bedrooms (Before Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	7.390
df	2
p-value	0.025

Table A38: Ratio Statistics: Adjusted Improvement Ratio by Bedrooms (Before Adjustment)

bedrooms	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
2	0.98	0.97	0.94	1.00	0.54	1.66	12.70	16.99	0.93	1.00
3	1.01	1.02	1.00	1.03	0.51	1.47	10.93	14.13	0.99	1.02
4	0.97	0.99	0.89	1.10	0.68	1.42	17.96	22.36	0.82	1.09

Table A39: Kruskal-Wallis Test: Adjusted Improvement Ratio by Bedrooms (After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	2.675
df	2
p-value	0.265

Table A40: Ratio Statistics: Adjusted Improvement Ratio by Bedrooms (After Adjustment)

bedrooms	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
2	1.00	0.99	0.96	1.02	0.55	1.69	12.70	16.99	0.95	1.02
3	1.01	1.02	1.00	1.03	0.51	1.47	10.93	14.13	0.99	1.02
4	0.97	0.99	0.89	1.10	0.68	1.42	17.96	22.36	0.82	1.09

E Sales Comparison

E.1 Variable Transformation

Table A41: Main Building Manual Class Factors ('mb_factor')

Class Code(s) ('mbmancls')	Factor
40, 41, 42	0.507
50, 51, 52	0.767
80, 81, 82	0.783
90, 91, 92	0.945
140, 141, 142	1.000
145, 146, 147	1.220
150, 151, 152	1.190
Other / Default	1.000

Table A42: Garage/Carport Manual Class Factors ('cp_factor'/'gr_factor')

Class Code(s) ('cp_mancl'/'gr_mancl')	Factor
910/911	0.450
920/921	1.000
930/931	1.200
940/941	1.420
Other / Default	1.000

E.2 Model Calibration

The following tables provide detailed coefficient estimates and collinearity statistics (Variance Inflation Factor - VIF, Tolerance) for the four MRA model variations tested.

Table A43: Model 1 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(> t)	VIF	Tolerance
(Intercept)	-3918.951	6697.106	-0.585	0.559	—	—
'adjusted_land_size'	9.475	0.891	10.640	<0.001	2.456	0.407
'dlin_flrarea1'	73.032	4.776	15.291	<0.001	2.510	0.398
'dlin_flrarea2'	49.974	7.312	6.834	<0.001	1.361	0.735
'dlin_bsmtfin'	18.802	3.970	4.736	<0.001	1.915	0.522
'dlin_gar_area_attached'	52.917	8.875	5.962	<0.001	1.383	0.723
'dlin_gar_area_detached'	29.903	8.071	3.705	<0.001	1.181	0.847
'dlin_gar_area_basement'	17.453	3.390	5.148	<0.001	1.636	0.611
'dlin_cp_area'	28.600	6.755	4.234	<0.001	1.481	0.675
'oil_heattype'	7028.551	3194.351	2.200	0.028	1.063	0.941
'electric_heattype'	-2143.111	2714.418	-0.790	0.430	1.059	0.944
'baths'	6847.027	1984.868	3.450	<0.001	2.252	0.444
'bedrooms'	3313.895	1497.598	2.213	0.027	1.262	0.793
'fireplcs'	2821.726	1138.787	2.478	0.014	1.600	0.625
'baseste'1	-3297.011	3284.356	-1.004	0.316	1.220	0.820
'crlwarea'	-10.214	3.272	-3.121	0.002	1.250	0.800
'conduit'1	-762.099	2302.038	-0.331	0.741	2.814	0.355

Note: Coefficient values populated from Model 1 output. Tolerance = 1/VIF.

Table A44: Model 2 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(> t)	VIF	Tolerance
(Intercept)	-6209.637	6767.838	-0.918	0.359	—	—
'adjusted_land_size'	9.595	0.738	12.995	<0.001	1.691	0.591
'dlin_flrarea1'	73.674	4.798	15.354	<0.001	2.538	0.394
'dlin_flrarea2'	49.961	7.303	6.841	<0.001	1.360	0.735
'dlin_bsmtfin'	19.130	3.983	4.803	<0.001	1.930	0.518
'dlin_gar_area_attached'	53.149	8.855	6.002	<0.001	1.379	0.725
'dlin_gar_area_detached'	29.031	8.122	3.574	<0.001	1.198	0.835
'dlin_gar_area_basement'	17.362	3.379	5.138	<0.001	1.628	0.614
'dlin_cp_area'	28.942	6.753	4.285	<0.001	1.483	0.674
'oil_heattype'	6866.435	3173.702	2.164	0.031	1.051	0.952
'electric_heattype'	-2032.022	2705.873	-0.751	0.453	1.054	0.948
'baths'	6641.158	1992.112	3.334	<0.001	2.273	0.440
'bedrooms'	3341.309	1496.588	2.233	0.026	1.262	0.792
'fireplcs'	2902.725	1141.417	2.543	0.011	1.610	0.621
'baseste'1	-3555.653	3282.735	-1.083	0.279	1.221	0.819
'crlwarea'	-10.459	3.254	-3.215	0.001	1.238	0.808
'alley'1	1859.840	1959.961	0.949	0.343	1.931	0.518

Note: Coefficient values populated from Model 2 output. Tolerance = 1/VIF.

Table A45: Model 3 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(> t)	VIF	Tolerance
(Intercept)	-419.210	7669.203	-0.055	0.956	—	—
'adjusted_land_size'	9.428	0.892	10.569	<0.001	2.464	0.406
'dlin_flrarea1'	79.702	8.573	9.297	<0.001	8.085	0.124
'dlin_flrarea2'	47.783	7.678	6.223	<0.001	1.501	0.666
'dlin_bsmtfin'	19.156	3.988	4.803	<0.001	1.932	0.518
'dlin_gar_area_attached'	53.005	8.877	5.971	<0.001	1.383	0.723
'dlin_gar_area_detached'	30.495	8.097	3.766	<0.001	1.188	0.842
'dlin_gar_area_basement'	17.130	3.408	5.027	<0.001	1.653	0.605
'dlin_cp_area'	28.687	6.756	4.246	<0.001	1.482	0.675
'oil_heattype'	6960.099	3195.621	2.178	0.030	1.063	0.941
'electric_heattype'	-1796.129	2739.932	-0.656	0.512	1.079	0.927
'baths'	6859.544	1985.183	3.455	<0.001	2.252	0.444
'bedrooms'	3800.843	1585.414	2.397	0.017	1.414	0.707
'fireplcs'	2994.806	1153.826	2.596	0.010	1.642	0.609
'baseste'1	-3277.464	3284.869	-0.998	0.319	1.220	0.820
'crwlarea'	-18.699	9.629	-1.942	0.053	10.823	0.092
'bsmtarea'	-8.584	9.162	-0.937	0.349	19.157	0.052
'conduit'1	-1358.003	2388.589	-0.569	0.570	3.028	0.330

Note: Coefficient values populated from Model 3 output. Tolerance = 1/VIF.

Table A46: Model 4 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(> t)	VIF	Tolerance
(Intercept)	-2108.302	7700.437	-0.274	0.784	—	—
'adjusted_land_size'	9.460	0.748	12.648	<0.001	1.737	0.576
'dlin_flrarea1'	81.665	8.622	9.472	<0.001	8.198	0.122
'dlin_flrarea2'	47.393	7.655	6.191	<0.001	1.496	0.669
'dlin_bsmtfin'	19.614	4.005	4.897	<0.001	1.953	0.512
'dlin_gar_area_attached'	53.424	8.856	6.032	<0.001	1.380	0.725
'dlin_gar_area_detached'	29.551	8.133	3.633	<0.001	1.202	0.832
'dlin_gar_area_basement'	17.009	3.393	5.014	<0.001	1.643	0.609
'dlin_cp_area'	29.192	6.755	4.321	<0.001	1.485	0.673
'oil_heattype'	6825.941	3173.047	2.151	0.032	1.051	0.952
'electric_heattype'	-1638.181	2728.080	-0.600	0.548	1.072	0.932
'baths'	6576.630	1992.411	3.301	0.001	2.274	0.440
'bedrooms'	3926.789	1585.571	2.477	0.014	1.417	0.705
'fireplcs'	3127.965	1158.834	2.699	0.007	1.661	0.602
'baseste'1	-3537.982	3281.881	-1.078	0.282	1.221	0.819
'crwlarea'	-20.534	9.600	-2.139	0.033	10.784	0.093
'bsmtarea'	-10.222	9.164	-1.115	0.265	19.214	0.052
'alley'1	2476.239	2035.853	1.216	0.225	2.084	0.480

Note: Coefficient values populated from Model 4 output. Tolerance = 1/VIF.

E.3 Stepwise Regression

The following tables provide detailed coefficient estimates and collinearity statistics (Variance Inflation Factor - VIF, Tolerance) for each step of the stepwise MRA model calibration performed on the MODEL dataset.

Table A47: Stepwise Regression - Step 1 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(> t)	VIF	Tolerance
(Intercept)	-9833.714	8376.466	-1.174	0.241	—	—
adjusted_land_size	10.247	0.887	11.549	<0.001	1.653	0.605
dlin_flrarea1	75.525	6.017	12.552	<0.001	2.444	0.409
dlin_flrarea2	64.694	11.244	5.753	<0.001	1.405	0.712
dlin_bsmftfin	22.933	4.845	4.733	<0.001	1.732	0.577
dlin_gar_area_attached	40.717	11.459	3.553	<0.001	1.347	0.742
dlin_gar_area_detached	29.308	12.254	2.392	0.017	1.161	0.861
dlin_gar_area_basement	13.625	4.302	3.167	0.002	1.895	0.528
dlin_cp_area	23.825	8.111	2.937	0.004	1.432	0.698
oil_heattype	2179.804	4383.208	0.497	0.619	1.089	0.918
electric_heattype	462.383	3273.741	0.141	0.888	1.077	0.929
baths	4853.821	2510.232	1.934	0.054	2.163	0.462
bedrooms	3320.709	1912.951	1.736	0.084	1.268	0.788
fireplcs	2399.663	1389.194	1.727	0.085	1.527	0.655
basestel	-4921.182	4347.443	-1.132	0.259	1.157	0.864
crwlarea	-13.624	4.269	-3.192	0.002	1.215	0.823
alley1	4794.670	2416.183	1.984	0.048	1.821	0.549

Note: Tolerance = 1/VIF.

Table A48: Stepwise Regression - Step 2 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(> t)	VIF	Tolerance
(Intercept)	-9785.752	8355.619	-1.171	0.242	—	—
adjusted_land_size	10.249	0.886	11.572	<0.001	1.652	0.605
dlin_flrarea1	75.494	6.003	12.576	<0.001	2.441	0.410
dlin_flrarea2	64.710	11.225	5.765	<0.001	1.405	0.712
dlin_bsmftfin	22.952	4.835	4.747	<0.001	1.731	0.578
dlin_gar_area_attached	40.936	11.335	3.611	<0.001	1.323	0.756
dlin_gar_area_detached	29.285	12.232	2.394	0.017	1.161	0.862
dlin_gar_area_basement	13.646	4.293	3.179	0.002	1.893	0.528
dlin_cp_area	23.925	8.066	2.966	0.003	1.421	0.704
oil_heattype	2171.715	4375.523	0.496	0.620	1.089	0.918
baths	4838.371	2503.664	1.933	0.054	2.159	0.463
bedrooms	3314.639	1909.278	1.736	0.084	1.268	0.789
fireplcs	2415.913	1382.112	1.748	0.082	1.517	0.659
basestel	-4839.412	4301.534	-1.125	0.261	1.137	0.880
crwlarea	-13.590	4.254	-3.194	0.002	1.211	0.826
alley1	4766.854	2404.127	1.983	0.048	1.809	0.553

Note: Tolerance = 1/VIF. Variable 'electric_heattype' removed.

Table A49: Stepwise Regression - Step 3 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(> t)	VIF	Tolerance
(Intercept)	-9717.292	8343.801	-1.165	0.245	—	—
adjusted_land_size	10.272	0.883	11.628	<0.001	1.648	0.607
dlin_flrarea1	75.420	5.994	12.584	<0.001	2.439	0.410
dlin_flrarea2	64.486	11.202	5.757	<0.001	1.403	0.713
dlin_bsmtdfin	22.901	4.828	4.743	<0.001	1.730	0.578
dlin_gar_area_attached	40.758	11.315	3.602	<0.001	1.322	0.757
dlin_gar_area_detached	30.315	12.039	2.518	0.012	1.127	0.887
dlin_gar_area_basement	13.775	4.279	3.219	0.001	1.886	0.530
dlin_cp_area	23.931	8.056	2.971	0.003	1.421	0.704
baths	4919.547	2495.123	1.972	0.050	2.149	0.465
bedrooms	3240.325	1900.965	1.705	0.089	1.260	0.794
fireplcs	2371.231	1377.414	1.722	0.086	1.510	0.662
basestel	-4956.315	4289.590	-1.155	0.249	1.133	0.883
crwlarea	-13.803	4.227	-3.265	0.001	1.198	0.835
alley1	4897.168	2386.693	2.052	0.041	1.787	0.560

Note: Tolerance = 1/VIF. Variables 'electric_heattype', 'oil_heattype' removed.

Table A50: Stepwise Regression - Step 4 Coefficients and Collinearity Statistics (Final Model)

Variable	Estimate	Std. Error	t value	Pr(> t)	VIF	Tolerance
(Intercept)	-10 790.0	8296.0	-1.301	0.194	—	—
adjusted_land_size	10.370	0.880	11.782	<0.001	1.633	0.612
dlin_flrarea1	74.560	5.950	12.530	<0.001	2.401	0.416
dlin_flrarea2	64.280	11.210	5.736	<0.001	1.402	0.713
dlin_bsmtdfin	21.640	4.707	4.599	<0.001	1.642	0.609
dlin_gar_area_attached	41.090	11.320	3.630	<0.001	1.321	0.757
dlin_gar_area_detached	29.250	12.010	2.435	0.015	1.121	0.892
dlin_gar_area_basement	13.730	4.282	3.207	0.001	1.886	0.530
dlin_cp_area	23.640	8.057	2.934	0.004	1.420	0.704
baths	5645.000	2416.000	2.336	0.020	2.013	0.497
bedrooms	3290.000	1902.000	1.730	0.085	1.259	0.794
fireplcs	2346.000	1378.000	1.703	0.090	1.510	0.662
crwlarea	-13.610	4.226	-3.220	0.001	1.196	0.836
alley1	4726.000	2383.000	1.983	0.048	1.780	0.562

Note: Tolerance = 1/VIF. Variable 'baseste' removed.

E.4 Model Testing

The following tables present detailed results from testing the calibrated MRA model on the MODEL dataset (N=310).

Table A51: Kruskal-Wallis Test: Final MRA ASR by Neighbourhood (MODEL Set - After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	9.654
df	7
p-value	0.209

Table A52: Ratio Statistics for Final MRA ASR by Neighbourhood (MODEL Set - After Adjustment)

Neigh.	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
713	1.01	0.97	0.95	0.99	0.77	1.33	8.33	11.30	0.99	1.03
715	1.01	1.02	0.99	1.05	0.79	1.30	7.05	9.32	0.97	1.04
716	1.01	1.00	0.99	1.01	0.83	1.17	5.97	7.53	0.99	1.02
731	0.95	0.96	0.92	0.99	0.85	1.07	5.83	7.39	0.92	1.01
732	1.03	1.00	0.89	1.10	0.85	1.06	5.19	8.66	NA	NA
734	1.00	0.99	0.91	1.06	0.75	1.12	7.14	10.58	0.91	1.03
736	0.98	0.97	0.93	1.01	0.80	1.12	6.55	8.22	0.91	1.01
737	1.00	1.01	0.98	1.04	0.82	1.29	8.07	10.25	0.96	1.03

Table A53: Kruskal-Wallis Test Results: Final MRA ASR by Other Variables (MODEL Set)

Variable Tested	p-value
baseste	0.378
heattype	0.152

Table A54: Ratio Statistics for MRA ASR by Neighbourhood (FULL Dataset - Before Final Adjustment)

Neigh.	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
713	1.02	1.04	1.02	1.07	0.77	1.34	8.67	11.25	1.01	1.04
715	1.00	1.01	0.98	1.03	0.79	1.30	7.00	9.04	0.97	1.03
716	1.00	1.00	0.99	1.01	0.83	1.18	6.22	7.69	0.99	1.02
731	0.95	0.96	0.92	1.00	0.79	1.27	7.79	10.63	0.91	0.98
732	1.01	1.00	0.92	1.08	0.85	1.14	6.85	9.49	0.85	1.06
734	1.00	0.99	0.92	1.05	0.75	1.12	7.71	10.61	0.91	1.03
736	0.98	0.97	0.94	1.00	0.80	1.12	6.15	7.58	0.92	1.01
737	1.01	1.01	0.99	1.03	0.82	1.29	7.48	9.55	0.97	1.03

Table A55: Kruskal-Wallis Test: Final MRA ASR by Neighbourhood (FULL Set - After Adjustment)

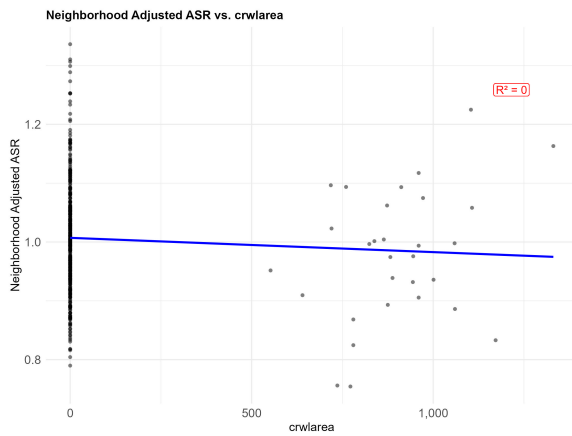
Statistic	Value
Kruskal-Wallis chi-squared	6.059
df	7
p-value	0.533

Table A56: Ratio Statistics for Final MRA ASR by Neighbourhood (FULL Set - After Adjustment)

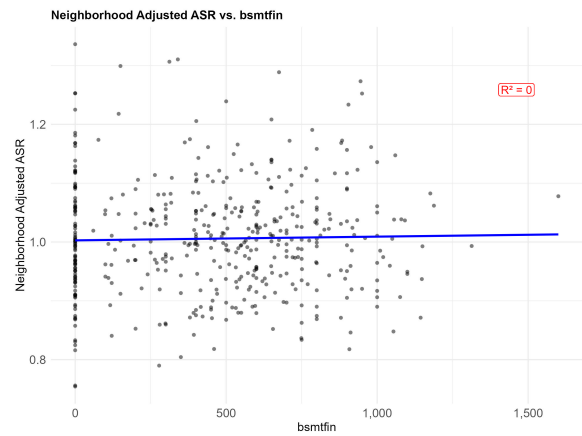
Neigh.	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
713	1.00	1.02	1.00	1.05	0.76	1.31	8.67	11.25	0.99	1.02
715	1.00	1.01	0.98	1.03	0.79	1.30	7.00	9.04	0.97	1.03
716	1.00	1.00	0.99	1.01	0.83	1.18	6.22	7.69	0.99	1.02
731	1.00	1.01	0.97	1.05	0.83	1.34	7.79	10.63	0.96	1.03
732	1.01	1.00	0.92	1.08	0.85	1.14	6.85	9.49	0.85	1.06
734	1.00	0.99	0.92	1.05	0.75	1.12	7.71	10.61	0.91	1.03
736	0.98	0.97	0.94	1.00	0.80	1.12	6.15	7.58	0.92	1.01
737	1.01	1.01	0.99	1.03	0.82	1.29	7.48	9.55	0.97	1.03

Table A57: Kruskal-Wallis Test Results: Final MRA ASR by Other Variables (FULL Set)

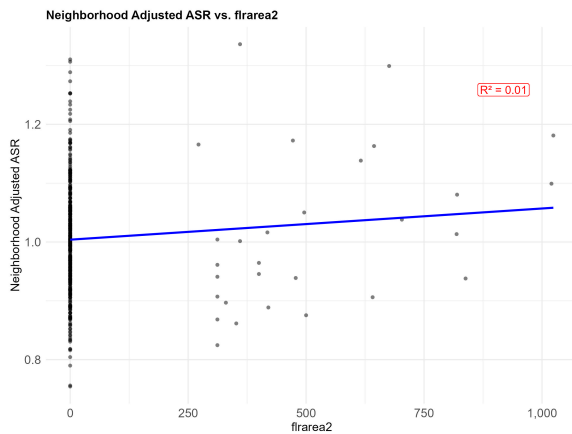
Variable Tested	p-value
heattype	0.157
gar_type	0.771
conduit	0.704
baths	0.824
bedrooms	0.100
fireplcs	0.620
baseste	0.389
carptfl	0.310
gr_manc1	0.132
cp_manc1	0.469



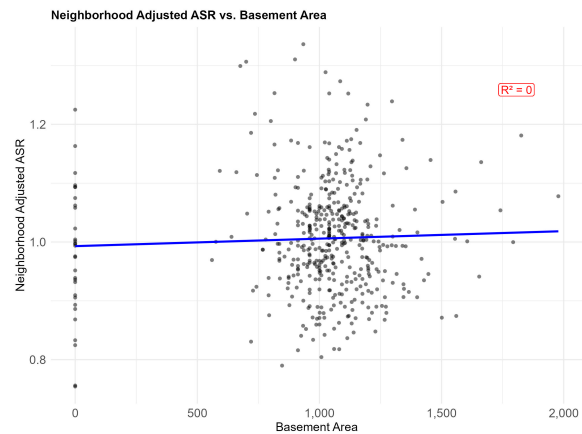
(a) ASR (Adjusted) vs. Crawl Area



(b) ASR (Adjusted) vs. Finished Basement Area

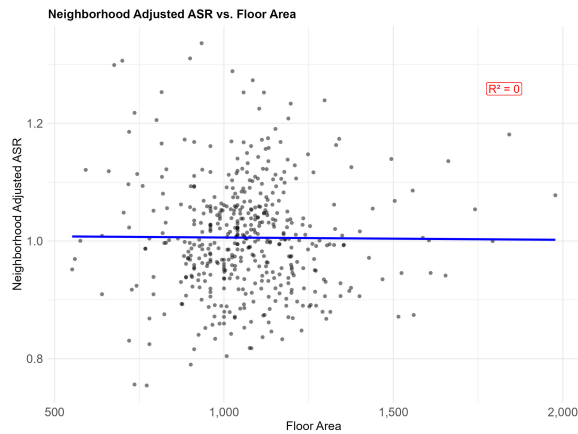


(c) ASR (Adjusted) vs. Floor Area 2

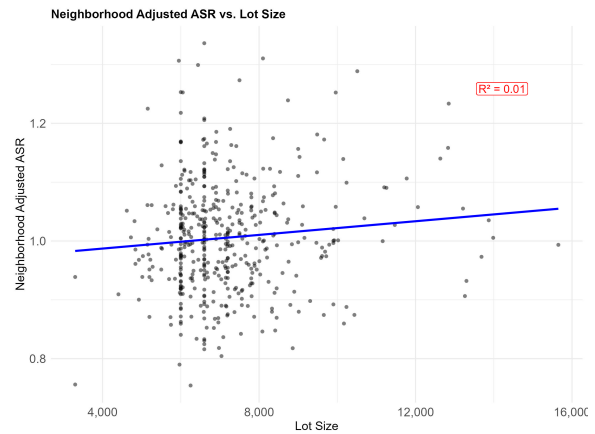


(d) ASR (Adjusted) vs. Basement Area

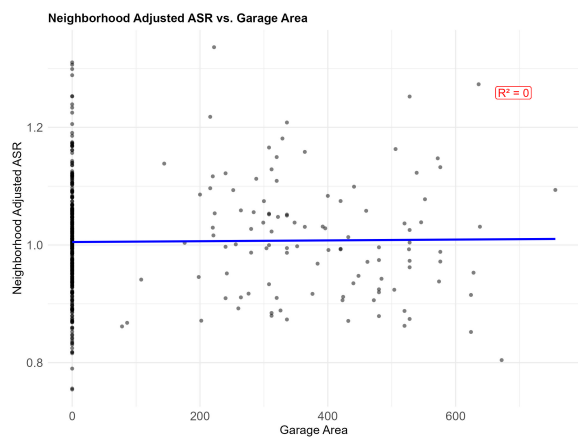
Figure A1: Scatter Plots of Assessed Sale Ratio (Neighborhood 2) vs. Various Area Variables



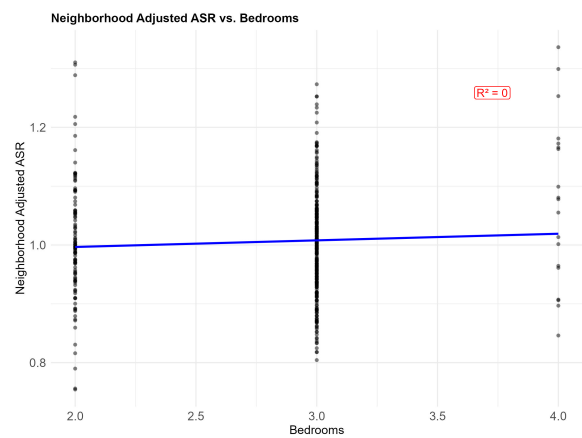
(a) ASR (Adjusted) vs. Floor Area 1



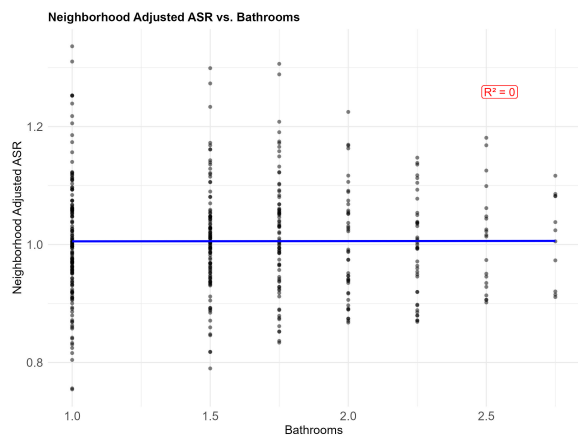
(b) ASR (Adjusted) vs. Lot Size



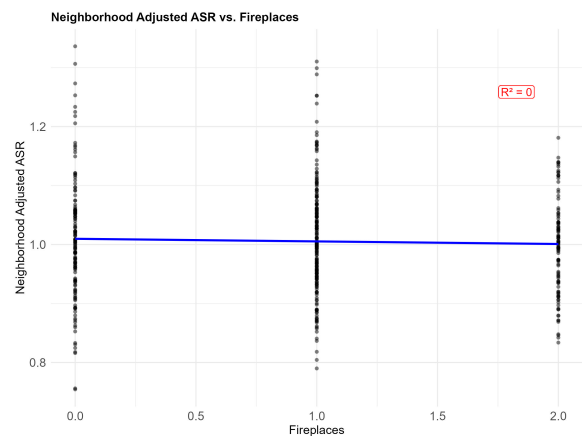
(c) ASR (Adjusted) vs. Garage Area



(d) ASR (Adjusted) vs. Bedrooms



(e) ASR (Adjusted) vs. Baths



(f) ASR (Neighborhood Adjusted) vs. Fireplaces

Figure A2: Scatter Plots of Assessed Sale Ratio (Neighborhood 2) vs. Various Property Characteristics

F Syntax

Full R syntax is attached separately; this section only focuses on the variable transformation that is referred in the main text.

F.1 Transformation

```
### Variable Transformation ###
sales_data%<>%mutate(adjusted_land_size = land_value_final
/ median_size_factor,
dlin_flrarea1 = MB_factor * flrarea1 * pctgood,
dlin_flrarea2 = MB_factor * flrarea2 * pctgood,
dlin_bsmtfin = MB_factor * bsmtfin * pctgood,
dlin_gar_area_attached = CP_factor * gar_area * (gar_type == 1) * pctgood,
dlin_gar_area_detached = CP_factor * gar_area * (gar_type == 2) * pctgood,
dlin_gar_area_detached = CP_factor * flrarea1 * (gar_type == 3) * pctgood,
dlin_cp_area = CP_factor * cp_area * pctgood,
gas_heattype = ifelse(heattype %in% c(1,2,7),1,0),
oil_heattype = ifelse(heattype %in% c(3,8),1,0),
electric_heattype = ifelse(heattype %in% c(4,5,6),1,0))
```